

TERRAFORM

Assignment 28 – Azure Provider: Authenticating using a Service Principal with a Client Secret

1. Identify your subscription (Used AZURE Cloudshell)

```
adrina [ ~ ]$ az account show --query id -o tsv
4aaefc3f-e730-4757-944c-df72e636bfe6
```

2. Create Service Principal (SP) with RBAC (Used AZURE Cloudshell)

```
adrina [ ~ ]$ SUBSCRIPTION_ID="4aaefc3f-e730-4757-944c-df72e636bfe6"
az ad sp create-for-rbac \
  --name "sp-terraform-epicbook" \
  --role "Contributor" \
  --scopes "/subscriptions/$SUBSCRIPTION_ID" \
  --years 1 \
  --query "[appId,password:password,tenant:tenant]" -o json
Creating 'Contributor' role assignment under scope '/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6'
The output includes credentials that you must protect. Be sure that you do not include these credentials in your code or check the credentials into your source control. For more information
, see https://aka.ms/azadsp-cli
{
  "appId": "607657a7-42d5-48c2-a2ee-a7c6b70789f6",
  "password": "kGz8Q~8WT~dgxR5XwCodkR8T_1fpEqFaISKYTawV",
  "tenant": "6e357e34-1d9d-4755-aea0-13cd3ef3d3da"
}
adrina [ ~ ]$
```

3. Export credentials for Terraform (Used Powershell)

```
PS C:\Program Files\Terraform> $env:ARM_CLIENT_ID="607657a7-42d5-48c2-a2ee-a7c6b70789f6"
PS C:\Program Files\Terraform> $env:ARM_CLIENT_SECRET="kGz8Q~8WT~dgxR5XwCodkR8T_1fpEqFaISKYTawV"
PS C:\Program Files\Terraform> $env:ARM_TENANT_ID="6e357e34-1d9d-4755-aea0-13cd3ef3d3da"
PS C:\Program Files\Terraform> $env:ARM_SUBSCRIPTION_ID="4aaefc3f-e730-4757-944c-df72e636bfe6"
PS C:\Program Files\Terraform>
```

4. Log out Azure CLI (to prove Terraform uses SP)

This step was not required as I used two different shells.

5. Create a minimal Terraform config to test (Used Powershell)

```
PS C:\WINDOWS\system32> cd\
PS C:\> cd 'C:\Program Files\Terraform'
PS C:\Program Files\Terraform> New-Item -Path .\main.tf -ItemType File

Directory: C:\Program Files\Terraform

Mode                LastWriteTime         Length Name
----                -
-a----             10/5/2025  10:51 PM              0 main.tf

PS C:\Program Files\Terraform> notepad .\main.tf
PS C:\Program Files\Terraform>
```

```
main.tf - Notepad
File Edit Format View Help
provider "azurerm" {
  features {}
  # Do NOT put subscription id.
}

resource "azurerm_resource_group" "rg" {
  name      = "rg-tf-sp-demo"
  location  = "East US"
}
```

terraform init

```
PS C:\Program Files\Terraform> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/azurerm...
- Installing hashicorp/azurerm v4.47.0...
- Installed hashicorp/azurerm v4.47.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Program Files\Terraform>
```

terraform plan

```
PS C:\Program Files\Terraform> terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

# azurerm_resource_group.rg will be created
+ resource "azurerm_resource_group" "rg" {
  + id           = (known after apply)
  + location     = "eastus"
  + name         = "rg-tf-sp-demo"
}

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + rg_name = "rg-tf-sp-demo"

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
PS C:\Program Files\Terraform>
```

terraform apply -auto-approve

```
PS C:\Program Files\Terraform> terraform apply -auto-approve

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # azurerm_resource_group.rg will be created
  + resource "azurerm_resource_group" "rg" {
    + id           = (known after apply)
    + location     = "eastus"
    + name         = "rg-tf-sp-demo"
  }

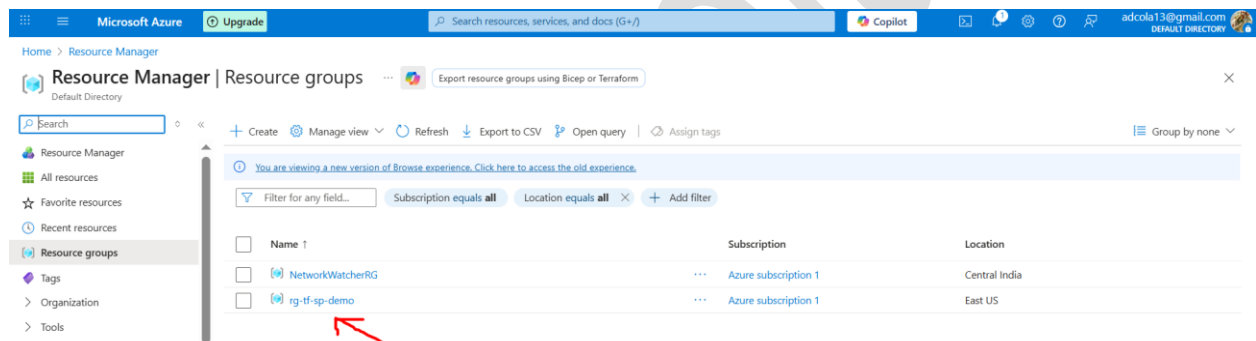
Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + rg_name = "rg-tf-sp-demo"
azurerm_resource_group.rg: Creating...
azurerm_resource_group.rg: Still creating... [00m10s elapsed]
azurerm_resource_group.rg: Creation complete after 15s [id=/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/rg-tf-sp-demo]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:
rg_name = "rg-tf-sp-demo"
PS C:\Program Files\Terraform>
```

6. Verification that resource group was created in my Azure.



Assignment 29 – Multi-Cloud + Multi-Region Deployment with Terraform (Azure + AWS)

1. Plan your names (write them down before you start). Ensure S3 names are globally unique.

Cloud	Region	Resource Type	Name
AWS	us-east-1	S3 Bucket	adrina-dev-assets-use1
AWS	eu-central-1	S3 Bucket	adrina-dev-assets-euc1
Azure	East US	Storage Account	adrinadevassetsuse1
Azure	West Europe	Storage Account	adrinadevassetsweu

2. AWS (two folders)

```
PS C:\Program Files\Terraform> mkdir devops_aws_us_east1
```

Directory: C:\Program Files\Terraform

Mode	LastWriteTime	Length	Name
d-----	10/6/2025 11:25 PM		devops_aws_us_east1

```
PS C:\Program Files\Terraform> mkdir devops_aws_eu_cent1
```

Directory: C:\Program Files\Terraform

Mode	LastWriteTime	Length	Name
d-----	10/6/2025 11:25 PM		devops_aws_eu_cent1

```
PS C:\Program Files\Terraform> cd .\devops_aws_us_east1\  
PS C:\Program Files\Terraform\devops_aws_us_east1> New-Item "main.tf" -ItemType File
```

Directory: C:\Program Files\Terraform\devops_aws_us_east1

Mode	LastWriteTime	Length	Name
-a----	10/6/2025 11:33 PM	0	main.tf

```
PS C:\Program Files\Terraform\devops_aws_us_east1> notepad .\main.tf  
PS C:\Program Files\Terraform\devops_aws_us_east1> notepad .\main.tf  
PS C:\Program Files\Terraform\devops_aws_us_east1> notepad .\main.tf  
PS C:\Program Files\Terraform\devops_aws_us_east1>
```

main.tf - Notepad

File Edit Format View Help

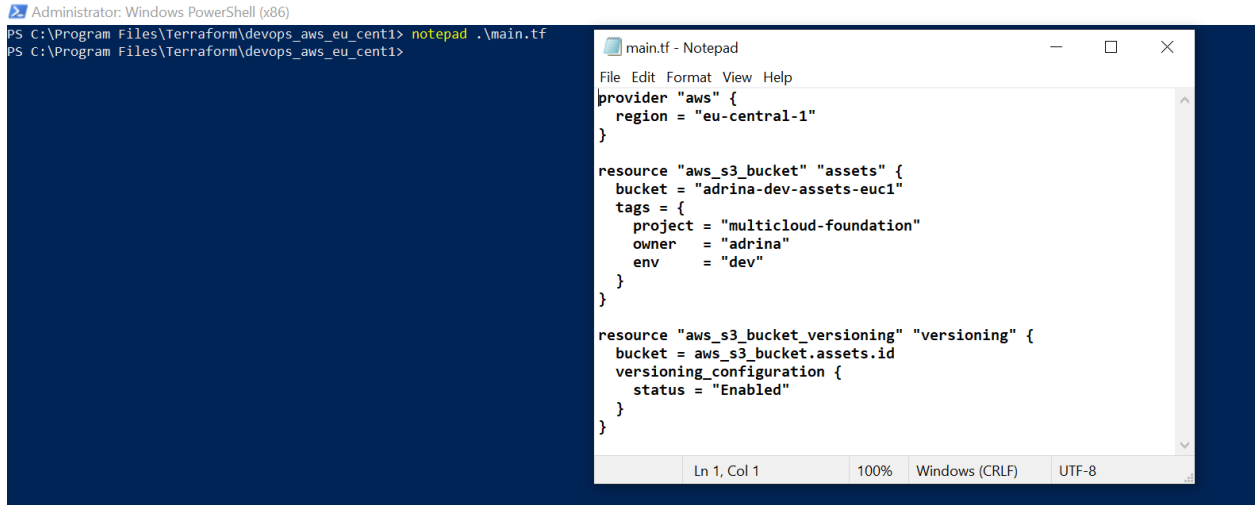
```
provider "aws" {  
  region = "us-east-1"  
}  
  
resource "aws_s3_bucket" "assets" {  
  bucket = "adrina-dev-assets-use1"  
  tags = {  
    project = "multicloud-foundation"  
    owner   = "adrina"  
    env     = "dev"  
  }  
}  
  
resource "aws_s3_bucket_versioning" "versioning" {  
  bucket = aws_s3_bucket.assets.id  
  versioning_configuration {  
    status = "Enabled"  
  }  
}
```

Ln 1, Col 1

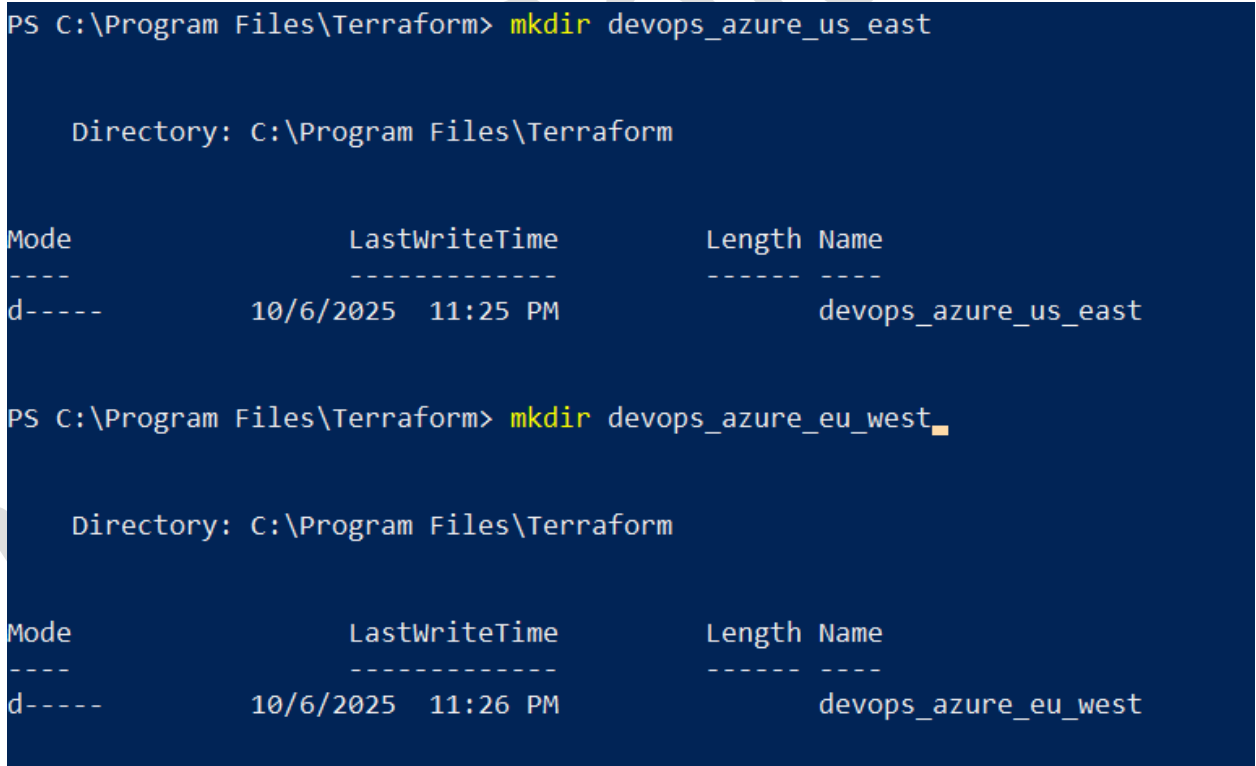
100%

Windows (CRLF)

UTF-8



3. Azure (two folders)



```
Select Administrator: Windows PowerShell (x86)
PS C:\Program Files\Terraform> cd .\devops_azure_us_east\
PS C:\Program Files\Terraform\devops_azure_us_east> notepad .\main.tf
PS C:\Program Files\Terraform\devops_azure_us_east>

main.tf - Notepad
File Edit Format View Help
terraform {
  required_providers {
    azurerm = {
      source = "hashicorp/azurerm"
      version = "~> 4.0"
    }
  }
}

provider "azurerm" {
  features {}
}

resource "azurerm_resource_group" "rg" {
  name     = "rg-dev-assets-eastus"
  location = "East US"
  tags = {
    project = "multicloud-foundation"
    owner   = "adrina"
    env     = "dev"
  }
}

resource "azurerm_storage_account" "storage" {
  name                        = "adrinadevassetsuse1"
  resource_group_name        = azurerm_resource_group.rg.name
  location                   = azurerm_resource_group.rg.location
  account_tier               = "Standard"
  account_replication_type   = "LRS"
  tags = {
    project = "multicloud-foundation"
    owner   = "adrina"
    env     = "dev"
  }
}
```

```
Administrator: Windows PowerShell (x86)
PS C:\Program Files\Terraform\devops_azure_eu_west> notepad .\main.tf
PS C:\Program Files\Terraform\devops_azure_eu_west>

main.tf - Notepad
File Edit Format View Help
terraform {
  required_providers {
    azurerm = {
      source = "hashicorp/azurerm"
      version = "~> 4.0"
    }
  }
}

provider "azurerm" {
  features {}
}

resource "azurerm_resource_group" "rg" {
  name     = "rg-dev-assets-westeu"
  location = "westeurope"
  tags = {
    project = "multicloud-foundation"
    owner   = "adrina"
    env     = "dev"
  }
}

resource "azurerm_storage_account" "storage" {
  name                        = "adrinadevassetseu1"
  resource_group_name        = azurerm_resource_group.rg.name
  location                   = azurerm_resource_group.rg.location
  account_tier               = "Standard"
  account_replication_type   = "LRS"
  tags = {
    project = "multicloud-foundation"
    owner   = "adrina"
    env     = "dev"
  }
}
```

4. Apply safely
 - a. AWS US EAST

```
PS C:\Program Files\Terraform\devops_aws_us_east1> aws configure
AWS Access Key ID [*****U6K3]: AKIAJ4R0433D02119CM02
AWS Secret Access Key [*****MHju]: G16DUTmE51w8V9w4y4dew4157gjb35-i1W04
Default region name [us-east-1]: us-east-1
Default output format [json]: json
```

i. Terraform Init

 Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\devops_aws_us_east1> terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.15.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

ii. Terraform plan

```
PS C:\Program Files\Terraform\devops_aws_us_east1> terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  ~ create
Terraform will perform the following actions:

# aws_s3_bucket.assets will be created
resource "aws_s3_bucket" "assets" {
  ~ acceleration_status = (known after apply)
  ~ acl                 = (known after apply)
  ~ arn                 = (known after apply)
  ~ bucket              = "adrina-dev-assets-us-east-1"
  ~ bucket_domain_name = (known after apply)
  ~ bucket_prefix       = (known after apply)
  ~ bucket_region       = (known after apply)
  ~ bucket_regional_domain_name = (known after apply)
  ~ force_destroy       = false
  ~ hosted_zone_id      = (known after apply)
  ~ id                  = (known after apply)
  ~ object_lock_enabled = (known after apply)
  ~ policy              = (known after apply)
  ~ region              = "us-east-1"
  ~ request_payer       = (known after apply)
  ~ tags                = {
    ~ "env"      = "dev"
    ~ "owner"    = "adrina"
    ~ "project"  = "multicloud-foundation"
  }
  ~ tags_all = {
    ~ "env"      = "dev"
    ~ "owner"    = "adrina"
    ~ "project"  = "multicloud-foundation"
  }
  ~ website_domain      = (known after apply)
  ~ website_endpoint    = (known after apply)

  ~ cors_rule (known after apply)
  ~ grant (known after apply)
  ~ lifecycle_rule (known after apply)
  ~ logging (known after apply)
```

```

+ object_lock_configuration (known after apply)
+ replication_configuration (known after apply)
+ server_side_encryption_configuration (known after apply)
+ versioning (known after apply)
+ website (known after apply)
}

# aws_s3_bucket.versioning will be created
+ resource "aws_s3_bucket_versioning" "versioning" {
+   bucket = (known after apply)
+   id     = (known after apply)
+   region = "us-east-1"

+   versioning_configuration {
+     mfa_delete = (known after apply)
+     status     = "Enabled"
+   }
+ }

Plan: 2 to add, 0 to change, 0 to destroy.

```

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
PS C:\Program Files\Terraform\devops_aws_us_east1>

iii. Terraform apply -auto-approve

Administrator: Windows PowerShell (x86)

PS C:\Program Files\Terraform\devops_aws_us_east1> Terraform apply -auto-approve

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```

# aws_s3_bucket.assets will be created
+ resource "aws_s3_bucket" "assets" {
+   acceleration_status = (known after apply)
+   acl                 = (known after apply)
+   arn                 = (known after apply)
+   bucket              = "adrina-dev-assets-use1"
+   bucket_domain_name = (known after apply)
+   bucket_prefix       = (known after apply)
+   bucket_region       = (known after apply)
+   bucket_regional_domain_name = (known after apply)
+   force_destroy       = false
+   hosted_zone_id      = (known after apply)
+   id                  = (known after apply)
+   object_lock_enabled = (known after apply)
+   policy              = (known after apply)
+   region              = "us-east-1"
+   request_payer       = (known after apply)
+   tags                = {
+     "env"      = "dev"
+     "owner"    = "adrina"
+     "project"  = "multicloud-foundation"
+   }
+   tags_all          = {
+     "env"      = "dev"
+     "owner"    = "adrina"
+     "project"  = "multicloud-foundation"
+   }
+   website_domain     = (known after apply)
+   website_endpoint   = (known after apply)

+ cors_rule (known after apply)

+ grant (known after apply)

+ lifecycle_rule (known after apply)

+ logging (known after apply)

```

```

+ object_lock_configuration (known after apply)
+ replication_configuration (known after apply)
+ server_side_encryption_configuration (known after apply)
+ versioning (known after apply)
+ website (known after apply)
}

# aws_s3_bucket_versioning.versioning will be created
+ resource "aws_s3_bucket_versioning" "versioning" {
+   bucket = (known after apply)
+   id     = (known after apply)
+   region = "us-east-1"

+   versioning_configuration {
+     mfa_delete = (known after apply)
+     status     = "Enabled"
+   }
+ }

Plan: 2 to add, 0 to change, 0 to destroy.
aws_s3_bucket.assets: Creating...
aws_s3_bucket.assets: Creation complete after 7s [id=adrina-dev-assets-use1]
aws_s3_bucket_versioning.versioning: Creating...
aws_s3_bucket_versioning.versioning: Creation complete after 2s [id=adrina-dev-assets-use1]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
PS C:\Program Files\Terraform\devops_aws_us_east1>

```

b. AWS EU CENTRAL

i. Terraform init

 Administrator: Windows PowerShell (x86)

```

PS C:\Program Files\Terraform> cd .\devops_aws_eu_cent1\
PS C:\Program Files\Terraform\devops_aws_eu_cent1> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.15.0...
- Installed hashicorp/aws v6.15.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

```

ii. Terraform plan

Administrator: Windows PowerShell (x86)

PS C:\Program Files\Terraform\devops_aws_eu_cent1> terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# aws_s3_bucket.assets will be created
+ resource "aws_s3_bucket" "assets" {
+   acceleration_status      = (known after apply)
+   acl                      = (known after apply)
+   arn                     = (known after apply)
+   bucket                  = "adrina-dev-assets-euc1"
+   bucket_domain_name      = (known after apply)
+   bucket_prefix           = (known after apply)
+   bucket_region           = (known after apply)
+   bucket_regional_domain_name = (known after apply)
+   force_destroy           = false
+   hosted_zone_id          = (known after apply)
+   id                      = (known after apply)
+   object_lock_enabled      = (known after apply)
+   policy                   = (known after apply)
+   region                  = "eu-central-1"
+   request_payer            = (known after apply)
+   tags                    = {
+     "env"      = "dev"
+     "owner"    = "adrina"
+     "project"  = "multicloud-foundation"
+   }
+   tags_all              = {
+     "env"      = "dev"
+     "owner"    = "adrina"
+     "project"  = "multicloud-foundation"
+   }
+   website_domain        = (known after apply)
+   website_endpoint      = (known after apply)

+   cors_rule (known after apply)

+   grant (known after apply)

+   lifecycle_rule (known after apply)

+   logging (known after apply)
```

```

+ object_lock_configuration (known after apply)
+ replication_configuration (known after apply)
+ server_side_encryption_configuration (known after apply)
+ versioning (known after apply)
+ website (known after apply)
}

# aws_s3_bucket_versioning.versioning will be created
+ resource "aws_s3_bucket_versioning" "versioning" {
+   bucket = (known after apply)
+   id     = (known after apply)
+   region = "eu-central-1"

+   versioning_configuration {
+     mfa_delete = (known after apply)
+     status     = "Enabled"
+   }
+ }

```

Plan: 2 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply".
 PS C:\Program Files\Terraform\devops_aws_eu_cent1>

iii. Terraform apply -auto-approve

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\devops_aws_eu_cent1> Terraform apply -auto-approve
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# aws_s3_bucket.assets will be created
+ resource "aws_s3_bucket" "assets" {
  + acceleration_status      = (known after apply)
  + acl                      = (known after apply)
  + arn                      = (known after apply)
  + bucket                  = "adrina-dev-assets-euc1"
  + bucket_domain_name      = (known after apply)
  + bucket_prefix           = (known after apply)
  + bucket_region           = (known after apply)
  + bucket_regional_domain_name = (known after apply)
  + force_destroy           = false
  + hosted_zone_id          = (known after apply)
  + id                      = (known after apply)
  + object_lock_enabled      = (known after apply)
  + policy                  = (known after apply)
  + region                  = "eu-central-1"
  + request_payer            = (known after apply)
  + tags                    = {
    + "env"      = "dev"
    + "owner"    = "adrina"
    + "project" = "multicloud-foundation"
  }
  + tags_all              = {
    + "env"      = "dev"
    + "owner"    = "adrina"
    + "project" = "multicloud-foundation"
  }
  + website_domain        = (known after apply)
  + website_endpoint      = (known after apply)

  + cors_rule (known after apply)

  + grant (known after apply)

  + lifecycle_rule (known after apply)

  + logging (known after apply)
```

```
  + object_lock_configuration (known after apply)

  + replication_configuration (known after apply)

  + server_side_encryption_configuration (known after apply)

  + versioning (known after apply)

  + website (known after apply)
}
```

```
# aws_s3_bucket_versioning.versioning will be created
+ resource "aws_s3_bucket_versioning" "versioning" {
  + bucket = (known after apply)
  + id     = (known after apply)
  + region = "eu-central-1"

  + versioning_configuration {
    + mfa_delete = (known after apply)
    + status     = "Enabled"
  }
}
```

Plan: 2 to add, 0 to change, 0 to destroy.

aws_s3_bucket.assets: Creating...

aws_s3_bucket.assets: Creation complete after 4s [id=adrina-dev-assets-euc1]

aws_s3_bucket_versioning.versioning: Creating...

aws_s3_bucket_versioning.versioning: Creation complete after 2s [id=adrina-dev-assets-euc1]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

```
PS C:\Program Files\Terraform\devops_aws_eu_cent1>
```

c. AZURE US EAST

i. Terraform init

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform> cd .\devops_azure_us_east\
PS C:\Program Files\Terraform\devops_azure_us_east> $env:ARM_CLIENT_ID="607657a7-42d5-48c2-a2ee-a7c6b70789f6"
PS C:\Program Files\Terraform\devops_azure_us_east> $env:ARM_CLIENT_SECRET="kGz8Q~8WT~dgXR5XwCodkR8T_1fpEqFaISKYTawv"
PS C:\Program Files\Terraform\devops_azure_us_east> $env:ARM_TENANT_ID="6e357e34-1d9d-4755-aea0-13cd3ef3ddda"
PS C:\Program Files\Terraform\devops_azure_us_east> $env:ARM_SUBSCRIPTION_ID="4aaefc3f-e730-4757-944c-df72e636bfe6"
PS C:\Program Files\Terraform\devops_azure_us_east> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 4.0"...
- Installing hashicorp/azurerm v4.47.0...
- Installed hashicorp/azurerm v4.47.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Program Files\Terraform\devops_azure_us_east>
```

ii. Terraform plan

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\devops_azure_us_east> terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# azurerm_resource_group.rg will be created
+ resource "azurerm_resource_group" "rg" {
  + id           = (known after apply)
  + location     = "eastus"
  + name        = "rg-dev-assets-eastus"
  + tags        = {
    + "env"      = "dev"
    + "owner"    = "adrina"
    + "project"  = "multicloud-foundation"
  }
}

# azurerm_storage_account.storage will be created
+ resource "azurerm_storage_account" "storage" {
  + access_tier                = (known after apply)
  + account_kind               = "StorageV2"
  + account_replication_type   = "LRS"
  + account_tier               = "Standard"
  + allow_nested_items_to_be_public = true
  + cross_tenant_replication_enabled = false
  + default_to_oauth_authentication = false
  + dns_endpoint_type          = "Standard"
  + https_traffic_only_enabled = true
  + id                         = (known after apply)
  + infrastructure_encryption_enabled = false
  + is_hns_enabled            = false
  + large_file_share_enabled   = (known after apply)
  + local_user_enabled         = true
  + location                   = "eastus"
  + min_tls_version            = "TLS1_2"
  + name                       = "adrinadevassetsuse1"
  + nfsv3_enabled              = false
  + primary_access_key         = (sensitive value)
  + primary_blob_connection_string = (sensitive value)
  + primary_blob_endpoint      = (known after apply)
  + primary_blob_host          = (known after apply)
  + primary_blob_internet_endpoint = (known after apply)
}
```

```
+ primary_blob_internet_host = (known after apply)
+ primary_blob_microsoft_endpoint = (known after apply)
+ primary_blob_microsoft_host = (known after apply)
+ primary_connection_string = (sensitive value)
+ primary_dfs_endpoint = (known after apply)
+ primary_dfs_host = (known after apply)
+ primary_dfs_internet_endpoint = (known after apply)
+ primary_dfs_internet_host = (known after apply)
+ primary_dfs_microsoft_endpoint = (known after apply)
+ primary_dfs_microsoft_host = (known after apply)
+ primary_file_endpoint = (known after apply)
+ primary_file_host = (known after apply)
+ primary_file_internet_endpoint = (known after apply)
+ primary_file_internet_host = (known after apply)
+ primary_file_microsoft_endpoint = (known after apply)
+ primary_file_microsoft_host = (known after apply)
+ primary_location = (known after apply)
+ primary_queue_endpoint = (known after apply)
+ primary_queue_host = (known after apply)
+ primary_queue_microsoft_endpoint = (known after apply)
+ primary_queue_microsoft_host = (known after apply)
+ primary_table_endpoint = (known after apply)
+ primary_table_host = (known after apply)
+ primary_table_microsoft_endpoint = (known after apply)
+ primary_table_microsoft_host = (known after apply)
+ primary_web_endpoint = (known after apply)
+ primary_web_host = (known after apply)
+ primary_web_internet_endpoint = (known after apply)
+ primary_web_internet_host = (known after apply)
+ primary_web_microsoft_endpoint = (known after apply)
+ primary_web_microsoft_host = (known after apply)
+ public_network_access_enabled = true
+ queue_encryption_key_type = "Service"
+ resource_group_name = "rg-dev-assets-eastus"
+ secondary_access_key = (sensitive value)
+ secondary_blob_connection_string = (sensitive value)
+ secondary_blob_endpoint = (known after apply)
+ secondary_blob_host = (known after apply)
+ secondary_blob_internet_endpoint = (known after apply)
+ secondary_blob_internet_host = (known after apply)
+ secondary_blob_microsoft_endpoint = (known after apply)
+ secondary_blob_microsoft_host = (known after apply)
+ secondary_connection_string = (sensitive value)
+ secondary_dfs_endpoint = (known after apply)
```

```
+ secondary_dfs_host = (known after apply)
+ secondary_dfs_internet_endpoint = (known after apply)
+ secondary_dfs_internet_host = (known after apply)
+ secondary_dfs_microsoft_endpoint = (known after apply)
+ secondary_dfs_microsoft_host = (known after apply)
+ secondary_file_endpoint = (known after apply)
+ secondary_file_host = (known after apply)
+ secondary_file_internet_endpoint = (known after apply)
+ secondary_file_internet_host = (known after apply)
+ secondary_file_microsoft_endpoint = (known after apply)
+ secondary_file_microsoft_host = (known after apply)
+ secondary_location = (known after apply)
+ secondary_queue_endpoint = (known after apply)
+ secondary_queue_host = (known after apply)
+ secondary_queue_microsoft_endpoint = (known after apply)
+ secondary_queue_microsoft_host = (known after apply)
+ secondary_table_endpoint = (known after apply)
+ secondary_table_host = (known after apply)
+ secondary_table_microsoft_endpoint = (known after apply)
+ secondary_table_microsoft_host = (known after apply)
+ secondary_web_endpoint = (known after apply)
+ secondary_web_host = (known after apply)
+ secondary_web_internet_endpoint = (known after apply)
+ secondary_web_internet_host = (known after apply)
+ secondary_web_microsoft_endpoint = (known after apply)
+ secondary_web_microsoft_host = (known after apply)
+ sftp_enabled = false
+ shared_access_key_enabled = true
+ table_encryption_key_type = "Service"
+ tags = {
+   "env" = "dev"
+   "owner" = "adrina"
+   "project" = "multicloud-foundation"
+ }

+ blob_properties (known after apply)

+ network_rules (known after apply)

+ queue_properties (known after apply)

+ routing (known after apply)

+ share_properties (known after apply)

+ static_website (known after apply)
}

Plan: 2 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply"
PS C:\Program Files\Terraform\devops_azure_us_east>
```

iii. Terraform apply -auto-approve

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\devops_azure_us_east> Terraform apply -auto-approve
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# azurerm_resource_group.rg will be created
+ resource "azurerm_resource_group" "rg" {
  + id           = (known after apply)
  + location     = "eastus"
  + name         = "rg-dev-assets-eastus"
  + tags        = {
    + "env"      = "dev"
    + "owner"    = "adrina"
    + "project" = "multicloud-foundation"
  }
}

# azurerm_storage_account.storage will be created
+ resource "azurerm_storage_account" "storage" {
  + access_tier                = (known after apply)
  + account_kind               = "StorageV2"
  + account_replication_type   = "LRS"
  + account_tier               = "Standard"
  + allow_nested_items_to_be_public = true
  + cross_tenant_replication_enabled = false
  + default_to_oauth_authentication = false
  + dns_endpoint_type          = "Standard"
  + https_traffic_only_enabled   = true
  + id                         = (known after apply)
  + infrastructure_encryption_enabled = false
  + is_hns_enabled             = false
  + large_file_share_enabled    = (known after apply)
  + local_user_enabled         = true
  + location                   = "eastus"
  + min_tls_version            = "TLS1_2"
  + name                       = "adrinadevassetsuse1"
  + nfsv3_enabled              = false
  + primary_access_key         = (sensitive value)
  + primary_blob_connection_string = (sensitive value)
  + primary_blob_endpoint      = (known after apply)
  + primary_blob_host          = (known after apply)
  + primary_blob_internet_endpoint = (known after apply)
}
```

```
+ primary_blob_host = (known after apply)
+ primary_blob_internet_endpoint = (known after apply)
+ primary_blob_internet_host = (known after apply)
+ primary_blob_microsoft_endpoint = (known after apply)
+ primary_blob_microsoft_host = (known after apply)
+ primary_connection_string = (sensitive value)
+ primary_dfs_endpoint = (known after apply)
+ primary_dfs_host = (known after apply)
+ primary_dfs_internet_endpoint = (known after apply)
+ primary_dfs_internet_host = (known after apply)
+ primary_dfs_microsoft_endpoint = (known after apply)
+ primary_dfs_microsoft_host = (known after apply)
+ primary_file_endpoint = (known after apply)
+ primary_file_host = (known after apply)
+ primary_file_internet_endpoint = (known after apply)
+ primary_file_internet_host = (known after apply)
+ primary_file_microsoft_endpoint = (known after apply)
+ primary_file_microsoft_host = (known after apply)
+ primary_location = (known after apply)
+ primary_queue_endpoint = (known after apply)
+ primary_queue_host = (known after apply)
+ primary_queue_microsoft_endpoint = (known after apply)
+ primary_queue_microsoft_host = (known after apply)
+ primary_table_endpoint = (known after apply)
+ primary_table_host = (known after apply)
+ primary_table_microsoft_endpoint = (known after apply)
+ primary_table_microsoft_host = (known after apply)
+ primary_web_endpoint = (known after apply)
+ primary_web_host = (known after apply)
+ primary_web_internet_endpoint = (known after apply)
+ primary_web_internet_host = (known after apply)
+ primary_web_microsoft_endpoint = (known after apply)
+ primary_web_microsoft_host = (known after apply)
+ public_network_access_enabled = true
+ queue_encryption_key_type = "Service"
+ resource_group_name = "rg-dev-assets-eastus"
+ secondary_access_key = (sensitive value)
+ secondary_blob_connection_string = (sensitive value)
+ secondary_blob_endpoint = (known after apply)
+ secondary_blob_host = (known after apply)
+ secondary_blob_internet_endpoint = (known after apply)
+ secondary_blob_internet_host = (known after apply)
+ secondary_blob_microsoft_endpoint = (known after apply)
+ secondary_blob_microsoft_host = (known after apply)
```

```

+ secondary_blob_microsoft_host = (known after apply)
+ secondary_connection_string    = (sensitive value)
+ secondary_dfs_endpoint         = (known after apply)
+ secondary_dfs_host             = (known after apply)
+ secondary_dfs_internet_endpoint = (known after apply)
+ secondary_dfs_internet_host    = (known after apply)
+ secondary_dfs_microsoft_endpoint = (known after apply)
+ secondary_dfs_microsoft_host   = (known after apply)
+ secondary_file_endpoint        = (known after apply)
+ secondary_file_host            = (known after apply)
+ secondary_file_internet_endpoint = (known after apply)
+ secondary_file_internet_host   = (known after apply)
+ secondary_file_microsoft_endpoint = (known after apply)
+ secondary_file_microsoft_host  = (known after apply)
+ secondary_location             = (known after apply)
+ secondary_queue_endpoint       = (known after apply)
+ secondary_queue_host           = (known after apply)
+ secondary_queue_microsoft_endpoint = (known after apply)
+ secondary_queue_microsoft_host  = (known after apply)
+ secondary_table_endpoint       = (known after apply)
+ secondary_table_host           = (known after apply)
+ secondary_table_microsoft_endpoint = (known after apply)
+ secondary_table_microsoft_host  = (known after apply)
+ secondary_web_endpoint         = (known after apply)
+ secondary_web_host             = (known after apply)
+ secondary_web_internet_endpoint = (known after apply)
+ secondary_web_internet_host    = (known after apply)
+ secondary_web_microsoft_endpoint = (known after apply)
+ secondary_web_microsoft_host   = (known after apply)
+ sftp_enabled                   = false
+ shared_access_key_enabled      = true
+ table_encryption_key_type      = "Service"
+ tags                           = {
  + "env"      = "dev"
  + "owner"    = "adrina"
  + "project"  = "multicloud-foundation"
}

+ blob_properties (known after apply)

+ network_rules (known after apply)

+ queue_properties (known after apply)

```

```

+ routing (known after apply)

+ share_properties (known after apply)

+ static_website (known after apply)
}

Plan: 2 to add, 0 to change, 0 to destroy.
azurerm_resource_group.rg: Creating...
azurerm_resource_group.rg: Still creating... [00m10s elapsed]
azurerm_resource_group.rg: Creation complete after 17s [id=/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/rg-dev-assets-eastus]
azurerm_storage_account.storage: Creating...
azurerm_storage_account.storage: Still creating... [00m10s elapsed]
azurerm_storage_account.storage: Still creating... [00m20s elapsed]
azurerm_storage_account.storage: Still creating... [00m30s elapsed]
azurerm_storage_account.storage: Still creating... [00m40s elapsed]
azurerm_storage_account.storage: Still creating... [00m50s elapsed]
azurerm_storage_account.storage: Still creating... [01m00s elapsed]
azurerm_storage_account.storage: Still creating... [01m10s elapsed]
azurerm_storage_account.storage: Creation complete after 1m14s [id=/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/rg-dev-assets-eastus/providers/Microsoft.Storage/storageAccounts/adrinadevassetsuse1]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
PS C:\Program Files\Terraform\devops_azure_us_east>

```

d. AZURE EU WEST

i. Terraform init

```
PS C:\Program Files\Terraform> cd .\devops_azure_eu_west\  
PS C:\Program Files\Terraform\devops_azure_eu_west> terraform init  
Initializing the backend...  
Initializing provider plugins...  
- Finding hashicorp/azurerm versions matching "~> 4.0"...  
- Installing hashicorp/azurerm v4.47.0...  
- Installed hashicorp/azurerm v4.47.0 (signed by HashiCorp)  
Terraform has created a lock file .terraform.lock.hcl to record the provider  
selections it made above. Include this file in your version control repository  
so that Terraform can guarantee to make the same selections by default when  
you run "terraform init" in the future.  
  
Terraform has been successfully initialized!  
  
You may now begin working with Terraform. Try running "terraform plan" to see  
any changes that are required for your infrastructure. All Terraform commands  
should now work.  
  
If you ever set or change modules or backend configuration for Terraform,  
rerun this command to reinitialize your working directory. If you forget, other  
commands will detect it and remind you to do so if necessary.  
PS C:\Program Files\Terraform\devops_azure_eu_west>
```

ii. Terraform plan

```
PS C:\Program Files\Terraform\devops_azure_eu_west> terraform plan
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```
# azure_rm_resource_group.rg will be created
+ resource "azure_rm_resource_group" "rg" {
  + id            = (known after apply)
  + location      = "westeurope"
  + name          = "rg-dev-assets-westeu"
  + tags          = {
    + "env"       = "dev"
    + "owner"     = "adrina"
    + "project"   = "multicloud-foundation"
  }
}

# azure_rm_storage_account.storage will be created
+ resource "azure_rm_storage_account" "storage" {
  + access_tier                = (known after apply)
  + account_kind                = "StorageV2"
  + account_replication_type    = "LRS"
  + account_tier                = "Standard"
  + allow_nested_items_to_be_public = true
  + cross_tenant_replication_enabled = false
  + default_to_oauth_authentication = false
  + dns_endpoint_type           = "Standard"
  + https_traffic_only_enabled   = true
  + id                          = (known after apply)
  + infrastructure_encryption_enabled = false
  + is_hns_enabled              = false
  + large_file_share_enabled     = (known after apply)
  + local_user_enabled           = true
  + location                    = "westeurope"
  + min_tls_version              = "TLS1_2"
  + name                        = "adrinadevasetseuw1"
  + nfsv3_enabled               = false
  + primary_access_key           = (sensitive value)
  + primary_blob_connection_string = (sensitive value)
  + primary_blob_endpoint        = (known after apply)
  + primary_blob_host             = (known after apply)
  + primary_blob_internet_endpoint = (known after apply)
}
```

```
+ primary_blob_internet_host = (known after apply)
+ primary_blob_microsoft_endpoint = (known after apply)
+ primary_blob_microsoft_host = (known after apply)
+ primary_connection_string = (sensitive value)
+ primary_dfs_endpoint = (known after apply)
+ primary_dfs_host = (known after apply)
+ primary_dfs_internet_endpoint = (known after apply)
+ primary_dfs_internet_host = (known after apply)
+ primary_dfs_microsoft_endpoint = (known after apply)
+ primary_dfs_microsoft_host = (known after apply)
+ primary_file_endpoint = (known after apply)
+ primary_file_host = (known after apply)
+ primary_file_internet_endpoint = (known after apply)
+ primary_file_internet_host = (known after apply)
+ primary_file_microsoft_endpoint = (known after apply)
+ primary_file_microsoft_host = (known after apply)
+ primary_location = (known after apply)
+ primary_queue_endpoint = (known after apply)
+ primary_queue_host = (known after apply)
+ primary_queue_microsoft_endpoint = (known after apply)
+ primary_queue_microsoft_host = (known after apply)
+ primary_table_endpoint = (known after apply)
+ primary_table_host = (known after apply)
+ primary_table_microsoft_endpoint = (known after apply)
+ primary_table_microsoft_host = (known after apply)
+ primary_web_endpoint = (known after apply)
+ primary_web_host = (known after apply)
+ primary_web_internet_endpoint = (known after apply)
+ primary_web_internet_host = (known after apply)
+ primary_web_microsoft_endpoint = (known after apply)
+ primary_web_microsoft_host = (known after apply)
+ public_network_access_enabled = true
+ queue_encryption_key_type = "Service"
+ resource_group_name = "rg-dev-assets-westeu"
+ secondary_access_key = (sensitive value)
+ secondary_blob_connection_string = (sensitive value)
+ secondary_blob_endpoint = (known after apply)
+ secondary_blob_host = (known after apply)
+ secondary_blob_internet_endpoint = (known after apply)
+ secondary_blob_internet_host = (known after apply)
+ secondary_blob_microsoft_endpoint = (known after apply)
+ secondary_blob_microsoft_host = (known after apply)
+ secondary_connection_string = (sensitive value)
+ secondary_dfs_endpoint = (known after apply)
```

```

+ secondary_dfs_host = (known after apply)
+ secondary_dfs_internet_endpoint = (known after apply)
+ secondary_dfs_internet_host = (known after apply)
+ secondary_dfs_microsoft_endpoint = (known after apply)
+ secondary_dfs_microsoft_host = (known after apply)
+ secondary_file_endpoint = (known after apply)
+ secondary_file_host = (known after apply)
+ secondary_file_internet_endpoint = (known after apply)
+ secondary_file_internet_host = (known after apply)
+ secondary_file_microsoft_endpoint = (known after apply)
+ secondary_file_microsoft_host = (known after apply)
+ secondary_location = (known after apply)
+ secondary_queue_endpoint = (known after apply)
+ secondary_queue_host = (known after apply)
+ secondary_queue_microsoft_endpoint = (known after apply)
+ secondary_queue_microsoft_host = (known after apply)
+ secondary_table_endpoint = (known after apply)
+ secondary_table_host = (known after apply)
+ secondary_table_microsoft_endpoint = (known after apply)
+ secondary_table_microsoft_host = (known after apply)
+ secondary_web_endpoint = (known after apply)
+ secondary_web_host = (known after apply)
+ secondary_web_internet_endpoint = (known after apply)
+ secondary_web_internet_host = (known after apply)
+ secondary_web_microsoft_endpoint = (known after apply)
+ secondary_web_microsoft_host = (known after apply)
+ sftp_enabled = false
+ shared_access_key_enabled = true
+ table_encryption_key_type = "Service"
+ tags = {
  + "env" = "dev"
  + "owner" = "adrina"
  + "project" = "multicloud-foundation"
}

+ blob_properties (known after apply)

+ network_rules (known after apply)

+ queue_properties (known after apply)

+ routing (known after apply)

+ share_properties (known after apply)

+ static_website (known after apply)
}

```

Plan: 2 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
PS C:\Program Files\Terraform\devops_azure_eu_west>

iii. Terraform apply -auto-approve

```
PS C:\Program Files\Terraform\devops_azure_eu_west> Terraform apply -auto-approve
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# azurerm_resource_group.rg will be created
+ resource "azurerm_resource_group" "rg" {
+   id           = (known after apply)
+   location     = "westeurope"
+   name         = "rg-dev-assets-westeu"
+   tags         = {
+     "env"      = "dev"
+     "owner"    = "adrina"
+     "project"  = "multicloud-foundation"
+   }
+ }

# azurerm_storage_account.storage will be created
+ resource "azurerm_storage_account" "storage" {
+   access_tier                = (known after apply)
+   account_kind               = "StorageV2"
+   account_replication_type   = "LRS"
+   account_tier               = "Standard"
+   allow_nested_items_to_be_public = true
+   cross_tenant_replication_enabled = false
+   default_to_oauth_authentication = false
+   dns_endpoint_type          = "Standard"
+   https_traffic_only_enabled   = true
+   id                         = (known after apply)
+   infrastructure_encryption_enabled = false
+   is_hns_enabled             = false
+   large_file_share_enabled    = (known after apply)
+   local_user_enabled         = true
+   location                   = "westeurope"
+   min_tls_version            = "TLS1_2"
+   name                       = "adrinadevassetseuw1"
+   nfsv3_enabled              = false
+   primary_access_key         = (sensitive value)
+   primary_blob_connection_string = (sensitive value)
+   primary_blob_endpoint      = (known after apply)
+   primary_blob_host          = (known after apply)
+   primary_blob_internet_endpoint = (known after apply)
+ }
```

```
+ primary_blob_internet_host      = (known after apply)
+ primary_blob_microsoft_endpoint = (known after apply)
+ primary_blob_microsoft_host     = (known after apply)
+ primary_connection_string       = (sensitive value)
+ primary_dfs_endpoint            = (known after apply)
+ primary_dfs_host                 = (known after apply)
+ primary_dfs_internet_endpoint   = (known after apply)
+ primary_dfs_internet_host       = (known after apply)
+ primary_dfs_microsoft_endpoint  = (known after apply)
+ primary_dfs_microsoft_host      = (known after apply)
+ primary_file_endpoint           = (known after apply)
+ primary_file_host                = (known after apply)
+ primary_file_internet_endpoint  = (known after apply)
+ primary_file_internet_host      = (known after apply)
+ primary_file_microsoft_endpoint = (known after apply)
+ primary_file_microsoft_host     = (known after apply)
+ primary_location                = (known after apply)
+ primary_queue_endpoint          = (known after apply)
+ primary_queue_host              = (known after apply)
+ primary_queue_microsoft_endpoint = (known after apply)
+ primary_queue_microsoft_host    = (known after apply)
+ primary_table_endpoint          = (known after apply)
+ primary_table_host              = (known after apply)
+ primary_table_microsoft_endpoint = (known after apply)
+ primary_table_microsoft_host    = (known after apply)
+ primary_web_endpoint            = (known after apply)
+ primary_web_host                 = (known after apply)
+ primary_web_internet_endpoint   = (known after apply)
+ primary_web_internet_host       = (known after apply)
+ primary_web_microsoft_endpoint  = (known after apply)
+ primary_web_microsoft_host      = (known after apply)
+ public_network_access_enabled    = true
+ queue_encryption_key_type       = "Service"
+ resource_group_name             = "rg-dev-assets-westeu"
+ secondary_access_key             = (sensitive value)
+ secondary_blob_connection_string = (sensitive value)
+ secondary_blob_endpoint          = (known after apply)
+ secondary_blob_host              = (known after apply)
+ secondary_blob_internet_endpoint = (known after apply)
+ secondary_blob_internet_host     = (known after apply)
+ secondary_blob_microsoft_endpoint = (known after apply)
+ secondary_blob_microsoft_host    = (known after apply)
+ secondary_connection_string      = (sensitive value)
+ secondary_dfs_endpoint           = (known after apply)
```

```

+ secondary_dfs_host = (known after apply)
+ secondary_dfs_internet_endpoint = (known after apply)
+ secondary_dfs_internet_host = (known after apply)
+ secondary_dfs_microsoft_endpoint = (known after apply)
+ secondary_dfs_microsoft_host = (known after apply)
+ secondary_file_endpoint = (known after apply)
+ secondary_file_host = (known after apply)
+ secondary_file_internet_endpoint = (known after apply)
+ secondary_file_internet_host = (known after apply)
+ secondary_file_microsoft_endpoint = (known after apply)
+ secondary_file_microsoft_host = (known after apply)
+ secondary_location = (known after apply)
+ secondary_queue_endpoint = (known after apply)
+ secondary_queue_host = (known after apply)
+ secondary_queue_microsoft_endpoint = (known after apply)
+ secondary_queue_microsoft_host = (known after apply)
+ secondary_table_endpoint = (known after apply)
+ secondary_table_host = (known after apply)
+ secondary_table_microsoft_endpoint = (known after apply)
+ secondary_table_microsoft_host = (known after apply)
+ secondary_web_endpoint = (known after apply)
+ secondary_web_host = (known after apply)
+ secondary_web_internet_endpoint = (known after apply)
+ secondary_web_internet_host = (known after apply)
+ secondary_web_microsoft_endpoint = (known after apply)
+ secondary_web_microsoft_host = (known after apply)
+ sftp_enabled = false
+ shared_access_key_enabled = true
+ table_encryption_key_type = "Service"
+ tags = {
+   "env" = "dev"
+   "owner" = "adrina"
+   "project" = "multicloud-foundation"
+ }

+ blob_properties (known after apply)

+ network_rules (known after apply)

+ queue_properties (known after apply)

+ routing (known after apply)

+ share_properties (known after apply)

+ static_website (known after apply)
}

Plan: 2 to add, 0 to change, 0 to destroy.
azurerm_resource_group.rg: Creating...
azurerm_resource_group.rg: Still creating... [00m10s elapsed]
azurerm_resource_group.rg: Creation complete after 18s [id=/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/rg-dev-assets-westeu]
azurerm_storage_account.storage: Creating...
azurerm_storage_account.storage: Still creating... [00m10s elapsed]
azurerm_storage_account.storage: Still creating... [00m20s elapsed]
azurerm_storage_account.storage: Still creating... [00m30s elapsed]
azurerm_storage_account.storage: Still creating... [00m40s elapsed]
azurerm_storage_account.storage: Still creating... [00m50s elapsed]
azurerm_storage_account.storage: Still creating... [01m00s elapsed]
azurerm_storage_account.storage: Still creating... [01m10s elapsed]
azurerm_storage_account.storage: Still creating... [01m20s elapsed]
azurerm_storage_account.storage: Creation complete after 1m26s [id=/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/rg-dev-assets-westeu/providers/Microsoft.Storage/storageAccounts/adrinadevassetseuwl]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
PS C:\Program Files\Terraform\devops_azure_eu_west>

```

5. Validation

a. AWS US EAST

us-east-1.console.aws.amazon.com/s3/buckets?region=us-east-1#

Amazon S3 > Buckets

General purpose buckets (5) info

Buckets are containers for data stored in S3.

Search: asset 1 match

Name	AWS Region	Creation date
adrina-dev-assets-use1	US East (N. Virginia) us-east-1	October 7, 2025, 09:00:16 (UTC+05:30)

Account snapshot info View dashboard

Updated daily

Storage Lens provides visibility into storage usage and activity trends.

External access summary - new info

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

b. AWS EU CENTRAL

us-east-1.console.aws.amazon.com/s3/buckets?region=us-east-1#

Amazon S3 > Buckets

General purpose buckets (6) info

Buckets are containers for data stored in S3.

Search: asset 2 matches

Name	AWS Region	Creation date
adrina-dev-assets-euc1	Europe (Frankfurt) eu-central-1	October 7, 2025, 09:27:35 (UTC+05:30)
adrina-dev-assets-use1	US East (N. Virginia) us-east-1	October 7, 2025, 09:00:16 (UTC+05:30)

Account snapshot info View dashboard

Updated daily

Storage Lens provides visibility into storage usage and activity trends.

External access summary - new info

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

c. AZURE US EAST

Microsoft Azure Upgrade Search resources, services, and docs (G+/J) Copilot adcola13@gmail.com DEFAULT DIRECTORY (ADCOLA13...)

Home > Resource groups >

Resource groups

Default Directory (adcola13@gmail.onmicrosoft.com)

+ Create Manage view ...

rg-dev-assets-eastus Resource group

Search + Create Manage view Delete resource group Refresh Export to CSV Open query Group by none

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Cost Management
- Monitoring
- Automation
- Help

Essentials

Resources Recommendations

Filter for any field... Type equals all Location equals all + Add filter

Name	Type	Location
adrinadevassetsuse1	Storage account	East US

Home > Resource groups > rg-dev-assets-eastus > adrinadevassetsuse1

adrinadevassetsuse1 | Tags

Storage account

Search Delete all Feedback

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Tag names are case insensitive, but tag values are case sensitive. [Learn more about tags?](#)

Do not enter names or values that could make your resources less secure or that contain personal/sensitive information because tag data will be replicated globally.

Name	Value
env	dev
owner	adrina
project	multicloud-foundation

adrinadevassetsuse1 (Storage account)

env : dev owner : adrina project : multicloud-foundation

No changes

d. AZURE EU WEST

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot adcola13@gmail.com

Home > Resource groups Export resource groups using Bicep or Terraform

+ Create Manage view Refresh Export to CSV Open query Assign tags Group by none

You are viewing a new version of Browse experience. Click here to access the old experience.

west Subscription equals all Location equals all Add filter

Name	Subscription	Location
rg-dev-assets-westeu	Azure subscription 1	West Europe

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot adcola13@gmail.com

Home > Resource groups << rg-dev-assets-westeu Help me generate Terraform for this resource group configuration. +2

Resource group

+ Create Manage view Delete resource group Refresh Export to CSV Open query Group by none

Overview Activity log Access control (IAM) Tags Resource visualizer Events Settings Cost Management Monitoring Automation Help

Essentials Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name	Type	Location
adrinadevassetseuw1	Storage account	West Europe

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot adcola13@gmail.com

Home > Resource groups > rg-dev-assets-westeu > adrinadevassetseuw1

Storage account adrinadevassetseuw1 | Tags

Search Delete all Feedback

Overview Activity log Tags Diagnose and solve problems Access Control (IAM) Data migration Events Storage browser Storage Mover Partner solutions Resource visualizer Data storage Security + networking

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Tag names are case insensitive, but tag values are case sensitive. Learn more about tags.

Do not enter names or values that could make your resources less secure or that contain personal/sensitive information because tag data will be replicated globally.

Name	Value
env	dev
owner	adrina
project	multicloud-foundation

adrinadevassetseuw1 (Storage account)

env : dev owner : adrina project : multicloud-foundation

No changes

6. Cleanup

terraform destroy -auto-approve

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\devops_aws_eu_cent1> terraform destroy -auto-approve
aws_s3_bucket.assets: Refreshing state... [id=adrina-dev-assets-euc1]
aws_s3_bucket.versioning.versioning: Refreshing state... [id=adrina-dev-assets-euc1]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  ~ destroy

Terraform will perform the following actions:

# aws_s3_bucket.assets will be destroyed
~ resource "aws_s3_bucket" "assets" {
  ~ arn              = "arn:aws:s3:::adrina-dev-assets-euc1" -> null
  ~ bucket           = "adrina-dev-assets-euc1" -> null
  ~ bucket_domain_name = "adrina-dev-assets-euc1.s3.amazonaws.com" -> null
  ~ bucket_region     = "eu-central-1" -> null
  ~ bucket_regional_domain_name = "adrina-dev-assets-euc1.s3.eu-central-1.amazonaws.com" -> null
  ~ force_destroy     = false -> null
  ~ hosted_zone_id    = "Z21DNDUJVLQW6Q" -> null
  ~ id               = "adrina-dev-assets-euc1" -> null
  ~ object_lock_enabled = false -> null
  ~ region            = "eu-central-1" -> null
  ~ request_payer      = "BucketOwner" -> null
  ~ tags              = {
    ~ "env"      = "dev"
    ~ "owner"    = "adrina"
    ~ "project"  = "multicloud-foundation"
  } -> null
  ~ tags_all       = {
    ~ "env"      = "dev"
    ~ "owner"    = "adrina"
    ~ "project"  = "multicloud-foundation"
  } -> null
  # (3 unchanged attributes hidden)

  ~ grant {
    ~ id          = "b3ec478e2c6a74e67c8960389ed951eacca825a2d98ec0900844c97f5a4f5e52" -> null
    ~ permissions = [
      "FULL_CONTROL",
    ] -> null
    ~ type        = "CanonicalUser" -> null
    # (1 unchanged attribute hidden)
  }

  ~ server_side_encryption_configuration {
```

```

- rule {
  bucket_key_enabled = false -> null

  apply_server_side_encryption_by_default {
    sse_algorithm = "AES256" -> null
    # (1 unchanged attribute hidden)
  }
}

- versioning {
  enabled = true -> null
  mfa_delete = false -> null
}

# aws_s3_bucket_versioning.versioning will be destroyed
- resource "aws_s3_bucket_versioning" "versioning" {
  bucket = "adrina-dev-assets-euc1" -> null
  id = "adrina-dev-assets-euc1" -> null
  region = "eu-central-1" -> null
  # (1 unchanged attribute hidden)

  versioning_configuration {
    status = "Enabled" -> null
    # (1 unchanged attribute hidden)
  }
}

Plan: 0 to add, 0 to change, 2 to destroy.
aws_s3_bucket_versioning.versioning: Destroying... [id=adrina-dev-assets-euc1]
aws_s3_bucket_versioning.versioning: Destruction complete after 4s
aws_s3_bucket.assets: Destroying... [id=adrina-dev-assets-euc1]
aws_s3_bucket.assets: Destruction complete after 1s

Destroy complete! Resources: 2 destroyed.
PS C:\Program Files\Terraform\devops_aws_eu_cent1>

```

- Terraform Registry

The Terraform Registry pages (for each provider & resource) list:

- The **schema**: what arguments the resource takes
- Example usage
- Attributes exposed (for referencing)
- When some arguments are optional or have defaults, the docs often mention that.
- If a property seems confusing (e.g. replication types in Azure), read the “Argument Reference” section and any notes or caveats.

Assignment 30 – Team-Ready State—Remote Backends & Locking (Azure + AWS)

Part A — Azure Remote Backend (Azure Storage)

1. Provision Backend Infrastructure (via AZURE Cloudshell)

Creation:

Resource Group (for backend)

Storage Account

Blob Container inside that storage account

Variables

RG_BACKEND="rg-tf-backend-devops-adrina"

STORAGE_ACC="tfstatedevopsadrina" # storage account names must be lowercase, 3–24 chars

CONTAINER="tfstate-container"

Create resource group

az group create --name \$RG_BACKEND --location eastus

Create storage account

az storage account create \

--name \$STORAGE_ACC \

--resource-group \$RG_BACKEND \

--sku Standard_LRS \

--encryption-services blob

Create blob container

az storage container create \

--account-name \$STORAGE_ACC \

--name \$CONTAINER

```
Microsoft Azure Upgrade Search resources, services, and docs (Ctrl+) Copilot adcolia13@gmail.com
Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

adriana [ ~ ] $ # Variables
RG_BACKEND="rg-tf-backend-devops-adriana"
STORAGE_ACC="tfstatedevopsadriana" # storage account names must be lowercase, 3-24 chars
CONTAINER="tfstate-container"

# Create resource group
az group create --name $RG_BACKEND --location eastus

# Create storage account
az storage account create \
  --name $STORAGE_ACC \
  --resource-group $RG_BACKEND \
  --sku Standard_LRS \
  --encryption-services blob

# Create blob container
az storage container create \
  --account-name $STORAGE_ACC \
  --name $CONTAINER

{
  "id": "/subscriptions/4aaefc3f-e730-4757-944c-df72e63bfe6/resourceGroups/rg-tf-backend-devops-adriana",
  "location": "eastus",
  "managedBy": null,
  "name": "rg-tf-backend-devops-adriana",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}

{
  "accessTier": "Hot",
  "accountMigrationInProgress": null,
  "allowBlobPublicAccess": false,
  "allowCrossTenantReplication": false,
  "allowSharedKeyAccess": null,
  "allowedCopyScope": null,
  "azureFilesIdentityBasedAuthentication": null,
  "blobFeatureStatus": null,
  "creationTime": "2025-10-08T04:18:10.584139+00:00",
  "customDomain": null,
  "defaultToOAuthAuthentication": null,
  "dnsEndpointType": null,
  "enableExtendedGroups": null,
  "enableHttpsTrafficOnly": true,
  "enableHNSV3": null,
  "encryption": {
    "encryptionIdentity": null,
    "keySource": "Microsoft.Storage",
    "keyVaultProperties": null,
    "requireInfrastructureEncryption": null,
    "services": {
      "blob": {
        "enabled": true,
        "keyType": "Account",
        "lastEnabledTime": "2025-10-08T04:18:10.771642+00:00"
      },
      "file": {
        "enabled": true,
        "keyType": "Account",
        "lastEnabledTime": "2025-10-08T04:18:10.771642+00:00"
      },
      "queue": null,
      "table": null
    }
  },
  "extendedLocation": null,
  "failoverInProgress": null,
  "geoReplicationStatus": null,
  "id": "/subscriptions/4aaefc3f-e730-4757-944c-df72e63bfe6/resourceGroups/rg-tf-backend-devops-adriana/providers/Microsoft.Storage/storageAccounts/tfstatedevopsadriana",
  "identity": null,
  "immutableStorageWithVersioning": null,
  "isDisabled": null,
  "isLocalUserEnabled": null,
  "isSftpEnabled": null,
  "isSkuConversionBlocked": null,
  "keyCreationTime": {
    "key1": "2025-10-08T04:18:10.756011+00:00",
    "key2": "2025-10-08T04:18:10.756011+00:00"
  },
  "keyPolicy": null,
  "kind": "StorageV2",
  "largeFileSharesState": null,
  "lastGeoFailoverTime": null,
  "location": "eastus",
  "minimumTlsVersion": "TLS1_0",
  "name": "tfstatedevopsadriana",
  "networkRuleSet": {
    "bypass": "AzureServices",
    "defaultAction": "Allow",
    "ipRules": [],
    "ipV6Rules": [],
    "resourceAccessRules": null,
    "virtualNetworkRules": []
  },
  "primaryEndpoints": {
    "blob": "https://tfstatedevopsadriana.blob.core.windows.net/",
    "dfs": "https://tfstatedevopsadriana.dfs.core.windows.net/",
    "file": "https://tfstatedevopsadriana.file.core.windows.net/",
    "internetEndpoints": null,
    "microsoftEndpoints": null,
    "queue": "https://tfstatedevopsadriana.queue.core.windows.net/",
    "table": "https://tfstatedevopsadriana.table.core.windows.net/",
    "web": "https://tfstatedevopsadriana.z13.web.core.windows.net/"
  },
  "primaryLocation": "eastus",
  "privateEndpointConnections": [],
  "provisioningState": "Succeeded",
  "publicNetworkAccess": null,
  "resourceGroup": "rg-tf-backend-devops-adriana",
  "routingPreference": null,
  "saPolicy": null,
  "secondaryEndpoints": null,
  "secondaryLocation": null,
  "sku": {
    "name": "Standard_LRS",
    "tier": "Standard"
  },
  "statusOfPrimary": "available",
  "statusOfSecondary": null,
  "storageAccountSkuConversionStatus": null,
  "tags": {},
  "type": "Microsoft.Storage/storageAccounts"
}

There are no credentials provided in your command and environment, we will query for account key for your storage account.
It is recommended to provide --connection-string, --account-key or --sas-token in your command as credentials.

You also can add --auth-mode login in your command to use Azure Active Directory (Azure AD) for authorization if your login account is assigned required RBAC roles.
For more information about RBAC roles in storage, visit https://learn.microsoft.com/azure/storage/common/storage-auth-aad-rbac-cli.

In addition, setting the corresponding environment variables can avoid inputting credentials in your command. Please use --help to get more information about environment variable usage.
{
  "created": true
}

adriana [ ~ ] $
```

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > Resource groups > rg-tf-backend-devops-adrina

Resource groups

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Essentials

Resources

Filter for any field...

Type equals all

Location equals all

Add filter

Name	Type	Location
tfstatedevopsadrina	Storage account	East US

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > Resource groups > rg-tf-backend-devops-adrina > tfstatedevopsadrina

Storage account

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

File shares

Essentials

Resource group (move)

Location

Subscription (move)

Subscription ID

Disk state

Tags (edit)

Performance

Replication

Account kind

Provisioning state

Created

Properties

Monitoring

Capabilities (7)

Recommendations (0)

Tutorials

Tools + SDKs

Blob service

Hierarchical namespace

Default access tier

Blob anonymous access

Blob soft delete

Security

Require secure transfer for REST API operations

Storage account key access

Minimum TLS version

Infrastructure encryption

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > Resource groups > rg-tf-backend-devops-adrina > tfstatedevopsadrina | Containers

Storage account

Containers

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Add container

Upload

Refresh

Delete

Change access level

Restore containers

Edit columns

Search containers by prefix

Only show active containers

Showing all 1 items

Name	Last modified	Anonymous access level	Lease state
tfstate-container	10/8/2025, 9:48:35 AM	Private	Available

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > Resource groups > rg-tf-backend-devops-adrina > tfstatedevopsadrina | Containers > tfstate-container

Container

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Add Directory

Upload

Change access level

Refresh

Delete

Copy

Paste

Rename

Acquire lease

Break lease

Edit columns

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 0 items

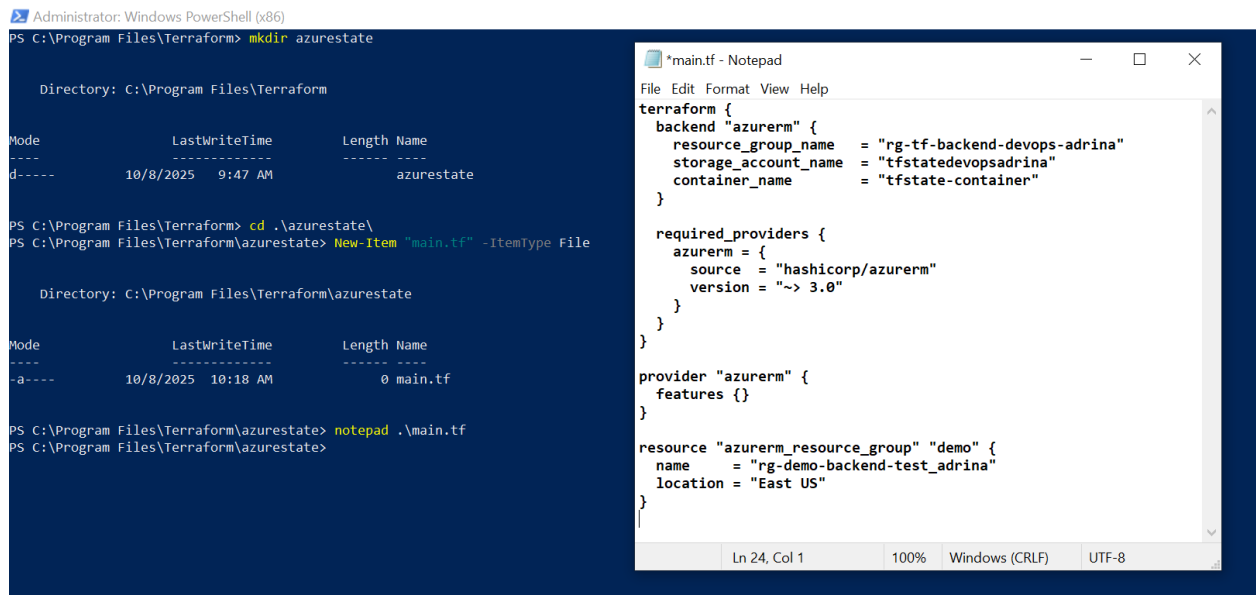
Name	Last modified	Access tier	Blob type	Size	Lease state
------	---------------	-------------	-----------	------	-------------

No items found

2. Create the Terraform configuration

Creation:

Main.tf file containing a tiny Terraform config that creates one innocuous resource (e.g., a Resource Group).



The screenshot shows two windows. The left window is a PowerShell prompt with the following commands and output:

```
PS C:\Program Files\Terraform> mkdir azurestate

Directory: C:\Program Files\Terraform

Mode                LastWriteTime         Length Name
----                -
d-----          10/8/2025   9:47 AM             azurestate

PS C:\Program Files\Terraform> cd .\azurestate\
PS C:\Program Files\Terraform\azurestate> New-Item "main.tf" -ItemType File

Directory: C:\Program Files\Terraform\azurestate

Mode                LastWriteTime         Length Name
----                -
-a-----          10/8/2025  10:18 AM              0 main.tf

PS C:\Program Files\Terraform\azurestate> notepad .\main.tf
PS C:\Program Files\Terraform\azurestate>
```

The right window is a Notepad editor titled "main.tf - Notepad" showing the following Terraform configuration:

```
terraform {
  backend "azurerm" {
    resource_group_name = "rg-tf-backend-devops-adrina"
    storage_account_name = "tfstatedevopsadrina"
    container_name       = "tfstate-container"
  }
}

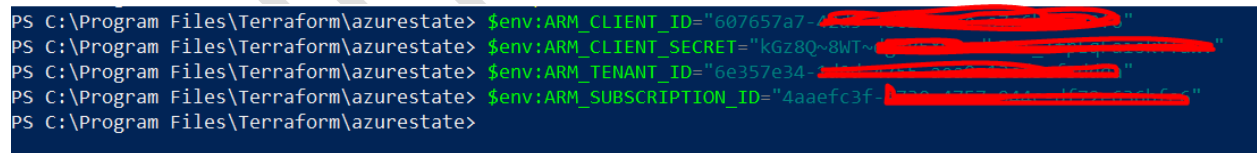
required_providers {
  azurerm = {
    source = "hashicorp/azurerm"
    version = "~> 3.0"
  }
}

provider "azurerm" {
  features {}
}

resource "azurerm_resource_group" "demo" {
  name     = "rg-demo-backend-test_adrina"
  location = "East US"
}
```

3. Configure the azurerm backend for this folder (you may pass sensitive bits via -backend-config during init).

Setup of environment variables



The screenshot shows a PowerShell prompt with the following commands and output:

```
PS C:\Program Files\Terraform\azurestate> $env:ARM_CLIENT_ID="607657a7-4203-4203-4203-4203"
PS C:\Program Files\Terraform\azurestate> $env:ARM_CLIENT_SECRET="kGz8Q~8WT~4203-4203-4203-4203"
PS C:\Program Files\Terraform\azurestate> $env:ARM_TENANT_ID="6e357e34-4203-4203-4203-4203"
PS C:\Program Files\Terraform\azurestate> $env:ARM_SUBSCRIPTION_ID="4aaefc3f-4203-4203-4203-4203"
PS C:\Program Files\Terraform\azurestate>
```

Terraform init

Terraform plan

4. terraform init → migrate (or create) state to Azure.

```

PS C:\Program Files\Terraform\azurestate> terraform init
Initializing the backend...
key
The blob key to use in the Storage Container.

Enter a value: rxFj+Lm/9hxrLwK+W1Yw...[REDACTED]...g==

Successfully configured the backend "azurerm"! Terraform will automatically
use this backend unless the backend configuration changes.
Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 3.0"...
- Installing hashicorp/azurerm v3.117.1...
- Installed hashicorp/azurerm v3.117.1 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Program Files\Terraform\azurestate>

```

Object created after running terraform plan and terraform apply

The screenshot shows the Microsoft Azure portal interface. The breadcrumb navigation indicates the path: Home > Resource groups > rg-tf-backend-devops-adrina > tfstatedevopsadrina | Containers > tfstate-container. The 'tfstate-container' container is selected, and the 'Overview' tab is active. A search bar shows 'rxFj+Lm'. Below the search bar, a table lists the blobs in the container. One blob is shown with the name 'rxFj+Lm/9hxrLwK+W1Yw...g==', a size of 181 B, and a state of 'Available'. Red arrows point to the container name and the blob name in the table.

Name	Last modified	Access tier	Blob type	Size	Lease state
rxFj+Lm/9hxrLwK+W1Yw...g==	10/8/2025, 11:24:36 AM	Hot (Inferred)	Block blob	181 B	Available

5. Locking test (Azure)

Running "terraform plan" and "terraform apply -auto-approve"

```
Administrator: Windows PowerShell (x86)
PS C:\Program Files\Terraform\azurestate> terraform plan
Acquiring state lock. This may take a few moments...

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  create

Terraform will perform the following actions:

# azurerm_resource_group.demo will be created
resource "azurerm_resource_group" "demo" {
  id       = (known after apply)
  location = "eastus"
  name     = "rg-demo-backend-test_adrina"
}

Plan: 1 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take
exactly these actions if you run "terraform apply" now.
Releasing state lock. This may take a few moments...
PS C:\Program Files\Terraform\azurestate> terraform apply
Acquiring state lock. This may take a few moments...
azurerm_resource_group.demo: Refreshing state... [id=/subscriptions/4aaefc3f-e730-4757-944c-d
f72e636bf66/resourceGroups/rg-demo-backend-test_adrina]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no
differences, so no changes are needed.
Releasing state lock. This may take a few moments...
PS C:\Program Files\Terraform\azurestate>

Administrator: Windows PowerShell (x86)
PS C:\Program Files\Terraform\azurestate> terraform apply -auto-approve
Acquiring state lock. This may take a few moments...

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  create

Terraform will perform the following actions:

# azurerm_resource_group.demo will be created
resource "azurerm_resource_group" "demo" {
  id       = (known after apply)
  location = "eastus"
  name     = "rg-demo-backend-test_adrina"
}

Plan: 1 to add, 0 to change, 0 to destroy.
azurerm_resource_group.demo: Creating...
azurerm_resource_group.demo: Still creating... [00m10s elapsed]
azurerm_resource_group.demo: Creation complete after 15s [id=/subscriptions/4aaefc3f-e730-475
7-944c-df72e636bf66/resourceGroups/rg-demo-backend-test_adrina]
Releasing state lock. This may take a few moments...

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Program Files\Terraform\azurestate> terraform apply -auto-approve
Acquiring state lock. This may take a few moments...

Error: Error acquiring the state lock

Error message: state blob is already locked
Lock Info:
ID:          5689eba2-4730-6065-9027-dba3f2dbe41
Path:        tfstate-container/rxFj+Lm/9hxrLwK+M1YwqsXXiJlB+N4d1qfujKvRRn+yIOs6RRml.ahPPI3Lk
gpAh86dL+XL2yAB+ASTaJZnE==
Operation:    OperationTypePlan
Who:          DESKTOP-TSHFOQK\alanm@DESKTOP-TSHFOQK
Version:      1.13.3
Created:      2025-10-08 06:14:15.8228447 +0000 UTC
Info:

Terraform acquires a state lock to protect the state from being written
by multiple users at the same time. Please resolve the issue above and try
again. For most commands, you can disable locking with the "-lock=false"
```

Demo resource group created

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/) | Copilot

Home > Resource groups > rg-demo-backend-test_adrina

Resource groups

Default Directory (adcola13@gmail.onmicrosoft.com)

+ Create Manage view ...

You are viewing a new version of Browse experience. Click here to access the old experience.

- ☐ Name ↑
- ☒ rg-demo-backend-test_adrina
- ☐ rg-tf-sp-demo

rg-demo-backend-test_adrina | Activity log

Search

Activity Edit columns Refresh Export Activity Logs Download as CSV Insights Feedback

Looking for Log Analytics? In Log Analytics you can search for performance, diagnostics, health logs, and more. [Visit Log Analytics](#)

Search Quick Insights

Management Group: None Subscription: Azure subscription 1 Event severity: All Timespan: Last 6 hours

Resource group: rg-demo-backend-test_adrina Add Filter

1 item.

Operation name	Status	Time	Time stamp	Subscription
Update resource group	Succeeded	11 minutes ...	Wed Oct 08 2025 11:42:40 ...	Azure subscription 1

Part B — AWS Remote Backend (S3 + DynamoDB)

1. Provision backend infrastructure

Creation:

S3 Bucket

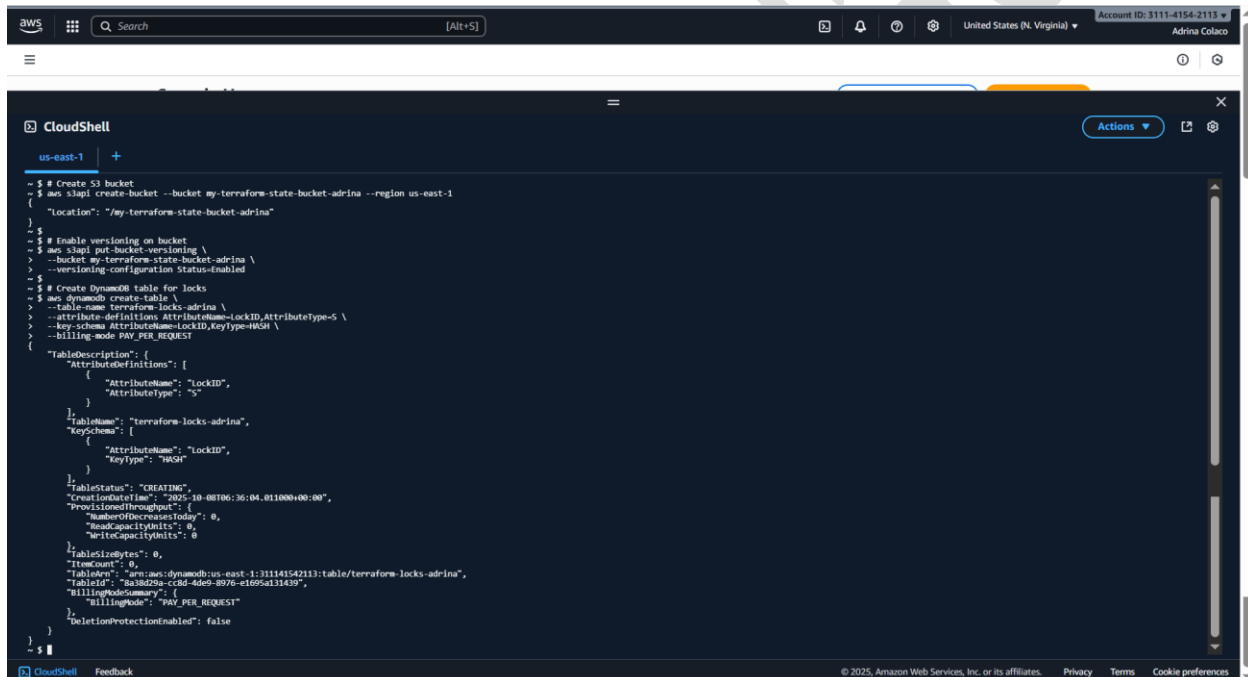
DynamoDB Table for locking

Create S3 bucket

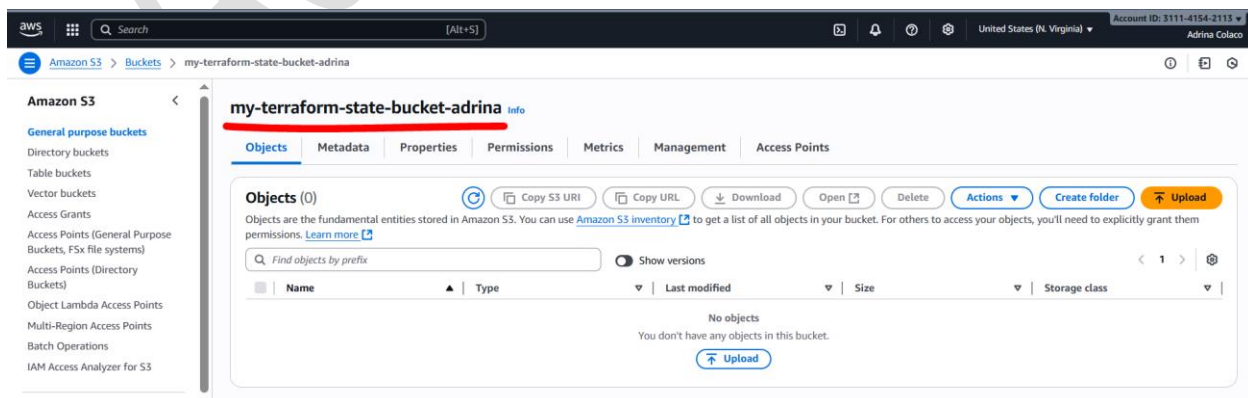
aws s3api create-bucket --bucket my-terraform-state-bucket-adrina --region us-east-1

```
# Enable versioning on bucket
aws s3api put-bucket-versioning \
  --bucket my-terraform-state-bucket-adrina \
  --versioning-configuration Status=Enabled
```

```
# Create DynamoDB table for locks
aws dynamodb create-table \
  --table-name terraform-locks-adrina \
  --attribute-definitions AttributeName=LockID,AttributeType=S \
  --key-schema AttributeName=LockID,KeyType=HASH \
  --billing-mode PAY_PER_REQUEST
```



```
us-east-1 +
~ $ # Create S3 bucket
~ $ aws s3api create-bucket --bucket my-terraform-state-bucket-adrina --region us-east-1
{
  "Location": "/my-terraform-state-bucket-adrina"
}
~ $ # Enable versioning on bucket
~ $ aws s3api put-bucket-versioning \
> --bucket my-terraform-state-bucket-adrina \
> --versioning-configuration Status=Enabled
~ $ # Create DynamoDB table for locks
~ $ aws dynamodb create-table \
> --table-name terraform-locks-adrina \
> --attribute-definitions AttributeName=LockID,AttributeType=S \
> --key-schema AttributeName=LockID,KeyType=HASH \
> --billing-mode PAY_PER_REQUEST
{
  "TableDescription": {
    "AttributeDefinitions": [
      {
        "AttributeName": "LockID",
        "AttributeType": "S"
      }
    ],
    "TableName": "terraform-locks-adrina",
    "KeySchema": [
      {
        "AttributeName": "LockID",
        "KeyType": "HASH"
      }
    ],
    "TableStatus": "CREATING",
    "CreationDateTime": "2025-10-08T06:36:04.011000+00:00",
    "ProvisionedThroughput": {
      "NumberOfDecreasesToday": 0,
      "ReadCapacityUnits": 0,
      "WriteCapacityUnits": 0
    },
    "TableSizeBytes": 0,
    "ItemCount": 0,
    "TableArn": "arn:aws:dynamodb:us-east-1:311141542113:table/terraform-locks-adrina",
    "TableId": "4a38d29a-cbd-4de0-8976-e1695a131439",
    "BillingModeSummary": {
      "BillingMode": "PAY_PER_REQUEST",
      "DeletionProtectionEnabled": false
    }
  }
}
~ $
```



2. Create the Terraform configuration to configure the s3 backend to point to your bucket and lock table.

Creation:

Main.tf file containing a tiny Terraform config that creates one innocuous resource (e.g., a S3 bucket).

```
Administrator: Windows PowerShell (x86)
PS C:\Program Files\Terraform> mkdir awsstate

Directory: C:\Program Files\Terraform

Mode                LastWriteTime         Length Name
----                -
d-----          10/8/2025 12:13 PM             awsstate

PS C:\Program Files\Terraform> cd .\awsstate\
PS C:\Program Files\Terraform\awsstate> New-Item "main.tf" -ItemType File

Directory: C:\Program Files\Terraform\awsstate

Mode                LastWriteTime         Length Name
----                -
-a-----          10/8/2025 12:13 PM              0 main.tf
```

```
Administrator: Windows PowerShell (x86)
PS C:\Program Files\Terraform\awsstate> notepad .\main.tf
PS C:\Program Files\Terraform\awsstate>
```

```
*main.tf - Notepad
File Edit Format View Help
terraform {
  backend "s3" {
    bucket = "my-terraform-state-bucket-adrina"
    key    = "terraform.aws.tfstate"
    region = "us-east-1"
    dynamodb_table = "terraform-locks-adrina"
    encrypt = true
  }

  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 4.0"
    }
  }
}

provider "aws" {
  region = "us-east-1"
}

resource "aws_s3_bucket" "demo" {
  bucket = "demo-tf-backend-test-adrina"
  tags = {
    project = "backend-test"
  }
}
```

3. terraform init → migrate (or create) state to S3.

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\awsstate> aws configure
AWS Access Key ID [*****MF30]: 
AWS Secret Access Key [*****X+Qd]: 
Default region name [us-east-1]: 
Default output format [json]: 
PS C:\Program Files\Terraform\awsstate> terraform init
Initializing the backend...

Successfully configured the backend "s3"! Terraform will automatically
use this backend unless the backend configuration changes.
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 4.0"...
- Installing hashicorp/aws v4.67.0...
- Installed hashicorp/aws v4.67.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Program Files\Terraform\awsstate>
```

Object created after running Terraform plan & Terraform apply -auto-approve

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\awsstate> terraform plan
Acquiring state lock. This may take a few moments...
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_s3_bucket.demo will be created
+ resource "aws_s3_bucket" "demo" {
  + acceleration_status      = (known after apply)
  + acl                      = (known after apply)
  + arn                      = (known after apply)
  + bucket                  = "demo-tf-backend-test-adrina"
  + bucket_domain_name      = (known after apply)
  + bucket_prefix           = (known after apply)
  + bucket_regional_domain_name = (known after apply)
  + force_destroy           = false
  + hosted_zone_id          = (known after apply)
  + id                      = (known after apply)
  + object_lock_enabled      = (known after apply)
  + policy                  = (known after apply)
  + region                  = (known after apply)
  + request_payer            = (known after apply)
  + tags                    = {
    + "project" = "backend-test"
  }
  + tags_all                = {
    + "project" = "backend-test"
  }
  + website_domain          = (known after apply)
  + website_endpoint        = (known after apply)

  + cors_rule (known after apply)

  + grant (known after apply)

  + lifecycle_rule (known after apply)

  + logging (known after apply)

  + object_lock_configuration (known after apply)

  + replication_configuration (known after apply)

  + server_side_encryption_configuration (known after apply)

  + versioning (known after apply)

  + website (known after apply)
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

Releasing state lock. This may take a few moments...

```
PS C:\Program Files\Terraform\awsstate>
```

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\awsstate> Terraform apply -auto-approve
Acquiring state lock. This may take a few moments...
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_s3_bucket.demo will be created
+ resource "aws_s3_bucket" "demo" {
  + acceleration_status      = (known after apply)
  + acl                      = (known after apply)
  + arn                      = (known after apply)
  + bucket                  = "demo-tf-backend-test-adrina"
  + bucket_domain_name      = (known after apply)
  + bucket_prefix           = (known after apply)
  + bucket_regional_domain_name = (known after apply)
  + force_destroy           = false
  + hosted_zone_id          = (known after apply)
  + id                      = (known after apply)
  + object_lock_enabled      = (known after apply)
  + policy                  = (known after apply)
  + region                  = (known after apply)
  + request_payer            = (known after apply)
  + tags                    = {
    + "project" = "backend-test"
  }
  + tags_all                = {
    + "project" = "backend-test"
  }
  + website_domain          = (known after apply)
  + website_endpoint        = (known after apply)

  + cors_rule (known after apply)

  + grant (known after apply)

  + lifecycle_rule (known after apply)

  + logging (known after apply)

  + object_lock_configuration (known after apply)

  + replication_configuration (known after apply)

  + server_side_encryption_configuration (known after apply)

  + versioning (known after apply)

  + website (known after apply)
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

aws_s3_bucket.demo: Creating...

aws_s3_bucket.demo: Creation complete after 7s [id=demo-tf-backend-test-adrina]

Releasing state lock. This may take a few moments...

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

```
PS C:\Program Files\Terraform\awsstate>
```

my-terraform-state-bucket-adrina [Info](#)

Objects | Metadata | Properties | Permissions | Metrics | Management | Access Points

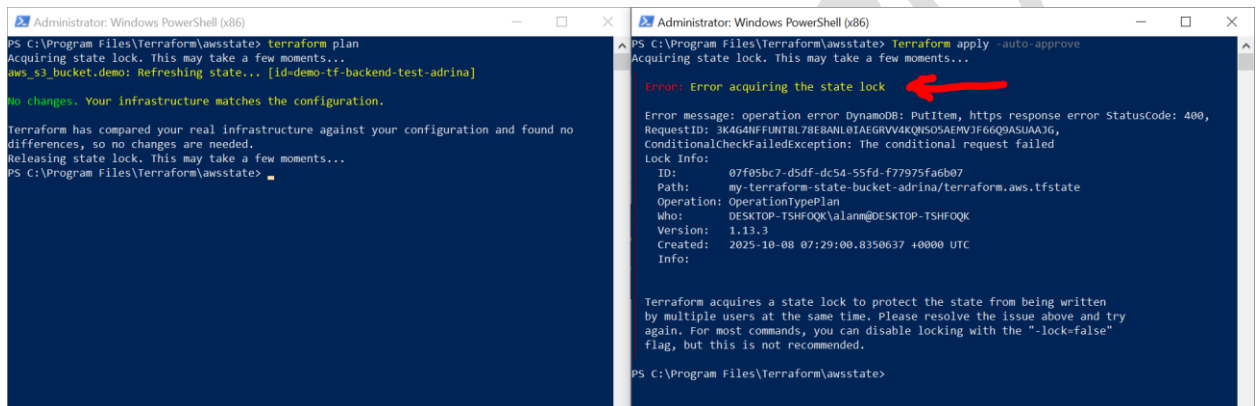
Objects (1) [Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☐ Show versions < 1 > [Settings](#)

☐	Name	Type	Last modified	Size	Storage class
☐	terraform.aws.tfstate	tfstate	October 8, 2025, 12:55:24 (UTC+05:30)	2.7 KB	Standard

4. Locking test (AWS)



The image shows two side-by-side Windows PowerShell terminal windows. The left window shows the output of the `terraform plan` command, which indicates that the infrastructure matches the configuration and no changes are needed. The right window shows the output of the `terraform apply -auto-approve` command, which results in an error: "Error: Error acquiring the state lock". A red arrow points to this error message. The error details include a message from DynamoDB, a request ID, and lock information for the file `my-terraform-state-bucket-adrina/terraform.aws.tfstate`. The error suggests that the state lock is already held by another process and provides instructions on how to resolve the issue.

```
PS C:\Program Files\Terraform\awsstate> terraform plan
Acquiring state lock. This may take a few moments...
aws_s3_bucket.demo: Refreshing state... [id=demo-tf-backend-test-adrina]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no
differences, so no changes are needed.
Releasing state lock. This may take a few moments...
PS C:\Program Files\Terraform\awsstate>

PS C:\Program Files\Terraform\awsstate> terraform apply -auto-approve
Acquiring state lock. This may take a few moments...

Error: Error acquiring the state lock

Error message: operation error DynamoDB: PutItem, https response error StatusCode: 400,
RequestID: 3K4G4NFFUN78L78E8AML0IAEGRV4KQNS0SAEMVJF66Q9ASUAAJ6,
ConditionalCheckFailedException: The conditional request failed
Lock Info:
  ID:         07f05bc7-d5df-dc54-55fd-f77975faeb07
  Path:       my-terraform-state-bucket-adrina/terraform.aws.tfstate
  Operation:  OperationTypePlan
  Who:        DESKTOP-TSHFOQK\alanm@DESKTOP-TSHFOQK
  Version:    1.13.3
  Created:    2025-10-08 07:29:00.8350637 +0000 UTC
  Info:

Terraform acquires a state lock to protect the state from being written
by multiple users at the same time. Please resolve the issue above and try
again. For most commands, you can disable locking with the "-lock=false"
flag, but this is not recommended.

PS C:\Program Files\Terraform\awsstate>
```

Assignment 31 – Deploy a React App with Terraform

Workspaces (dev & prod)

1. Fork & build the React app

```
git clone https://github.com/pravinmishraaws/my-react-app.git
cd my-react-app
npm install
npm run build
```

Select Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform> git clone https://github.com/pravinmishraaws/my-react-app.git
Cloning into 'my-react-app'...
remote: Enumerating objects: 69, done.
remote: Counting objects: 100% (69/69), done.
remote: Compressing objects: 100% (66/66), done.
remote: Total 69 (delta 28), reused 26 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (69/69), 188.95 KiB | 8.21 MiB/s, done.
Resolving deltas: 100% (28/28), done.
PS C:\Program Files\Terraform> cd my-react-app
```

MINGW64:/c/Program Files/Terraform/my-react-app

```
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program Files/Terraform/my-react-app (main)
$ npm install
npm warn deprecated stable@0.1.8: Modern JS already guarantees Array#sort() is a
stable sort, so this library is deprecated. See the compatibility table on MDN:
https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Object
s/Array/sort#browser_compatibility
npm warn deprecated w3c-hr-time@1.0.2: Use your platform's native performance.no
w() and performance.timeOrigin.
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supp
orted
npm warn deprecated rollup-plugin-terser@7.0.2: This package has been deprecated
and is no longer maintained. Please use @rollup/plugin-terser
npm warn deprecated sourcemap-codec@1.4.8: Please use @jridgewell/sourcemap-code
c instead
npm warn deprecated q@1.5.1: You or someone you depend on is using Q, the JavaSc
ript Promise library that gave JavaScript developers strong feelings about promi
ses. They can almost certainly migrate to the native JavaScript promise now. Tha
nk you literally everyone for joining me in this bet against the odds. Be excell
ent to each other.
npm warn deprecated
npm warn deprecated (For a CapTP with native promises, see @endo/eventual-send a
nd @endo/captp)
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memo
ry. Do not use it. Check out lru-cache if you want a good and tested way to coal
esce async requests by a key value, which is much more comprehensive and powerfu
l.
npm warn deprecated glob@7.2.3: Glob versions prior to v9 are no longer supporte
d
npm warn deprecated domexception@2.0.1: Use your platform's native DOMException
instead
npm warn deprecated workbox-cacheable-response@6.6.0: workbox-background-sync@6.
6.0
npm warn deprecated abab@2.0.6: Use your platform's native atob() and btoa() met
hods instead
npm warn deprecated workbox-google-analytics@6.6.0: It is not compatible with ne
wer versions of GA starting with v4, as long as you are using GAv3 it should be
ok, but the package is not longer being maintained
npm warn deprecated @humanwhocodes/config-array@0.13.0: Use @eslint/config-array
instead
npm warn deprecated @humanwhocodes/object-schema@2.0.3: Use @eslint/object-schem
a instead
npm warn deprecated @babel/plugin-proposal-numeric-separator@7.18.6: This propos
al has been merged to the ECMAScript standard and thus this plugin is no longer
maintained. Please use @babel/plugin-transform-numeric-separator instead.
npm warn deprecated @babel/plugin-proposal-class-properties@7.18.6: This proposa
l has been merged to the ECMAScript standard and thus this plugin is no longer m
aintained. Please use @babel/plugin-transform-class-properties instead.
npm warn deprecated @babel/plugin-proposal-private-methods@7.18.6: This proposal
has been merged to the ECMAScript standard and thus this plugin is no longer ma
intained. Please use @babel/plugin-transform-private-methods instead.
npm warn deprecated @babel/plugin-proposal-nullish-coalescing-operator@7.18.6: T
his proposal has been merged to the ECMAScript standard and thus this plugin is
no longer maintained. Please use @babel/plugin-transform-nullish-coalescing-oper
ator instead.
```

```
npm warn deprecated svgo@1.3.2: This SVGO version is no longer supported. Upgrade to v2.x.x.
npm warn deprecated eslint@8.57.1: This version is no longer supported. Please see https://eslint.org/version-support for other options.
```

```
added 1342 packages in 2m
```

```
268 packages are looking for funding
  run `npm fund` for details
```

```
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program Files/Terraform/my-react-app (main)
$ npm run build
```

```
> my-react-app@0.1.0 build
> react-scripts build
```

```
Creating an optimized production build...
```

```
Browserslist: browsers data (caniuse-lite) is 8 months old. Please run:
  npx update-browserslist-db@latest
```

```
Why you should do it regularly: https://github.com/browserslist/update-db#readme
```

```
Browserslist: browsers data (caniuse-lite) is 8 months old. Please run:
  npx update-browserslist-db@latest
```

```
Why you should do it regularly: https://github.com/browserslist/update-db#readme
```

```
Compiled successfully.
```

```
File sizes after gzip:
```

59.23 kB	build\static\js\main.b4f77641.js
1.77 kB	build\static\js\453.ad6eb26e.chunk.js
263 B	build\static\css\main.e6c13ad2.css

```
The project was built assuming it is hosted at /.
```

```
You can control this with the homepage field in your package.json.
```

```
The build folder is ready to be deployed.
```

```
You may serve it with a static server:
```

```
npm install -g serve
serve -s build
```

```
Find out more about deployment here:
```

```
https://cra.link/deployment
```

```
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program Files/Terraform/my-react-app (main)
$ ls
```

```
README.md  node_modules/  package.json  src/
build/     package-lock.json  public/
```

```
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program Files/Terraform/my-react-app (main)
$ |
```

2. Setup Terraform and Initialize workspaces

```
PS C:\Program Files\Terraform\my-react-app> ls

Directory: C:\Program Files\Terraform\my-react-app

Mode                LastWriteTime         Length Name
----                -
d-----          10/8/2025   6:17 PM                build
d-----          10/8/2025   6:17 PM            node_modules
d-----          10/8/2025   3:46 PM             public
d-----          10/8/2025   3:46 PM              src
-a-----          10/8/2025   3:46 PM           192 .gitignore
-a-----          10/8/2025   6:16 PM        646094 package-lock.json
-a-----          10/8/2025   3:46 PM           898 package.json
-a-----          10/8/2025   3:46 PM          3207 README.md

PS C:\Program Files\Terraform\my-react-app> New-Item "backend.tf" -ItemType File

Directory: C:\Program Files\Terraform\my-react-app

Mode                LastWriteTime         Length Name
----                -
-a-----          10/8/2025   6:55 PM              0 backend.tf

PS C:\Program Files\Terraform\my-react-app> New-Item "main.tf" -ItemType File

Directory: C:\Program Files\Terraform\my-react-app

Mode                LastWriteTime         Length Name
----                -
-a-----          10/8/2025   6:59 PM              0 main.tf

PS C:\Program Files\Terraform\my-react-app> New-Item "variables.tf" -ItemType File

Directory: C:\Program Files\Terraform\my-react-app
```

```
PS C:\Program Files\Terraform\my-react-app> New-Item "outputs.tf" -ItemType File
```

```
Directory: C:\Program Files\Terraform\my-react-app
```

Mode	LastWriteTime	Length	Name
-a----	10/8/2025 6:59 PM	0	outputs.tf

```
PS C:\Program Files\Terraform\my-react-app> New-Item "provider.tf" -ItemType File
```

```
Directory: C:\Program Files\Terraform\my-react-app
```

Mode	LastWriteTime	Length	Name
-a----	10/8/2025 6:59 PM	0	provider.tf

```
PS C:\Program Files\Terraform\my-react-app> New-Item "upload.tf" -ItemType File
```

```
Directory: C:\Program Files\Terraform\my-react-app
```

Mode	LastWriteTime	Length	Name
-a----	10/8/2025 6:59 PM	0	upload.tf

```
PS C:\Program Files\Terraform\my-react-app>
```

Backend.tf

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\my-react-app> notepad .\backend.tf  
PS C:\Program Files\Terraform\my-react-app>
```

```
"backend.tf - Notepad  
File Edit Format View Help  
terraform {  
  backend "s3" {  
    bucket = "my-terraform-state-bucket-adrina"  
    key     = "static-site/terraform.tfstate"  
    region  = "us-east-1"  
    dynamodb_table = "terraform-locks-adrina"  
    encrypt  = true  
  }  
}
```

Provider.tf

```
Administrator: Windows PowerShell (x86)
PS C:\Program Files\Terraform\my-react-app> notepad .\backend.tf
PS C:\Program Files\Terraform\my-react-app> notepad .\provider.tf
PS C:\Program Files\Terraform\my-react-app>

*provider.tf - Notepad
File Edit Format View Help
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 4.0"
    }
  }
}

provider "aws" {
  region = "us-east-1"
}
```

3. Setup of the workspaces

terraform init

```
Administrator: Windows PowerShell (x86)
PS C:\Program Files\Terraform\my-react-app> terraform init
Initializing the backend...

Successfully configured the backend "s3"! Terraform will automatically
use this backend unless the backend configuration changes.
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 4.0"...
- Installing hashicorp/aws v4.67.0...
- Installed hashicorp/aws v4.67.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Program Files\Terraform\my-react-app>
```

terraform workspace new dev

terraform workspace new prod

terraform workspace select dev

```
PS C:\Program Files\Terraform\my-react-app> terraform workspace new dev  
Created and switched to workspace "dev"!
```

You're now on a new, empty workspace. Workspaces isolate their state, so if you run "terraform plan" Terraform will not see any existing state for this configuration.

```
PS C:\Program Files\Terraform\my-react-app> terraform workspace new prod  
Created and switched to workspace "prod"!
```

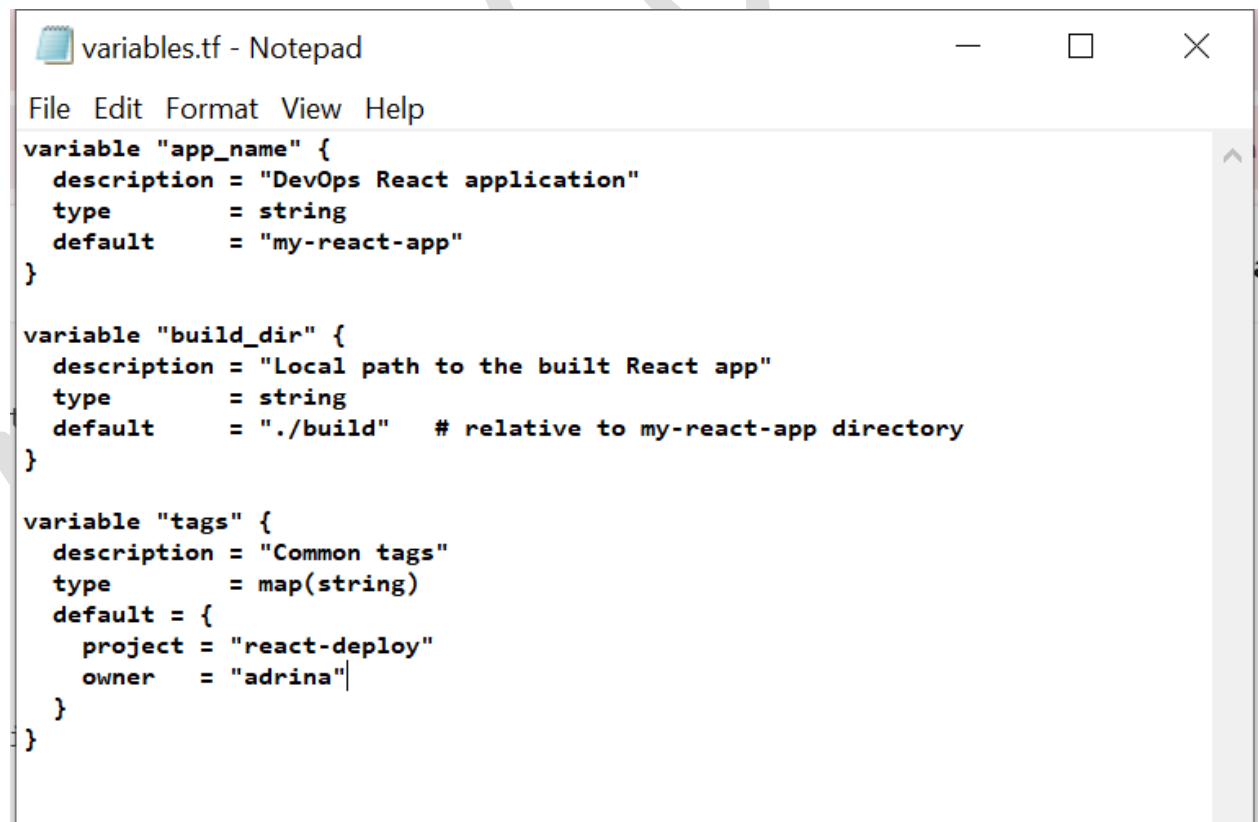
You're now on a new, empty workspace. Workspaces isolate their state, so if you run "terraform plan" Terraform will not see any existing state for this configuration.

```
PS C:\Program Files\Terraform\my-react-app> terraform workspace select dev  
Switched to workspace "dev".
```

```
PS C:\Program Files\Terraform\my-react-app>
```

4. Complete the remaining terraform configuration setup

Variables.tf



```
variable "app_name" {  
  description = "DevOps React application"  
  type        = string  
  default     = "my-react-app"  
}  
  
variable "build_dir" {  
  description = "Local path to the built React app"  
  type        = string  
  default     = "./build" # relative to my-react-app directory  
}  
  
variable "tags" {  
  description = "Common tags"  
  type        = map(string)  
  default = {  
    project = "react-deploy"  
    owner   = "adrina"  
  }  
}
```

Main.tf

 *main.tf - Notepad

File Edit Format View Help

```
locals {
  env = terraform.workspace

  tags = merge(
    var.tags,
    {
      env = local.env
    }
  )
}

# S3 bucket to host static assets
resource "aws_s3_bucket" "site" {
  bucket = "${var.app_name}-${terraform.workspace}-site"

  website {
    index_document = "index.html"
    error_document = "index.html"
  }

  tags = local.tags
}

resource "aws_s3_bucket_public_access_block" "site_block" {
  bucket = aws_s3_bucket.site.id

  block_public_acls   = false
  ignore_public_acls  = false
  block_public_policy = false
  restrict_public_buckets = false
}

# Bucket policy to allow public reads (for website)
resource "aws_s3_bucket_policy" "site_policy" {
  bucket = aws_s3_bucket.site.id

  policy = jsonencode({
    Version = "2012-10-17"
    Statement = [
      {
        Effect = "Allow"
        Principal = "*"
        Action = ["s3:GetObject"]
        Resource = ["${aws_s3_bucket.site.arn}/*"]
      }
    ]
  })
}
```

Upload.tf

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\my-react-app> notepad .\variables.tf
PS C:\Program Files\Terraform\my-react-app> notepad .\main.tf
PS C:\Program Files\Terraform\my-react-app> notepad .\upload.tf
PS C:\Program Files\Terraform\my-react-app>
```

```
upload.tf - Notepad
File Edit Format View Help

locals {
  files = fileset(var.build_dir, "**/*")
}

resource "aws_s3_bucket_object" "static_files" {
  for_each = { for file in local.files : file => file }

  bucket = aws_s3_bucket.site.bucket
  key    = each.key
  source = "${var.build_dir}/${each.value}"
  etag   = filemd5("${var.build_dir}/${each.value}")

  content_type = lookup(
    {
      "html" = "text/html",
      "css"  = "text/css",
      "js"   = "application/javascript"
    },
    regex("\\.([^.]+)$", each.key)[0],
    "application/octet-stream"
  )
}
```

Outputs.tf

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\my-react-app> notepad .\variables.tf
PS C:\Program Files\Terraform\my-react-app> notepad .\main.tf
PS C:\Program Files\Terraform\my-react-app> notepad .\upload.tf
PS C:\Program Files\Terraform\my-react-app> notepad .\outputs.tf
PS C:\Program Files\Terraform\my-react-app>
```

```
outputs.tf - Notepad
File Edit Format View Help

output "workspace" {
  value = terraform.workspace
}

output "site_url" {
  value = aws_s3_bucket.site.website_endpoint
}

output "cdn_url" {
  value = aws_cloudfront_distribution.cdn.domain_name
}
```

5. Deploy dev

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\my-react-app> terraform workspace select dev
```

```
PS C:\Program Files\Terraform\my-react-app> terraform init
```

Initializing the backend...

Initializing provider plugins...

- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v4.67.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
PS C:\Program Files\Terraform\my-react-app>
```

Terraform plan

Administrator: Windows PowerShell (x86)

```
PS C:\Program Files\Terraform\my-react-app> terraform plan
Acquiring state lock. This may take a few moments...
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# aws_cloudfront_distribution.cdn will be created
+ resource "aws_cloudfront_distribution" "cdn" {
+   arn                               = (known after apply)
+   caller_reference                   = (known after apply)
+   default_root_object               = "index.html"
+   domain_name                       = (known after apply)
+   enabled                           = true
+   etag                              = (known after apply)
+   hosted_zone_id                   = (known after apply)
+   http_version                     = "http2"
+   id                                = (known after apply)
+   in_progress_validation_batches    = (known after apply)
+   is_ipv6_enabled                   = false
+   last_modified_time                = (known after apply)
+   price_class                       = "PriceClass_All"
+   retain_on_delete                  = false
+   status                            = (known after apply)
+   tags                              = {
+     "env"      = "dev"
+     "owner"    = "adrina"
+     "project"  = "react-deploy"
+   }
+   tags_all                          = {
+     "env"      = "dev"
+     "owner"    = "adrina"
+     "project"  = "react-deploy"
+   }
+   trusted_key_groups                = (known after apply)
+   trusted_signers                   = (known after apply)
+   wait_for_deployment               = true

+   default_cache_behavior {
+     allowed_methods = [
+       "GET",
+       "HEAD",
+     ]
+   }
+ }
```

```
# aws_s3_bucket.site will be created
resource "aws_s3_bucket" "site" {
  + acceleration_status = (known after apply)
  + acl                  = "public-read"
  + arn                  = (known after apply)
  + bucket               = "my-react-app-dev-site"
  + bucket_domain_name  = (known after apply)
  + bucket_prefix        = (known after apply)
  + bucket_regional_domain_name = (known after apply)
  + force_destroy        = false
  + hosted_zone_id       = (known after apply)
  + id                   = (known after apply)
  + object_lock_enabled  = (known after apply)
  + policy               = (known after apply)
  + region               = (known after apply)
  + request_payer        = (known after apply)
  + tags                 = {
    + "env"      = "dev"
    + "owner"    = "adrina"
    + "project"  = "react-deploy"
  }
  + tags_all           = {
    + "env"      = "dev"
    + "owner"    = "adrina"
    + "project"  = "react-deploy"
  }
  + website_domain      = (known after apply)
  + website_endpoint    = (known after apply)

  + cors_rule (known after apply)

  + grant (known after apply)

  + lifecycle_rule (known after apply)

  + logging (known after apply)

  + object_lock_configuration (known after apply)

  + replication_configuration (known after apply)

  + server_side_encryption_configuration (known after apply)
}
```

```
# aws_s3_bucket_object.static_files["asset-manifest.json"] will be created
+ resource "aws_s3_bucket_object" "static_files" {
+   acl                = "private"
+   bucket              = "my-react-app-dev-site"
+   bucket_key_enabled = (known after apply)
+   content_type        = "application/octet-stream"
+   etag                = "69b14989bc433eaca14ad3993d5f18ac"
+   force_destroy       = false
+   id                  = (known after apply)
+   key                 = "asset-manifest.json"
+   kms_key_id          = (known after apply)
+   server_side_encryption = (known after apply)
+   source              = "../build/asset-manifest.json"
+   storage_class        = (known after apply)
+   tags_all            = (known after apply)
+   version_id          = (known after apply)
+ }

# aws_s3_bucket_object.static_files["favicon.ico"] will be created
+ resource "aws_s3_bucket_object" "static_files" {
+   acl                = "private"
+   bucket              = "my-react-app-dev-site"
+   bucket_key_enabled = (known after apply)
+   content_type        = "application/octet-stream"
+   etag                = "c92b85a5b907c70211f4ec25e29a8c4a"
+   force_destroy       = false
+   id                  = (known after apply)
+   key                 = "favicon.ico"
+   kms_key_id          = (known after apply)
+   server_side_encryption = (known after apply)
+   source              = "../build/favicon.ico"
+   storage_class        = (known after apply)
+   tags_all            = (known after apply)
+   version_id          = (known after apply)
+ }

# aws_s3_bucket_object.static_files["index.html"] will be created
+ resource "aws_s3_bucket_object" "static_files" {
+   acl                = "private"
+   bucket              = "my-react-app-dev-site"
+   bucket_key_enabled = (known after apply)
+   content_type        = "text/html"
+   etag                = "4985aa6649c0a8599f515d35e7cd3bf8"
+   force_destroy       = false
+ }
```

```
# aws_s3_bucket_object.static_files["logo192.png"] will be created
+ resource "aws_s3_bucket_object" "static_files" {
  + acl                = "private"
  + bucket             = "my-react-app-dev-site"
  + bucket_key_enabled = (known after apply)
  + content_type       = "application/octet-stream"
  + etag              = "33dbdd0177549353eeeb785d02c294af"
  + force_destroy      = false
  + id                = (known after apply)
  + key               = "logo192.png"
  + kms_key_id        = (known after apply)
  + server_side_encryption = (known after apply)
  + source             = "../build/logo192.png"
  + storage_class      = (known after apply)
  + tags_all          = (known after apply)
  + version_id        = (known after apply)
}

# aws_s3_bucket_object.static_files["logo512.png"] will be created
+ resource "aws_s3_bucket_object" "static_files" {
  + acl                = "private"
  + bucket             = "my-react-app-dev-site"
  + bucket_key_enabled = (known after apply)
  + content_type       = "application/octet-stream"
  + etag              = "917515db74ea8d1aee6a246cfbcc0b45"
  + force_destroy      = false
  + id                = (known after apply)
  + key               = "logo512.png"
  + kms_key_id        = (known after apply)
  + server_side_encryption = (known after apply)
  + source             = "../build/logo512.png"
  + storage_class      = (known after apply)
  + tags_all          = (known after apply)
  + version_id        = (known after apply)
}

# aws_s3_bucket_object.static_files["manifest.json"] will be created
+ resource "aws_s3_bucket_object" "static_files" {
  + acl                = "private"
  + bucket             = "my-react-app-dev-site"
  + bucket_key_enabled = (known after apply)
  + content_type       = "application/octet-stream"
  + etag              = "89be60895176466398d53bd4a693d9f6"
  + force_destroy      = false
}
```

```

# aws_s3_bucket_policy.site_policy will be created
+ resource "aws_s3_bucket_policy" "site_policy" {
+   bucket = (known after apply)
+   id     = (known after apply)
+   policy = (known after apply)
+ }

Plan: 17 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ cdn_url    = (known after apply)
+ site_url   = (known after apply)
+ workspace  = "dev"

Warning: Argument is deprecated

    with aws_s3_bucket.site,
    on main.tf line 13, in resource "aws_s3_bucket" "site":
    13: resource "aws_s3_bucket" "site" {

Use the aws_s3_bucket_website_configuration resource instead

(and 3 more similar warnings elsewhere)

Warning: Deprecated attribute

    on main.tf line 45, in resource "aws_cloudfront_distribution" "cdn":
    45:     domain_name = aws_s3_bucket.site.website_endpoint

The attribute "website_endpoint" is deprecated. Refer to the provider documentation for details.

(and one more similar warning elsewhere)

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform
Releasing state lock. This may take a few moments...
PS C:\Program Files\Terraform\my-react-app>

```

Terraform apply -auto-approve

```

PS C:\Program Files\Terraform\my-react-app> terraform apply -auto-approve
Acquiring state lock. This may take a few moments...
aws_s3_bucket.site: Refreshing state... [id=my-react-app-dev-site]
aws_s3_bucket_policy.site_policy: Refreshing state... [id=my-react-app-dev-site]
aws_s3_bucket_object.static_files["static/js/453.ad6eb26e.chunk.js.map"]: Refreshing state... [id=static/js/453.ad6eb26e.chunk.js.map]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js.map"]: Refreshing state... [id=static/js/main.b4f77641.js.map]
aws_s3_bucket_object.static_files["logo192.png"]: Refreshing state... [id=logo192.png]
aws_s3_bucket_object.static_files["static/js/453.ad6eb26e.chunk.js"]: Refreshing state... [id=static/js/453.ad6eb26e.chunk.js]
aws_s3_bucket_object.static_files["asset-manifest.json"]: Refreshing state... [id=asset-manifest.json]
aws_s3_bucket_object.static_files["index.html"]: Refreshing state... [id=index.html]
aws_s3_bucket_object.static_files["static/css/main.e6c13ad2.css.map"]: Refreshing state... [id=static/css/main.e6c13ad2.css.map]
aws_s3_bucket_object.static_files["manifest.json"]: Refreshing state... [id=manifest.json]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js"]: Refreshing state... [id=static/js/main.b4f77641.js]
aws_s3_bucket_object.static_files["robots.txt"]: Refreshing state... [id=robots.txt]
aws_s3_bucket_object.static_files["favicon.ico"]: Refreshing state... [id=favicon.ico]
aws_s3_bucket_object.static_files["logo512.png"]: Refreshing state... [id=logo512.png]
aws_s3_bucket_object.static_files["static/css/main.e6c13ad2.css"]: Refreshing state... [id=static/css/main.e6c13ad2.css]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js.LICENSE.txt"]: Refreshing state... [id=static/js/main.b4f77641.js.LICENSE.txt]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~ update in-place

Terraform will perform the following actions:

# aws_s3_bucket_object.static_files["static/js/main.b4f77641.js"] will be updated in-place
~ resource "aws_s3_bucket_object" "static_files" {
  ~ etag          = "9c392e374d4eabda1f98a7a4366e3252" -> "ecaa9a0f4635f8b64bca53887abed6e3"
    id            = "static/js/main.b4f77641.js"
    tags          = {}
    version_id    = (known after apply)
  # (19 unchanged attributes hidden)
}

Plan: 0 to add, 1 to change, 0 to destroy.
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js"]: Modifying... [id=static/js/main.b4f77641.js]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js"]: Modifications complete after 3s [id=static/js/main.b4f77641.js]

Warning: Argument is deprecated

  with aws_s3_bucket.site,
  on main.tf line 13, in resource "aws_s3_bucket" "site":
  13: resource "aws_s3_bucket" "site" {

Use the aws_s3_bucket_website_configuration resource instead

```

```

(and 31 more similar warnings elsewhere)

Warning: Deprecated attribute

  on outputs.tf line 6, in output "site_url":
   6:   value = aws_s3_bucket.site.website_endpoint

The attribute "website_endpoint" is deprecated. Refer to the provider documentation for details.

(and one more similar warning elsewhere)

```

Releasing state lock. This may take a few moments...

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

Outputs:

```

site_url = "my-react-app-dev-site.s3-website-us-east-1.amazonaws.com"
workspace = "dev"
PS C:\Program Files\Terraform\my-react-app>

```

Amazon S3

Buckets

my-react-app-dev-site

Amazon S3

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Access Grants

Access Points (General Purpose Buckets, FSx file systems)

Access Points (Directory Buckets)

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Tags (3)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

Key	Value
owner	adrina
env	dev
project	react-deploy

Amazon S3

Requester pays

When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. [Learn more](#)

Requester pays Disabled

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

We recommend using AWS Amplify Hosting for static website hosting

Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. [Learn more about Amplify Hosting](#) or [View your existing Amplify apps](#)

Create Amplify app

S3 static website hosting Enabled

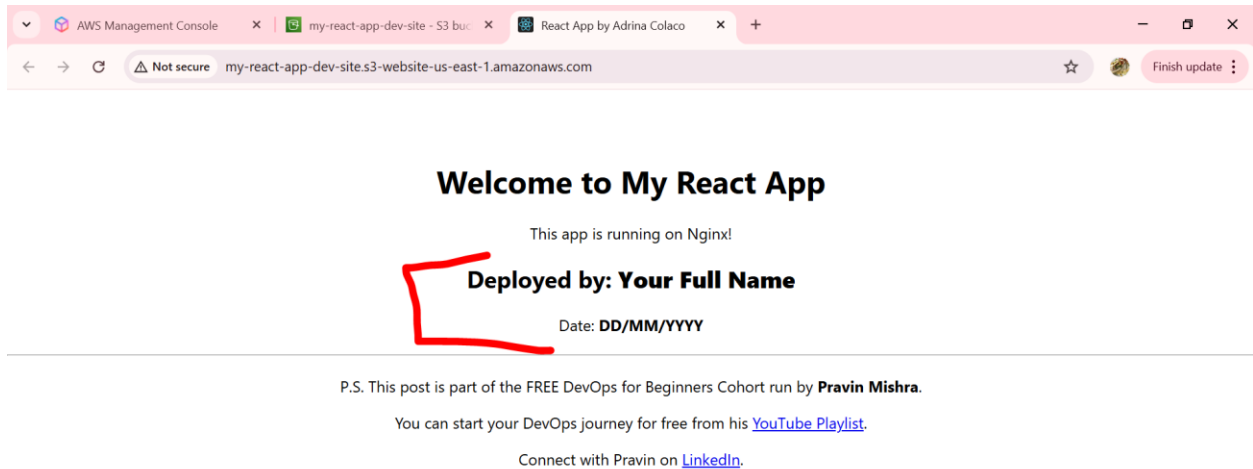
Hosting type Bucket hosting

Bucket website endpoint

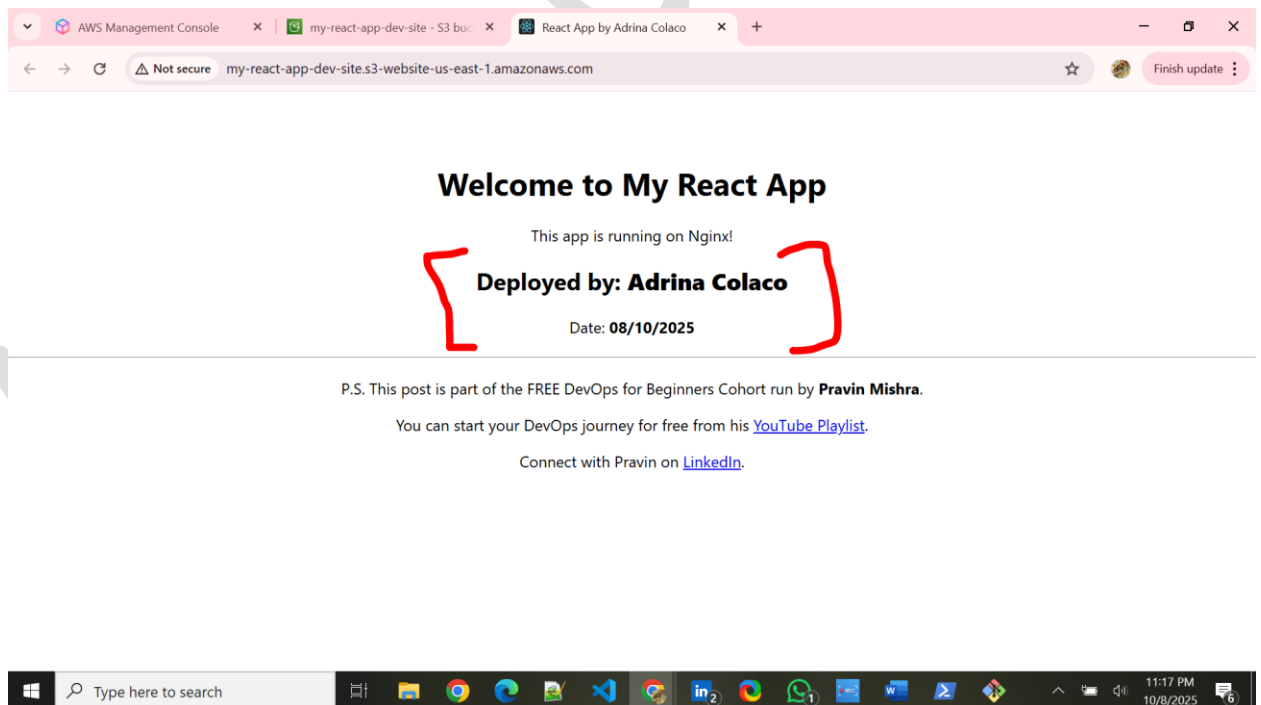
When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://my-react-app-dev-site.s3-website-us-east-1.amazonaws.com>

Initial deployment



After adding my changes



6. Deploy prod (separate state via workspace)
terraform init
terraform plan

Terraform apply -auto-approve

```
PS C:\Program Files\Terraform\my-react-app> terraform workspace select prod
switched to workspace "prod".
```

```
PS C:\Program Files\Terraform\my-react-app> terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v4.67.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Program Files\Terraform\my-react-app> terraform plan
Acquiring state lock. This may take a few moments...
aws_s3_bucket.site: Refreshing state... [id=my-react-app-prod-site]
aws_s3_bucket_public_access_block.site_block: Refreshing state... [id=my-react-app-prod-site]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js.map"]: Refreshing state... [id=static/js/main.b4f77641.js.map]
aws_s3_bucket_object.static_files["logo512.png"]: Refreshing state... [id=logo512.png]
aws_s3_bucket_object.static_files["logo192.png"]: Refreshing state... [id=logo192.png]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js"]: Refreshing state... [id=static/js/main.b4f77641.js]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js.LICENSE.txt"]: Refreshing state... [id=static/js/main.b4f77641.js.LICENSE.txt]
aws_s3_bucket_object.static_files["static/css/main.e6c13ad2.css.map"]: Refreshing state... [id=static/css/main.e6c13ad2.css.map]
aws_s3_bucket_object.static_files["asset-manifest.json"]: Refreshing state... [id=asset-manifest.json]
aws_s3_bucket_object.static_files["favicon.ico"]: Refreshing state... [id=favicon.ico]
aws_s3_bucket_object.static_files["static/css/main.e6c13ad2.css"]: Refreshing state... [id=static/css/main.e6c13ad2.css]
aws_s3_bucket_object.static_files["static/js/453.ad6eb26e.chunk.js"]: Refreshing state... [id=static/js/453.ad6eb26e.chunk.js]
aws_s3_bucket_object.static_files["manifest.json"]: Refreshing state... [id=manifest.json]
aws_s3_bucket_object.static_files["robots.txt"]: Refreshing state... [id=robots.txt]
aws_s3_bucket_object.static_files["static/js/453.ad6eb26e.chunk.js.map"]: Refreshing state... [id=static/js/453.ad6eb26e.chunk.js.map]
aws_s3_bucket_object.static_files["index.html"]: Refreshing state... [id=index.html]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_s3_bucket_policy.site_policy will be created
+ resource "aws_s3_bucket_policy" "site_policy" {
+   bucket = "my-react-app-prod-site"
```

```
PS C:\Program Files\Terraform\my-react-app> terraform apply -auto-approve
Acquiring state lock. This may take a few moments...
aws_s3_bucket.site: Refreshing state... [id=my-react-app-prod-site]
aws_s3_bucket_public_access_block.site_block: Refreshing state... [id=my-react-app-prod-site]
aws_s3_bucket_object.static_files["manifest.json"]: Refreshing state... [id=manifest.json]
aws_s3_bucket_object.static_files["logo192.png"]: Refreshing state... [id=logo192.png]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js.map"]: Refreshing state... [id=static/js/main.b4f77641.js.map]
aws_s3_bucket_object.static_files["static/js/453.ad6eb26e.chunk.js.map"]: Refreshing state... [id=static/js/453.ad6eb26e.chunk.js.map]
aws_s3_bucket_object.static_files["robots.txt"]: Refreshing state... [id=robots.txt]
aws_s3_bucket_object.static_files["static/css/main.e6c13ad2.css.map"]: Refreshing state... [id=static/css/main.e6c13ad2.css.map]
aws_s3_bucket_object.static_files["index.html"]: Refreshing state... [id=index.html]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js.LICENSE.txt"]: Refreshing state... [id=static/js/main.b4f77641.js.LICENSE.txt]
aws_s3_bucket_object.static_files["logo512.png"]: Refreshing state... [id=logo512.png]
aws_s3_bucket_object.static_files["favicon.ico"]: Refreshing state... [id=favicon.ico]
aws_s3_bucket_object.static_files["static/js/453.ad6eb26e.chunk.js"]: Refreshing state... [id=static/js/453.ad6eb26e.chunk.js]
aws_s3_bucket_object.static_files["static/css/main.e6c13ad2.css"]: Refreshing state... [id=static/css/main.e6c13ad2.css]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js"]: Refreshing state... [id=static/js/main.b4f77641.js]
aws_s3_bucket_object.static_files["asset-manifest.json"]: Refreshing state... [id=asset-manifest.json]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

# aws_s3_bucket_policy.site_policy will be created
+ resource "aws_s3_bucket_policy" "site_policy" {
  + bucket = "my-react-app-prod-site"
  + id      = (known after apply)
  + policy = jsonencode(
    {
      + Statement = [
        + {
          + Action = [
            + "s3:GetObject",
          ]
          + Effect   = "Allow"
          + Principal = "*"
          + Resource = [
            + "arn:aws:s3:::my-react-app-prod-site/*",
          ]
        },
      ],
    },
  )
  + Version = "2012-10-17"
}
```

```
Principal = "*"
Resource = [
  "arn:aws:s3::my-react-app-prod-site/*",
]
},
]
Version = "2012-10-17"
}
)
}

Plan: 1 to add, 0 to change, 0 to destroy.
aws_s3_bucket_policy.site_policy: Creating...
aws_s3_bucket_policy.site_policy: Creation complete after 1s [id=my-react-app-prod-site]

Warning: Argument is deprecated

    with aws_s3_bucket.site,
    on main.tf line 13, in resource "aws_s3_bucket" "site":
    13: resource "aws_s3_bucket" "site" {

Use the aws_s3_bucket_website_configuration resource instead

(and 29 more similar warnings elsewhere)

Warning: Deprecated attribute

    on outputs.tf line 6, in output "site_url":
     6:   value = aws_s3_bucket.site.website_endpoint

The attribute "website_endpoint" is deprecated. Refer to the provider documentation for details.

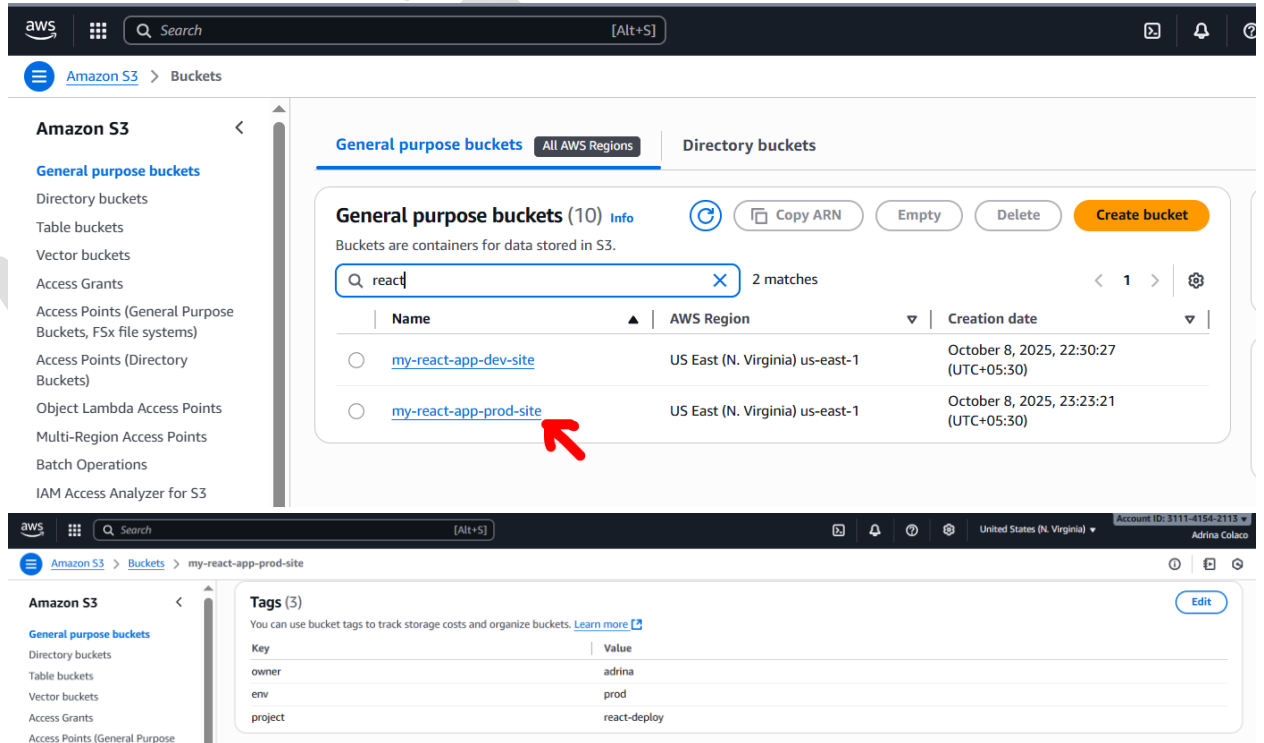
(and one more similar warning elsewhere)

Releasing state lock. This may take a few moments...

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

site_url = "my-react-app-prod-site.s3-website-us-east-1.amazonaws.com"
workspace = "prod"
PS C:\Program Files\Terraform\my-react-app>
```



Amazon S3 > Buckets

General purpose buckets (10) info

Buckets are containers for data stored in S3.

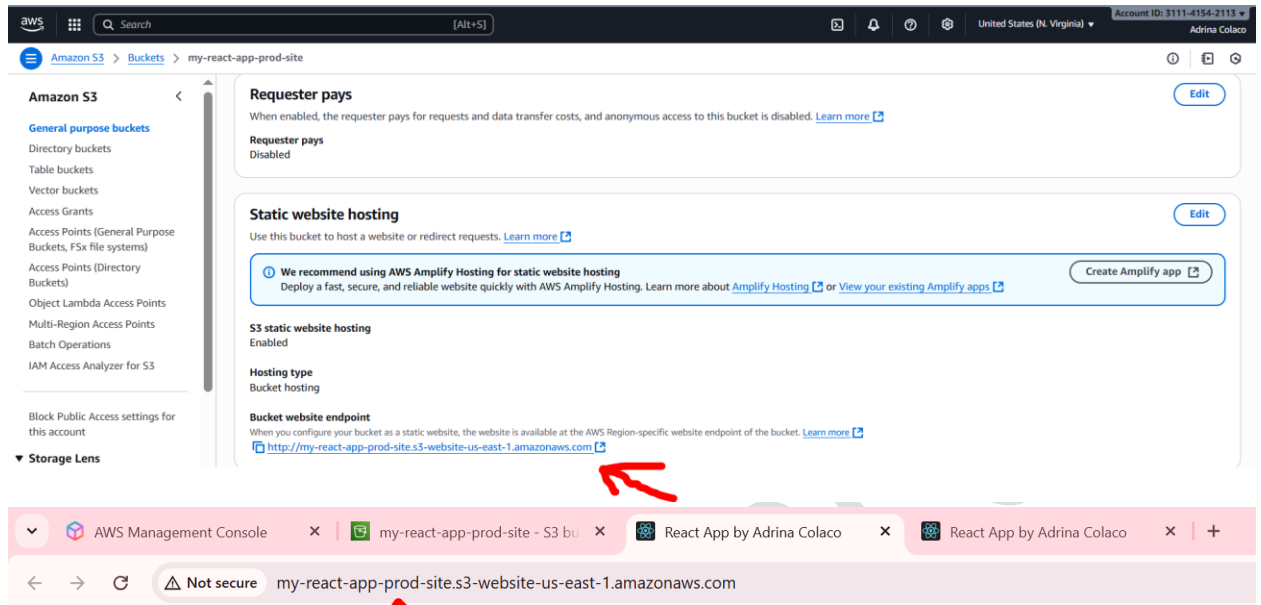
Search: react 2 matches

Name	AWS Region	Creation date
my-react-app-dev-site	US East (N. Virginia) us-east-1	October 8, 2025, 22:30:27 (UTC+05:30)
my-react-app-prod-site	US East (N. Virginia) us-east-1	October 8, 2025, 23:23:21 (UTC+05:30)

Tags (3)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

Key	Value
owner	adrina
env	prod
project	react-deploy



Welcome to My React App

This app is running on Nginx!

Deployed by: **Adrina Colaco**

Date: **08/10/2025**

P.S. This post is part of the FREE DevOps for Beginners Cohort run by **Pravin Mishra**.

You can start your DevOps journey for free from his [YouTube Playlist](#).

Connect with Pravin on [LinkedIn](#).

7. Prove isolation

a. In DEV

Make changes in JS.

Added profession to name and added a day to date.

terraform init

terraform plan

Terraform apply -auto-approve

```
PS C:\Program Files\Terraform\my-react-app> terraform workspace select dev
Switched to workspace "dev".
PS C:\Program Files\Terraform\my-react-app>
```

```
PS C:\Program Files\Terraform\my-react-app> terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v4.67.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Program Files\Terraform\my-react-app> terraform plan
Acquiring state lock. This may take a few moments...
aws_s3_bucket.site: Refreshing state... [id=my-react-app-dev-site]
aws_s3_bucket_policy.site_policy: Refreshing state... [id=my-react-app-dev-site]
aws_s3_bucket_object.static_files["manifest.json"]: Refreshing state... [id=manifest.json]
aws_s3_bucket_object.static_files["static/css/main.e6c13ad2.css.map"]: Refreshing state... [id=static/css/main.e6c13ad2.css.map]
aws_s3_bucket_object.static_files["logo512.png"]: Refreshing state... [id=logo512.png]
aws_s3_bucket_object.static_files["static/js/453.ad6eb26e.chunk.js"]: Refreshing state... [id=static/js/453.ad6eb26e.chunk.js]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js"]: Refreshing state... [id=static/js/main.b4f77641.js]
aws_s3_bucket_object.static_files["logo192.png"]: Refreshing state... [id=logo192.png]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js.map"]: Refreshing state... [id=static/js/main.b4f77641.js.map]
aws_s3_bucket_object.static_files["robots.txt"]: Refreshing state... [id=robots.txt]
aws_s3_bucket_object.static_files["favicon.ico"]: Refreshing state... [id=favicon.ico]
aws_s3_bucket_object.static_files["index.html"]: Refreshing state... [id=index.html]
aws_s3_bucket_object.static_files["asset-manifest.json"]: Refreshing state... [id=asset-manifest.json]
aws_s3_bucket_object.static_files["static/js/453.ad6eb26e.chunk.js.map"]: Refreshing state... [id=static/js/453.ad6eb26e.chunk.js.map]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js.LICENSE.txt"]: Refreshing state... [id=static/js/main.b4f77641.js.LICENSE.txt]
aws_s3_bucket_object.static_files["static/css/main.e6c13ad2.css"]: Refreshing state... [id=static/css/main.e6c13ad2.css]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
~ update in-place

Terraform will perform the following actions:

# aws_s3_bucket_object.static_files["static/js/main.b4f77641.js"] will be updated in-place
  id      = "static/js/main.b4f77641.js"
  tags    = {}
+ version_id = (known after apply)
```

```

}

# aws_s3_bucket_public_access_block.site_block will be created
+ resource "aws_s3_bucket_public_access_block" "site_block" {
  + block_public_acls      = false
  + block_public_policy    = false
  + bucket                 = "my-react-app-dev-site"
  + id                     = (known after apply)
  + ignore_public_acls     = false
  + restrict_public_buckets = false
}

Plan: 1 to add, 1 to change, 0 to destroy.

Warning: Argument is deprecated

  with aws_s3_bucket.site,
  on main.tf line 13, in resource "aws_s3_bucket" "site":
  13: resource "aws_s3_bucket" "site" {

Use the aws_s3_bucket_website_configuration resource instead

(and 2 more similar warnings elsewhere)

Warning: Deprecated attribute

  on outputs.tf line 6, in output "site_url":
   6:   value = aws_s3_bucket.site.website_endpoint

The attribute "website_endpoint" is deprecated. Refer to the provider documentation for details.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform a
Releasing state lock. This may take a few moments...
PS C:\Program Files\Terraform\my-react-app> terraform workspace select dev
PS C:\Program Files\Terraform\my-react-app> terraform apply -auto-approve
Acquiring state lock. This may take a few moments...
aws_s3_bucket.site: Refreshing state... [id=my-react-app-dev-site]
aws_s3_bucket_policy.site_policy: Refreshing state... [id=my-react-app-dev-site]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js.map"]: Refreshing state... [id=static/js/main.b4f77641.js.map]
aws_s3_bucket_object.static_files["robots.txt"]: Refreshing state... [id=robots.txt]

```

```
+ resource "aws_s3_bucket_public_access_block" "site_block" {
+   block_public_acls       = false
+   block_public_policy     = false
+   bucket                 = "my-react-app-dev-site"
+   id                     = (known after apply)
+   ignore_public_acls     = false
+   restrict_public_buckets = false
+ }

Plan: 1 to add, 1 to change, 0 to destroy.
aws_s3_bucket_public_access_block.site_block: Creating...
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js"]: Modifying... [id=static/js/main.b4f77641.js]
aws_s3_bucket_public_access_block.site_block: Creation complete after 3s [id=my-react-app-dev-site]
aws_s3_bucket_object.static_files["static/js/main.b4f77641.js"]: Modifications complete after 4s [id=static/js/main.b4f77641.js]

Warning: Argument is deprecated

    with aws_s3_bucket.site,
    on main.tf line 13, in resource "aws_s3_bucket" "site":
    13: resource "aws_s3_bucket" "site" {

Use the aws_s3_bucket_website_configuration resource instead

(and 31 more similar warnings elsewhere)

Warning: Deprecated attribute

    on outputs.tf line 6, in output "site_url":
     6:   value = aws_s3_bucket.site.website_endpoint

The attribute "website_endpoint" is deprecated. Refer to the provider documentation for details.

(and one more similar warning elsewhere)

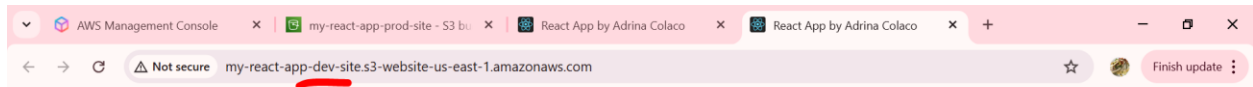
Releasing state lock. This may take a few moments...

Apply complete! Resources: 1 added, 1 changed, 0 destroyed.

Outputs:

site_url = "my-react-app-dev-site.s3-website-us-east-1.amazonaws.com"
workspace = "dev"
PS C:\Program Files\Terraform\my-react-app>
```





Welcome to My React App

This app is running on Nginx!

Deployed by: **Adrina Colaco - Business Analyst**

Date: **09/10/2025**

P.S. This post is part of the FREE DevOps for Beginners Cohort run by **Pravin Mishra**.

You can start your DevOps journey for free from his [YouTube Playlist](#).

Connect with Pravin on [LinkedIn](#).



b. In PROD:

Changes from DEV do not reflect in PROD



Welcome to My React App

This app is running on Nginx!

Deployed by: **Adrina Colaco**

Date: **08/10/2025**

P.S. This post is part of the FREE DevOps for Beginners Cohort run by **Pravin Mishra**.

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Connect with Pravin on [LinkedIn](#).



Video:



React_App_Terraform
.mp4

Assignment 32 – EpicBook on Azure/AWS with Production-Grade Terraform

Step 1 — Project Structure

```
terraform-epicbook/  
├─ main.tf  
├─ variables.tf  
├─ outputs.tf  
├─ backend.tf  
├─ locals.tf  
├─ modules/  
│   ├─ network/  
│   │   ├─ main.tf  
│   │   ├─ variables.tf  
│   │   └─ outputs.tf  
│   ├─ database/  
│   │   ├─ main.tf  
│   │   ├─ variables.tf  
│   │   └─ outputs.tf  
│   └─ compute/  
│       ├─ main.tf  
│       ├─ variables.tf  
│       └─ outputs.tf
```

Each module will represent one “layer” of your stack.

```
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook  
$ ls -l  
total 15  
-rw-r--r-- 1 alanm 197609 222 Oct 9 13:45 backend.tf  
-rw-r--r-- 1 alanm 197609 411 Oct 29 09:50 locals.tf  
-rw-r--r-- 1 alanm 197609 1427 Oct 9 21:21 main.tf  
drwxr-xr-x 1 alanm 197609 0 Oct 9 12:54 modules/  
-rw-r--r-- 1 alanm 197609 906 Oct 9 16:14 outputs.tf  
-rw-r--r-- 1 alanm 197609 161 Oct 9 16:09 terraform.tfvars  
-rw-r--r-- 1 alanm 197609 747 Oct 9 18:20 variables.tf
```

```
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/modules
$ ls -l
total 0
drwxr-xr-x 1 alanm 197609 0 Oct  9 15:01 compute/
drwxr-xr-x 1 alanm 197609 0 Oct  9 12:54 database/
drwxr-xr-x 1 alanm 197609 0 Oct  9 12:54 network/
```

```
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/modules
$ cd compute/

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/modules/comput
e
$ ls -l
total 10
-rw-r--r-- 1 alanm 197609 361 Oct  9 15:02 cloud-init.yml
-rw-r--r-- 1 alanm 197609 1396 Oct  9 14:59 main.tf
-rw-r--r-- 1 alanm 197609 520 Oct  9 16:21 outputs.tf
-rw-r--r-- 1 alanm 197609 984 Oct  9 16:57 variables.tf

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/modules/comput
e
$ cd ..

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/modules
$ cd database/

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/modules/databa
se
$ ls -l
total 9
-rw-r--r-- 1 alanm 197609 1013 Oct  9 22:11 main.tf
-rw-r--r-- 1 alanm 197609 417 Oct  9 16:21 outputs.tf
-rw-r--r-- 1 alanm 197609 798 Oct  9 17:12 variables.tf

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/modules/databa
se
$ cd ..

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/modules
$ cd network/

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/modules/networ
k
$ ls -l
total 9
-rw-r--r-- 1 alanm 197609 2854 Oct  9 19:48 main.tf
-rw-r--r-- 1 alanm 197609 433 Oct  9 18:04 outputs.tf
-rw-r--r-- 1 alanm 197609 1366 Oct  9 17:33 variables.tf

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/modules/networ
k
$ |
```

Clone the epicbooks repository on local system and verify the config.json for credentials.

```

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ ls
backend.tf  main.tf  outputs.tf  variables.tf
locals.tf  modules/  terraform.tfvars

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ git clone https://github.com/adcola13/theepicbook
Cloning into 'theepicbook'...
remote: Enumerating objects: 156, done.
remote: Counting objects: 100% (156/156), done.
remote: Compressing objects: 100% (146/146), done.
remote: Total 156 (delta 43), reused 93 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (156/156), 30.70 MiB | 831.00 KiB/s, done.
Resolving deltas: 100% (43/43), done.

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ cd theepicbook/

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/t
heepicbook (main)
$ ls
'Installation & Configuration Guide.md'  package.json
README.md                                public/
config/                                  routes/
db/                                       server.js
models/                                  views/
package-lock.json

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/t
heepicbook (main)
$ cd config/

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/t
heepicbook/config (main)
$ ls
config.json

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/t
heepicbook/config (main)
$ nano config.json

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook/t

```

Create a Key Pair for SSH

```

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ ssh-keygen -t rsa -b 4096 -f ~/.ssh/my-keypair
Generating public/private rsa key pair.
Enter passphrase for "/c/Users/alanm/.ssh/my-keypair" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/alanm/.ssh/my-keypair
Your public key has been saved in /c/Users/alanm/.ssh/my-keypair.pub
The key fingerprint is:
SHA256: [REDACTED] alanm@DESKTOP-TSHFOQK
The key's randomart image is:
+---[RSA 4096]-----+
|      oB+.          |
|      .ooo         |
|      .o+.         |
|      ..*..        |
|      S O .o E     |
|      *            |
|      [REDACTED]    |
|      [REDACTED]    |
|      ..+ . + *=B   |
|      ...o. . =o+o  |
+---[SHA256]-----+
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ |

```

Step 2 — Backend Setup (Remote State + Locking)

Before coding Terraform resources, create your **S3 bucket** and **DynamoDB table**.

Create S3 bucket and DynamoDB table

```
aws s3api create-bucket --bucket tfstate-epicbook --region us-east-1
```

```

us-east-1 +
~ $ aws s3api create-bucket --bucket tfstate-epicbook --region us-east-1
{
  "Location": "/tfstate-epicbook"
}
~ $

```

Amazon S3

General purpose buckets All AWS Regions Directory buckets

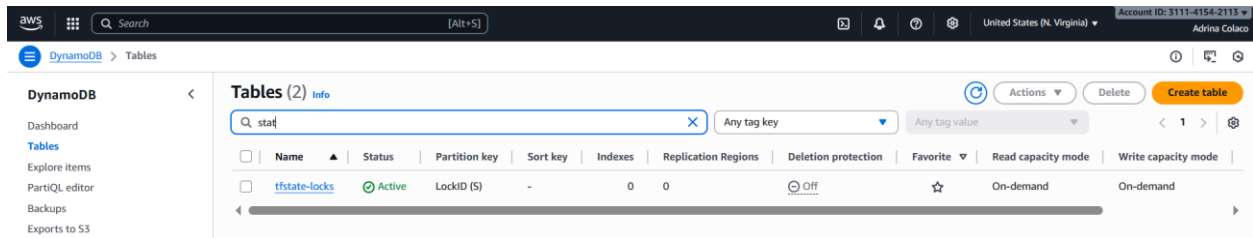
General purpose buckets (7) [Info](#)

Buckets are containers for data stored in S3.

Find buckets by name

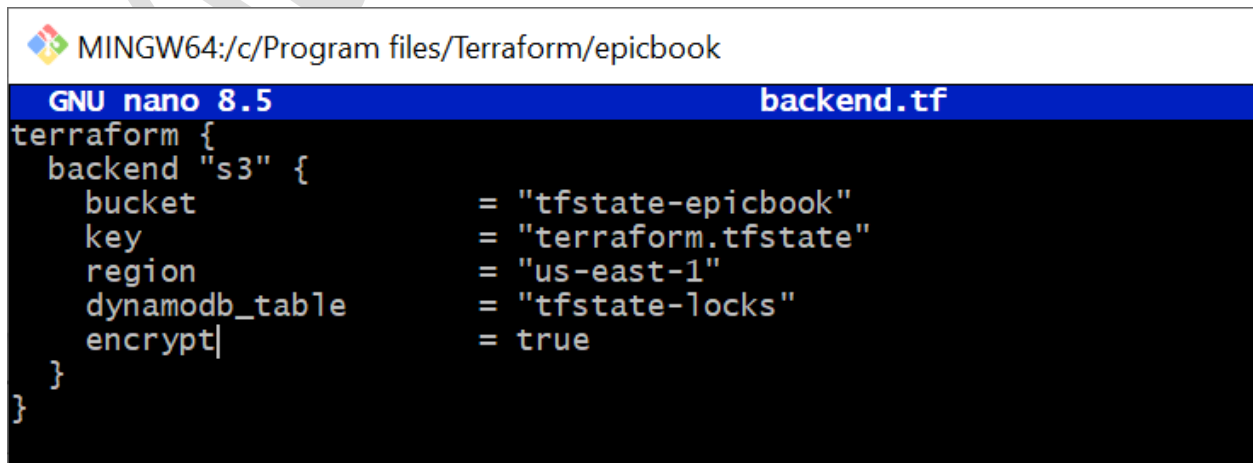
Name	AWS Region	Creation date
tfstate-epicbook	US East (N. Virginia) us-east-1	October 29, 2025, 11:07:02 (UTC+05:30)

```
aws dynamodb create-table \
--table-name tfstate-locks \
--attribute-definitions AttributeName=LockID,AttributeType=S \
--key-schema AttributeName=LockID,KeyType=HASH \
--billing-mode PAY_PER_REQUEST
```



backend.tf

```
terraform {
  backend "s3" {
    bucket      = "tfstate-epicbook"
    key         = "terraform.tfstate"
    region      = "us-east-1"
    dynamodb_table = "tfstate-locks"
    encrypt     = true
  }
}
```



Then initialize:

terraform init -reconfigure

```
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ nano backend.tf

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ terraform init -reconfigure
Initializing the backend...

Successfully configured the backend "s3"! Terraform will automatically
use this backend unless the backend configuration changes.
Initializing modules...
Initializing provider plugins...
- Reusing previous version of hashicorp/azurerm from the dependency lock file
- Using previously-installed hashicorp/azurerm v3.117.1

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ |
```

Step 3 — Define Locals and Variables

Use locals/maps to drive differences between dev and prod:

locals.tf

```
locals {
  env_config = {
    dev = {
      region    = "us-east-1"
      name_prefix = "dev-epicbook"
      vm_size    = "t3.micro"
      allowed_ip = "YOUR_PUBLIC_IP/32"
    }
    prod = {
      region    = "ap-southeast-1"
      name_prefix = "prod-epicbook"
      vm_size    = "t3.small"
    }
  }
}
```

```

    allowed_ip = "YOUR_PUBLIC_IP/32"
  }
}

current = local.env_config[terraform.workspace]
}

```

```

GNU nano 8.5 locals.tf
locals {
  env_config = {
    dev = {
      region      = "us-east-1"
      name_prefix = "dev-epicbook"
      vm_size     = "t3.micro"
      allowed_ip  = "YOUR_PUBLIC_IP/32"
    }
    prod = {
      region      = "ap-southeast-1"
      name_prefix = "prod-epicbook"
      vm_size     = "t3.small"
      allowed_ip  = "YOUR_PUBLIC_IP/32"
    }
  }

  current = local.env_config[terraform.workspace]
}

```

Step 4 — Network Module

modules/network/main.tf

```

resource "aws_vpc" "main" {
  cidr_block = "10.0.0.0/16"
  tags = { Name = "${var.name_prefix}-vpc" }
}

resource "aws_subnet" "public" {
  vpc_id            = aws_vpc.main.id
  cidr_block        = "10.0.1.0/24"
  map_public_ip_on_launch = true
  availability_zone = "${var.region}a"
}

```

```
tags = { Name = "${var.name_prefix}-public" }  
}
```

```
resource "aws_subnet" "mysql" {  
  vpc_id      = aws_vpc.main.id  
  cidr_block   = "10.0.2.0/24"  
  availability_zone = "${var.region}a"  
  tags = { Name = "${var.name_prefix}-mysql" }  
}
```

```
resource "aws_internet_gateway" "gw" {  
  vpc_id = aws_vpc.main.id  
}
```

```
resource "aws_route_table" "public" {  
  vpc_id = aws_vpc.main.id  
  route {  
    cidr_block = "0.0.0.0/0"  
    gateway_id = aws_internet_gateway.gw.id  
  }  
}
```

```
resource "aws_route_table_association" "public_assoc" {  
  subnet_id   = aws_subnet.public.id  
  route_table_id = aws_route_table.public.id  
}
```

Security Groups

```
resource "aws_security_group" "public_sg" {  
  name = "${var.name_prefix}-public-sg"  
  vpc_id = aws_vpc.main.id
```

```
  ingress {  
    from_port = 22  
    to_port   = 22  
    protocol  = "tcp"  
    cidr_blocks = [var.allowed_ip]  
  }  
}
```

```
ingress {
  from_port = 80
  to_port   = 80
  protocol  = "tcp"
  cidr_blocks = ["0.0.0.0/0"]
}

egress {
  from_port = 0
  to_port   = 0
  protocol  = "-1"
  cidr_blocks = ["0.0.0.0/0"]
}
}

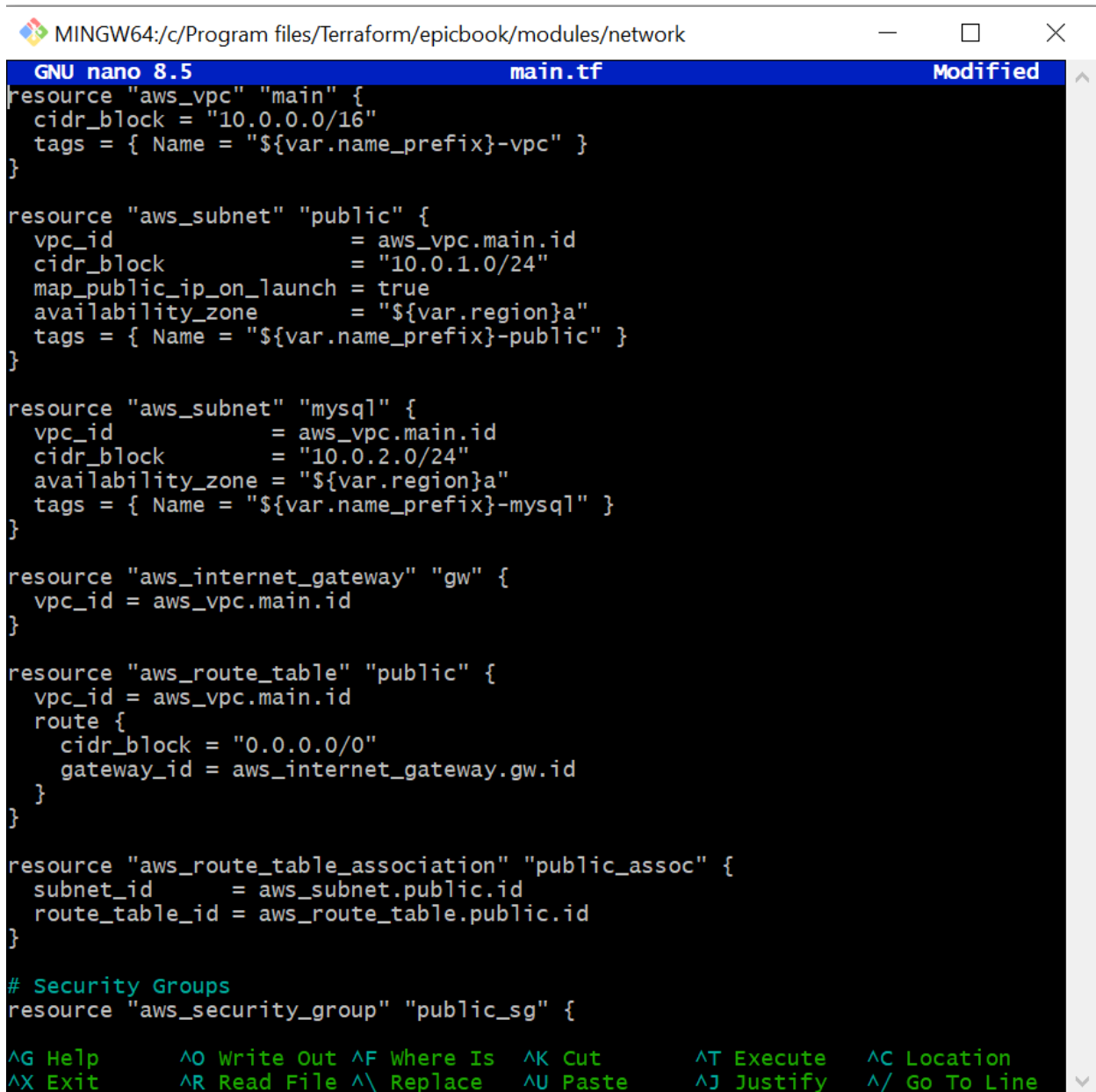
resource "aws_security_group" "private_sg" {
  name = "${var.name_prefix}-private-sg"
  vpc_id = aws_vpc.main.id

  ingress {
    from_port = 3306
    to_port   = 3306
    protocol  = "tcp"
    cidr_blocks = ["10.0.1.0/24"]
  }

  egress {
    from_port = 0
    to_port   = 0
    protocol  = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}

output "vpc_id" { value = aws_vpc.main.id }
output "public_subnet_id" { value = aws_subnet.public.id }
output "mysql_subnet_id" { value = aws_subnet.mysql.id }
```

```
output "public_sg_id" { value = aws_security_group.public_sg.id }
output "private_sg_id" { value = aws_security_group.private_sg.id }
```

A screenshot of a terminal window titled 'MINGW64:/c:/Program files/Terraform/epicbook/modules/network'. The terminal shows the 'main.tf' file being edited with 'GNU nano 8.5'. The configuration defines an AWS VPC, two subnets (public and mysql), an internet gateway, a route table, and a route table association. It also includes a section for Security Groups, starting with 'public_sg'. At the bottom of the terminal, there is a row of green keyboard shortcuts: ^G Help, ^O Write Out, ^F Where Is, ^K Cut, ^T Execute, ^C Location, ^X Exit, ^R Read File, ^\ Replace, ^U Paste, ^J Justify, and ^_ Go To Line.

```
resource "aws_vpc" "main" {
  cidr_block = "10.0.0.0/16"
  tags = { Name = "${var.name_prefix}-vpc" }
}

resource "aws_subnet" "public" {
  vpc_id            = aws_vpc.main.id
  cidr_block        = "10.0.1.0/24"
  map_public_ip_on_launch = true
  availability_zone  = "${var.region}a"
  tags = { Name = "${var.name_prefix}-public" }
}

resource "aws_subnet" "mysql" {
  vpc_id            = aws_vpc.main.id
  cidr_block        = "10.0.2.0/24"
  availability_zone  = "${var.region}a"
  tags = { Name = "${var.name_prefix}-mysql" }
}

resource "aws_internet_gateway" "gw" {
  vpc_id = aws_vpc.main.id
}

resource "aws_route_table" "public" {
  vpc_id = aws_vpc.main.id
  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.gw.id
  }
}

resource "aws_route_table_association" "public_assoc" {
  subnet_id      = aws_subnet.public.id
  route_table_id = aws_route_table.public.id
}

# Security Groups
resource "aws_security_group" "public_sg" {
```

modules/network/outputs.tf

```
output "vpc_id" {
  description = "ID of the created VPC"
  value       = aws_vpc.main.id
}
```

```
output "public_subnet_id" {
```

```
description = "ID of the public subnet where VM is deployed"
value      = aws_subnet.public.id
}
```

```
output "mysql_subnet_id" {
  description = "ID of the MySQL private subnet"
  value      = aws_subnet.mysql.id
}
```

```
output "public_sg_id" {
  description = "Security group ID for public subnet (VM access)"
  value      = aws_security_group.public_sg.id
}
```

```
output "private_sg_id" {
  description = "Security group ID for private subnet (DB access)"
  value      = aws_security_group.private_sg.id
}
```

```
output "vpc_cidr" {
  description = "CIDR block of the VPC"
  value      = aws_vpc.main.cidr_block
}
```

```
MINGW64:/c/Program files/Terraform/epicbook/modules/network
GNU nano 8.5 outputs.tf
output "vpc_id" {
  description = "ID of the created VPC"
  value      = aws_vpc.main.id
}

output "public_subnet_id" {
  description = "ID of the public subnet where VM is deployed"
  value      = aws_subnet.public.id
}

output "mysql_subnet_id" {
  description = "ID of the MySQL private subnet"
  value      = aws_subnet.mysql.id
}

output "public_sg_id" {
  description = "Security group ID for public subnet (VM access)"
  value      = aws_security_group.public_sg.id
}

output "private_sg_id" {
  description = "Security group ID for private subnet (DB access)"
  value      = aws_security_group.private_sg.id
}

output "vpc_cidr" {
  description = "CIDR block of the VPC"
  value      = aws_vpc.main.cidr_block
}
|
```

modules/network/variables.tf

```
variable "region" {}
```

```
variable "name_prefix" {}
```

```
variable "allowed_ip" {}
```

```
MINGW64:/c/Program files/Terraform/epicbook/modules/network
GNU nano 8.5 variables.tf
variable "region" {}
variable "name_prefix" {}
variable "allowed_ip" {}
|
```

Step 5 — Database Module (RDS MySQL)

modules/database/main.tf

```

resource "aws_db_subnet_group" "mysql" {
  name      = "${var.name_prefix}-mysql-subnetgrp"
  subnet_ids = [var.mysql_subnet_id]
}

resource "aws_db_instance" "mysql" {
  identifier      = "${var.name_prefix}-mysql"
  engine          = "mysql"
  engine_version  = "8.0"
  instance_class  = "db.t3.micro"
  allocated_storage = 20
  db_name         = var.db_name
  username        = var.db_user
  password        = var.db_password
  skip_final_snapshot = true
  vpc_security_group_ids = [var.private_sg_id]
  db_subnet_group_name = aws_db_subnet_group.mysql.name
  publicly_accessible = false
}

```

```

MINGW64:/c/Program files/Terraform/epicbook/modules/database
GNU nano 8.5 main.tf
resource "aws_db_subnet_group" "mysql" {
  name      = "${var.name_prefix}-mysql-subnetgrp"
  subnet_ids = [var.mysql_subnet_id]
}

resource "aws_db_instance" "mysql" {
  identifier      = "${var.name_prefix}-mysql"
  engine          = "mysql"
  engine_version  = "8.0"
  instance_class  = "db.t3.micro"
  allocated_storage = 20
  db_name         = var.db_name
  username        = var.db_user
  password        = var.db_password
  skip_final_snapshot = true
  vpc_security_group_ids = [var.private_sg_id]
  db_subnet_group_name = aws_db_subnet_group.mysql.name
  publicly_accessible = false
}

```

modules/database/outputs.tf

```
output "db_instance_id" {  
  description = "RDS instance ID"  
  value      = aws_db_instance.mysql.id  
}
```

```
output "db_endpoint" {  
  description = "RDS MySQL connection endpoint (private)"  
  value      = aws_db_instance.mysql.address  
}
```

```
output "db_name" {  
  description = "Database name created"  
  value      = aws_db_instance.mysql.db_name  
}
```

```
output "db_port" {  
  description = "Port for MySQL (usually 3306)"  
  value      = aws_db_instance.mysql.port  
}
```

```
output "db_arn" {  
  description = "ARN of the database instance"  
  value      = aws_db_instance.mysql.arn  
}
```

```
output "db_instance_id" {
  description = "RDS instance ID"
  value       = aws_db_instance.mysql.id
}

output "db_endpoint" {
  description = "RDS MySQL connection endpoint (private)"
  value       = aws_db_instance.mysql.address
}

output "db_name" {
  description = "Database name created"
  value       = aws_db_instance.mysql.db_name
}

output "db_port" {
  description = "Port for MySQL (usually 3306)"
  value       = aws_db_instance.mysql.port
}

output "db_arn" {
  description = "ARN of the database instance"
  value       = aws_db_instance.mysql.arn
}
```

modules/database/variables.tf

```
variable "name_prefix" {}
variable "mysql_subnet_id" {}
variable "private_sg_id" {}
variable "db_name" {}
variable "db_user" {}
variable "db_password" {
  sensitive = true
}
```

MINGW64:/c:/Program files/Terraform/epicbook/modules/database

GNU nano 8.5

variables.tf

```
variable "name_prefix" {}
variable "mysql_subnet_id" {}
variable "private_sg_id" {}
variable "db_name" {}
variable "db_user" {}
variable "db_password" {
  sensitive = true
}
```

Step 6 — Compute Module (EC2 + App Setup)

modules/compute/main.tf

```
resource "aws_instance" "vm" {
  ami          = data.aws_ami.ubuntu.id
  instance_type = var.vm_size
  subnet_id    = var.public_subnet_id
  vpc_security_group_ids = [var.public_sg_id]
  associate_public_ip_address = true

  user_data = file("${path.module}/cloud-init.sh")

  tags = { Name = "${var.name_prefix}-vm" }
}

data "aws_ami" "ubuntu" {
  most_recent = true
  owners      = ["099720109477"] # Canonical
  filter {
    name = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
  }
}

output "public_ip" { value = aws_instance.vm.public_ip }
```

```
MINGW64:/c/Program files/Terraform/epicbook/modules/compute
GNU nano 8.5 main.tf
resource "aws_instance" "vm" {
  ami           = data.aws_ami.ubuntu.id
  instance_type = var.vm_size
  subnet_id     = var.public_subnet_id
  vpc_security_group_ids = [var.public_sg_id]
  associate_public_ip_address = true

  user_data = file("${path.module}/cloud-init.sh")

  tags = { Name = "${var.name_prefix}-vm" }
}

data "aws_ami" "ubuntu" {
  most_recent = true
  owners      = ["099720109477"] # Canonical
  filter {
    name   = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
  }
}

output "public_ip" { value = aws_instance.vm.public_ip }
```

modules/compute/cloud-init.sh

```
#!/bin/bash
apt update -y
apt install -y nginx nodejs npm git mysql-client

cd /opt
git clone https://github.com/adcola13/theepicbook
cd epicbook
npm install
npm run build
```

Configure Nginx

```
cat >/etc/nginx/sites-available/default <<EOF
server {
    listen 80 default_server;
    root /opt/epicbook/frontend/build;
    index index.html;
    location / {
```

```

        try_files $uri /index.html;
    }
    location /api {
        proxy_pass http://localhost:3000;
    }
}
EOF

```

systemctl restart nginx

npm run start &

```

MINGW64:/c/Program files/Terraform/epicbook/modules/compute
GNU nano 8.5 cloud-init.sh
#!/bin/bash
apt update -y
apt install -y nginx nodejs npm git mysql-client

cd /opt
git clone https://github.com/yourusername/epicbook.git
cd epicbook
npm install
npm run build

# Configure Nginx
cat >/etc/nginx/sites-available/default <<EOF
server {
    listen 80 default_server;
    root /opt/epicbook/frontend/build;
    index index.html;
    location / {
        try_files $uri /index.html;
    }
    location /api {
        proxy_pass http://localhost:3000;
    }
}
EOF

systemctl restart nginx
npm run start &

```

modules/compute/outputs.tf

```

output "instance_id" {
    description = "EC2 instance ID"
}

```

```
    value    = aws_instance.vm.id
}
```

```
output "public_ip" {
  description = "Public IP address of the EC2 instance"
  value      = aws_instance.vm.public_ip
}
```

```
output "private_ip" {
  description = "Private IP address of the EC2 instance"
  value      = aws_instance.vm.private_ip
}
```

```
output "public_dns" {
  description = "Public DNS name of the instance"
  value      = aws_instance.vm.public_dns
}
```

```
output "ssh_command" {
  description = "SSH command for quick access"
  value      = "ssh ubuntu@${aws_instance.vm.public_ip}"
}
```



MINGW64:/c/Program files/Terraform/epicbook/modules/compute

GNU nano 8.5

outputs.tf

```
output "instance_id" {
  description = "EC2 instance ID"
  value      = aws_instance.vm.id
}

output "public_ip" {
  description = "Public IP address of the EC2 instance"
  value      = aws_instance.vm.public_ip
}

output "private_ip" {
  description = "Private IP address of the EC2 instance"
  value      = aws_instance.vm.private_ip
}

output "public_dns" {
  description = "Public DNS name of the instance"
  value      = aws_instance.vm.public_dns
}

output "ssh_command" {
  description = "SSH command for quick access"
  value      = "ssh ubuntu@${aws_instance.vm.public_ip}"
}

|
```

modules/compute/variables.tf

```
variable "public_subnet_id" {
  description = "The ID of the public subnet where the EC2 instance will be launched"
  type       = string
}

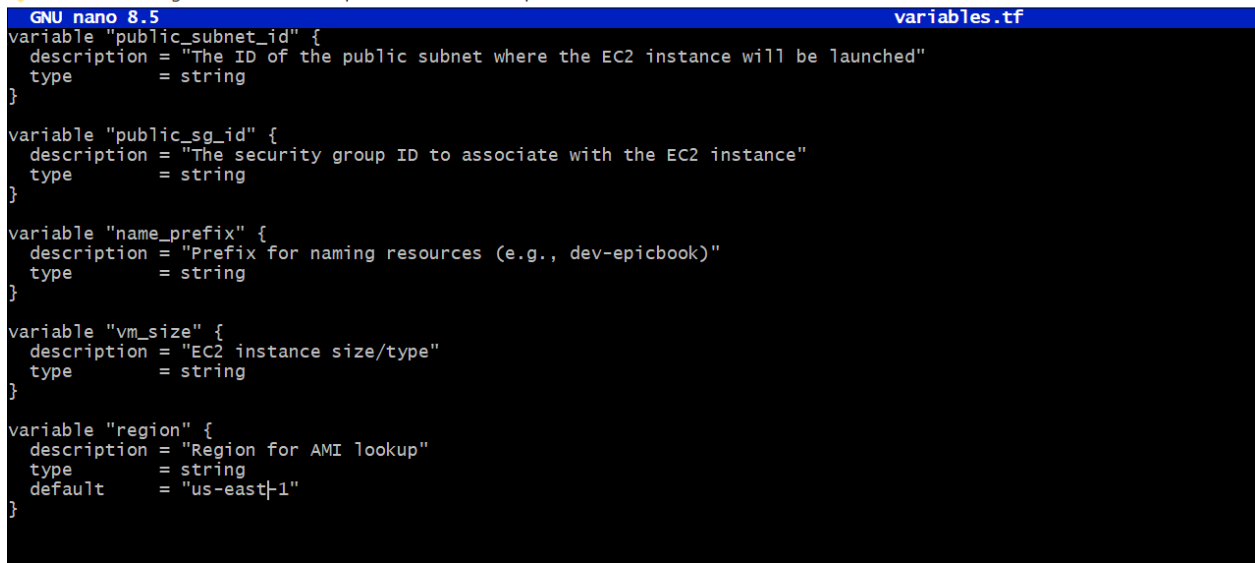
variable "public_sg_id" {
  description = "The security group ID to associate with the EC2 instance"
  type       = string
}

variable "name_prefix" {
  description = "Prefix for naming resources (e.g., dev-epicbook)"
  type       = string
}
```

```
variable "vm_size" {
  description = "EC2 instance size/type"
  type        = string
}
```

```
variable "region" {
  description = "Region for AMI lookup"
  type        = string
  default     = "ap-south-1"
}
```

MINGW64:/c:/Program files/Terraform/epicbook/modules/compute



```
variable "public_subnet_id" {
  description = "The ID of the public subnet where the EC2 instance will be launched"
  type        = string
}

variable "public_sg_id" {
  description = "The security group ID to associate with the EC2 instance"
  type        = string
}

variable "name_prefix" {
  description = "Prefix for naming resources (e.g., dev-epicbook)"
  type        = string
}

variable "vm_size" {
  description = "EC2 instance size/type"
  type        = string
}

variable "region" {
  description = "Region for AMI lookup"
  type        = string
  default     = "us-east-1"
}
```

Step 7 — Root Module Integration

main.tf

```
provider "aws" {
  region = local.current.region
}
```

```
module "network" {
  source      = "../modules/network"
  region      = local.current.region
  name_prefix = local.current.name_prefix
  allowed_ip  = local.current.allowed_ip
}
```

```
}
```

```
module "database" {  
  source      = "./modules/database"  
  name_prefix = local.current.name_prefix  
  mysql_subnet_id = module.network.mysql_subnet_id  
  private_sg_id  = module.network.private_sg_id  
  db_name        = var.db_name  
  db_user        = var.db_user  
  db_password    = var.db_password  
}
```

```
module "compute" {  
  source      = "./modules/compute"  
  name_prefix = local.current.name_prefix  
  vm_size     = local.current.vm_size  
  public_subnet_id = module.network.public_subnet_id  
  public_sg_id  = module.network.public_sg_id  
}
```

```
output "vm_public_ip" { value = module.compute.public_ip }  
output "db_endpoint" { value = module.database.db_endpoint }
```

```
GNU nano 8.5 main.tf
provider "aws" {
  region = local.current.region
}

module "network" {
  source      = "./modules/network"
  region      = local.current.region
  name_prefix = local.current.name_prefix
  allowed_ip  = local.current.allowed_ip
}

module "database" {
  source            = "./modules/database"
  name_prefix       = local.current.name_prefix
  mysql_subnet_id  = module.network.mysql_subnet_id
  private_sg_id    = module.network.private_sg_id
  db_name           = var.db_name
  db_user           = var.db_user
  db_password       = var.db_password
}

module "compute" {
  source            = "./modules/compute"
  name_prefix       = local.current.name_prefix
  vm_size           = local.current.vm_size
  public_subnet_id = module.network.public_subnet_id
  public_sg_id      = module.network.public_sg_id
}
```

Step 8 — Variables

variables.tf

```
variable "db_name" { default = "epicbook" }

variable "db_user" { default = "adminuser" }

variable "db_password" {

  description = "MySQL admin password"

  sensitive = true
}
```



MINGW64:/c/Program files/Terraform/epicbook

GNU nano 8.5

variables.tf

```
variable "db_name" { default = "epicbook" }
variable "db_user" { default = "adminuser" }
variable "db_password" {
  description = "MySQL admin password"
  sensitive   = true
}
```

Outputs.tf

----- Compute Outputs -----

```
output "vm_public_ip" {
  description = "Public IP of the EC2 instance hosting EpicBook"
  value       = module.compute.public_ip
}
```

```
output "vm_ssh_command" {
  description = "SSH command to access the EC2 instance"
  value       = module.compute.ssh_command
}
```

```
output "app_url" {
  description = "URL to access the EpicBook app"
  value       = "http://${module.compute.public_ip}"
}
```

----- Database Outputs -----

```
output "db_endpoint" {
```

```
description = "Private RDS MySQL endpoint"
value      = module.database.db_endpoint
}
```

```
output "db_name" {
  description = "Database name"
  value      = module.database.db_name
}
```

```
output "db_port" {
  description = "Database port"
  value      = module.database.db_port
}
```

```
# ----- Networking Outputs -----
```

```
output "vpc_id" {
  description = "VPC ID for reference"
  value      = module.network.vpc_id
}
```

```
output "public_subnet_id" {
  description = "Public subnet ID used by the EC2 instance"
  value      = module.network.public_subnet_id
}
```

```
output "mysql_subnet_id" {
```

```
description = "Private MySQL subnet ID"
value      = module.network.mysql_subnet_id
}

output "public_sg_id" {
  description = "Security group for public subnet"
  value      = module.network.public_sg_id
}

output "private_sg_id" {
  description = "Security group for private subnet"
  value      = module.network.private_sg_id
}

# ----- Meta -----
output "environment" {
  description = "Current Terraform workspace (environment)"
  value      = terraform.workspace
}
```

```
GNU nano 8.5                                outputs.tf
# ----- Compute Outputs -----
output "vm_public_ip" {
  description = "Public IP of the EC2 instance hosting EpicBook"
  value       = module.compute.public_ip
}

output "vm_ssh_command" {
  description = "SSH command to access the EC2 instance"
  value       = module.compute.ssh_command
}

output "app_url" {
  description = "URL to access the EpicBook app"
  value       = "http://${module.compute.public_ip}"
}

# ----- Database Outputs -----
output "db_endpoint" {
  description = "Private RDS MySQL endpoint"
  value       = module.database.db_endpoint
}

output "db_name" {
  description = "Database name"
  value       = module.database.db_name
}

output "db_port" {
  description = "Database port"
  value       = module.database.db_port
}

# ----- Networking Outputs -----
output "vpc_id" {
  description = "VPC ID for reference"
  value       = module.network.vpc_id
}

output "public_subnet_id" {
  [ Wrote 63 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Step 9 — Workspaces

terraform init -reconfigure

```
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ terraform init -reconfigure
Initializing the backend...

Successfully configured the backend "s3"! Terraform will automatically
use this backend unless the backend configuration changes.
Initializing modules...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v5.100.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
```

terraform workspace new dev

```
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ terraform workspace new dev
Created and switched to workspace "dev"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ terraform validate
```

terraform validate

```
alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ terraform validate
Success! The configuration is valid.
```

terraform plan

```
alann@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ terraform plan
Acquiring state lock. This may take a few moments...
var.db_password
  MySQL admin password

Enter a value:

module.compute.data.aws_ami.ubuntu: Reading...
module.compute.data.aws_ami.ubuntu: Read complete after 2s [id=ami-0c398cb65a93047f2]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# module.compute.aws_instance.vm will be created
+ resource "aws_instance" "vm" {
  + ami                        = "ami-0c398cb65a93047f2"
  + arn                      = (known after apply)
  + associate_public_ip_address = true
  + availability_zone         = (known after apply)
  + cpu_core_count            = (known after apply)
  + cpu_threads_per_core      = (known after apply)
  + disable_api_stop          = (known after apply)
  + disable_api_termination   = (known after apply)
  + ebs_optimized              = (known after apply)
  + enable_primary_ipv6        = (known after apply)
  + get_password_data          = false
  + host_id                   = (known after apply)
  + host_resource_group_arn    = (known after apply)
  + iam_instance_profile       = (known after apply)
  + id                        = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle         = (known after apply)
  + instance_state             = (known after apply)
  + instance_type              = "t3.micro"
  + ipv6_address_count         = (known after apply)
  + ipv6_addresses             = (known after apply)
  + key_name                   = (known after apply)
  + monitoring                 = (known after apply)
  + outpost_arn                = (known after apply)
  + password_data              = (known after apply)
  + placement_group            = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns                = (known after apply)
  + private_ip                 = (known after apply)
}

# module.network.aws_vpc.main will be created
+ resource "aws_vpc" "main" {
  + arn                        = (known after apply)
  + cidr_block                 = "10.0.0.0/16"
  + default_network_acl_id     = (known after apply)
  + default_route_table_id     = (known after apply)
  + default_security_group_id  = (known after apply)
  + dhcp_options_id            = (known after apply)
  + enable_dns_hostnames       = (known after apply)
  + enable_dns_support         = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                         = (known after apply)
  + instance_tenancy           = "default"
  + ipv6_association_id        = (known after apply)
  + ipv6_cidr_block            = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id        = (known after apply)
  + owner_id                   = (known after apply)
  + tags                       = {
    + "Name" = "dev-epicbook-vpc"
  }
  + tags_all                   = {
    + "Name" = "dev-epicbook-vpc"
  }
}

Plan: 11 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ app_url           = (known after apply)
+ db_endpoint       = (known after apply)
+ db_name           = "epicbook"
+ db_port           = (known after apply)
+ environment       = "dev"
+ mysql_subnet_id   = (known after apply)
+ private_sg_id     = (known after apply)
+ public_sg_id      = (known after apply)
+ public_subnet_id  = (known after apply)
+ vm_public_ip      = (known after apply)
+ vm_ssh_command    = (known after apply)
+ vpc_id            = (known after apply)

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
Releasing state lock. This may take a few moments...
alann@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$
```

terraform apply -auto-approve

```

alanm@DESKTOP-TSHFOQK MINGW64 ~/.ssh
$ chmod 400 "my-keypair"

alanm@DESKTOP-TSHFOQK MINGW64 ~/.ssh
$ ssh -i "my-keypair" ubuntu@44.202.166.111
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1040-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Oct 30 07:24:18 UTC 2025

System load:  0.08               Processes:            105
Usage of /:   37.8% of 7.57GB    Users logged in:     0
Memory usage: 29%               IPv4 address for ens5: 10.0.1.102
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

5 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Oct 30 07:07:34 2025 from 182.70.43.19
ubuntu@ip-10-0-1-102:~$ |

```

sudo apt update -y

sudo apt install -y mysql-server

sudo systemctl start mysql

sudo systemctl enable mysql

sudo systemctl status mysql

```

ubuntu@ip-10-0-1-102:~/app$ sudo systemctl start mysql
sudo systemctl enable mysql
sudo systemctl status mysql
Synchronizing state of mysql.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2025-10-30 08:45:28 UTC; 19s ago
     Main PID: 8240 (mysqld)
    Status: "Server is operational"
      Tasks: 38 (limit: 1073)
     Memory: 354.5M
        CPU: 962ms
    CGroup: /system.slice/mysql.service
            └─8240 /usr/sbin/mysqld

Oct 30 08:45:27 ip-10-0-1-102 systemd[1]: Starting MySQL Community Server...
Oct 30 08:45:28 ip-10-0-1-102 systemd[1]: Started MySQL Community Server.

```

```
ubuntu@ip-10-0-1-102:~/app$ mysql -h dev-epicbook-mysql.c4xii6s2qytn.us-east-1.rds.amazonaws.com -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 57
Server version: 8.0.42 Source distribution
```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> show databases
+-----+
| Database |
+-----+
| bookstore |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.01 sec)
```

```
mysql> /**
/*> On your local machine, replace 'bookstore' with 'bookstore_db'
/*> **/
mysql>
mysql> -- Use JAWS_DB Database
mysql> USE bookstore;
Database changed
mysql>
mysql> -- Create Table Author on DB
mysql> -- CREATE TABLE `bookstore`.`Author` (
mysql> --   `id` INT NOT NULL AUTO_INCREMENT,
mysql> --   `firstName` VARCHAR(45) NOT NULL,
mysql> --   `lastName` VARCHAR(45) NOT NULL,
mysql> --   PRIMARY KEY (`id`));
mysql>
mysql> -- Create Table Author on DB
mysql> CREATE TABLE `bookstore`.`Author` (
```

```
mysql>
mysql> -- Create Table Author on DB
mysql> CREATE TABLE `bookstore`.`Author` (
  ->   `id` INT NOT NULL AUTO_INCREMENT,
  ->   `firstName` VARCHAR(45) NOT NULL,
  ->   `lastName` VARCHAR(45) NOT NULL,
  ->   `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  ->   `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  ->   PRIMARY KEY (`id`));
-- Create Table Book after Author because of FK constraint to Author Tb1
CREATE TABLE `bookstore`.`Book` (
  `id` INT NOT NULL AUTO_INCREMENT,
Query OK, 0 rows affected (0.07 sec)
```

```
mysql>
mysql> -- Create Table Book after Author because of FK constraint to Author Tb1
mysql> CREATE TABLE `bookstore`.`Book` (
  ->   `id` INT NOT NULL AUTO_INCREMENT,
  ->   `title` VARCHAR(255) NOT NULL,
  ->   `genre` VARCHAR(255) NOT NULL,
  ->   `pubYear` INT NOT NULL,
  ->   `price` DECIMAL(13,2) NOT NULL,
  ->   `inventory` INT NOT NULL,
  ->   `bookDescription` TEXT NOT NULL,
  ->   `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  ->   `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  ->   `authorId` INT NOT NULL,
  ->   PRIMARY KEY (`id`),
  ->   INDEX `AuthorId_idx` (`AuthorId` ASC),
  ->   CONSTRAINT `AuthorId`
  ->     FOREIGN KEY (`AuthorId`)
  ->     REFERENCES `bookstore`.`Author` (`id`)
  ->     ON DELETE NO ACTION
  ->     ON UPDATE NO ACTION);
```

```
-- Create Table Cart
Query OK, 0 rows affected (0.03 sec)
```

```
mysql>
mysql>
mysql>
mysql> -- Create Table Cart
mysql> CREATE TABLE `bookstore`.`Cart` (
```

```
mysql> -- Create Table Cart
mysql> CREATE TABLE `bookstore`.`Cart` (
  -> `id` INT NOT NULL AUTO_INCREMENT,
  -> `quantity` INT NOT NULL,
  -> `price` DECIMAL(13,2) NOT NULL,
  -> `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  -> `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  -> PRIMARY KEY (`id`));
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> USE bookstore;
Database changed
mysql> -- Insert Books into Book Table
mysql> INSERT INTO Author (firstName,lastName,createdAt,updatedAt)
  -> VALUES
  -> ('Elin','Hildebrand',sysdate(),sysdate())
  -> ('Brit','Bennett',sysdate(),sysdate())
  -> ('John','Bolton',sysdate(),sysdate())
  -> ('Robin','DiAngelo',sysdate(),sysdate())
  -> ('Delia','Owens',sysdate(),sysdate())
  -> ('John','Grisham',sysdate(),sysdate())
  -> ('Ibram X.','Kendi',sysdate(),sysdate())
  -> ('Celeste','Ng',sysdate(),sysdate())
  -> ('Ijeoma','Oluo',sysdate(),sysdate())
  -> ('Harper','Lee',sysdate(),sysdate())
  -> ('L.M.','Montgomery',sysdate(),sysdate())
  -> ('Jack','London',sysdate(),sysdate())
  -> ('Jane','Austen',sysdate(),sysdate())
  -> ('F. Scott','Fitzgerald',sysdate(),sysdate())
  -> ('Emily','Bronte',sysdate(),sysdate())
  -> ('Oscar','Wilde',sysdate(),sysdate())
  -> ('George','Orwell',sysdate(),sysdate())
  -> ('J.D.','Salinger',sysdate(),sysdate())
  -> ('Shel','Silverstein',sysdate(),sysdate())
  -> ('Maurice','Sendak',sysdate(),sysdate())
  -> ('Dr.','Suess',sysdate(),sysdate())
  -> ('Beatrix','Potter',sysdate(),sysdate())
  -> ('C.S.','Lewis',sysdate(),sysdate())
  -> ('E.B.','White',sysdate(),sysdate())
  -> ('Eric','Carle',sysdate(),sysdate())
  -> ('Michael','Ende',sysdate(),sysdate())
  -> ('Howard','Zinn',sysdate(),sysdate())
  -> ('Norton','Juster',sysdate(),sysdate())
```

```
  -> ('Art','Spiegelman',sysdate(),sysdate())
  -> ('Margaret','Atwood',sysdate(),sysdate())
  -> ('Douglas','Adams',sysdate(),sysdate())
  -> ('David Foster','Wallace',sysdate(),sysdate())
  -> ('Philip D.','Morgan',sysdate(),sysdate())
  -> ('Hortense','Spillers',sysdate(),sysdate())
  -> ('Stephen','Best',sysdate(),sysdate())
  -> ('Anthony B.','Pinn',sysdate(),sysdate())
  -> ('Harvey','Young',sysdate(),sysdate())
  -> ('Michael A.','Gomez',sysdate(),sysdate())
  -> ('Sarah','Haley',sysdate(),sysdate())
  -> ('Daryl Michael','Scott',sysdate(),sysdate())
  -> ('Cheryl','Harris',sysdate(),sysdate())
  -> ('Frank','Herbert',sysdate(),sysdate())
  -> ('Andrzej','Sapkowski',sysdate(),sysdate())
  -> ('J.R.R.','Tolkein',sysdate(),sysdate())
  -> ('J.K.','Rowling',sysdate(),sysdate())
  -> ('Stephenie','Meyer',sysdate(),sysdate())
  -> ('George R. R.','Martin',sysdate(),sysdate())
  -> ('Diana','Gabaldon',sysdate(),sysdate())
  -> ('Robert','Jordan',sysdate(),sysdate())
  -> ('Cixin','Liu',sysdate(),sysdate())
  -> ;
Query OK, 53 rows affected (0.00 sec)
Records: 53 Duplicates: 0 Warnings: 0

mysql> -- Use JAWS_DB
mysql> USE bookstore;
Database changed
mysql> /**
  -> /* on your local machine, use the following:
  -> /* USE bookstore_db;
  -> /* */
mysql>
mysql> -- Insert Books into Book Table
mysql> INSERT INTO Book(title,genre,pubYear,price,inventory,bookDescription,createdAt,updatedAt,AuthorId)
  -> -- Values are || Title | Genre | Published Year | Price | Inventory | Description | Created DtTm | Updated DtTm | Author ID ||
  -> VALUES
  -> ('28 Summers','NVT',2020,'28',4,'A relationship that started in 1993 between Mallory Blessing and Jake McCloud comes to light while she is on her dea
thbed and his wife runs for president.',sysdate(),sysdate(),1)
  -> ('The Vanishing Half','NVT',2020,'27',8,'The lives of twin sisters who run away from a Southern Black community at age 16 diverge as one returns and
the other takes on a different racial identity but their fates intertwine.',sysdate(),sysdate(),2)
```

```

-> ,('Art','Spiegelman',sysdate(),sysdate())
-> ,('Margaret','Atwood',sysdate(),sysdate())
-> ,('Douglas','Adams',sysdate(),sysdate())
-> ,('David Foster','Wallace',sysdate(),sysdate())
-> ,('Philip D.','Morgan',sysdate(),sysdate())
-> ,('Hortense','Spillers',sysdate(),sysdate())
-> ,('Stephen','Best',sysdate(),sysdate())
-> ,('Anthony B.','Pinn',sysdate(),sysdate())
-> ,('Harvey','Young',sysdate(),sysdate())
-> ,('Michael A.','Gomez',sysdate(),sysdate())
-> ,('Sarah','Haley',sysdate(),sysdate())
-> ,('Daryl Michael','Scott',sysdate(),sysdate())
-> ,('Cheryl','Harris',sysdate(),sysdate())
-> ,('Frank','Herbert',sysdate(),sysdate())
-> ,('Andrzej','Sapkowski',sysdate(),sysdate())
-> ,('J.R.R.','Tolkein',sysdate(),sysdate())
-> ,('J.K.','Rowling',sysdate(),sysdate())
-> ,('Stephenie','Meyer',sysdate(),sysdate())
-> ,('George R. R.','Martin',sysdate(),sysdate())
-> ,('Diana','Gabaldon',sysdate(),sysdate())
-> ,('Robert','Jordan',sysdate(),sysdate())
-> ,('Cixin','Liu',sysdate(),sysdate())
-> ;
Query OK, 53 rows affected (0.00 sec)
Records: 53 Duplicates: 0 Warnings: 0

mysql> -- Use JAWS_DB
mysql> USE bookstore;
Database changed
mysql>
mysql> /**
/*> on your local machine, use the following:
/*> USE bookstore_db;
/*> **/
mysql>
mysql> -- Insert Books into Book Table
mysql> INSERT INTO Book(title,genre,pubYear,price,inventory,bookDescription,createdAt,updatedAt,AuthorId)
->
-> -- Values are || Title | Genre | Published Year | Price | Inventory | Description | Created DtTm | Updated DtTm | Author ID ||
-> VALUES
-> ('28 Summers','NYT',2020,'28',4,'A relationship that started in 1993 between Mallory Blessing and Jake McCloud comes to light while she is on her dea
thbed and his wife runs for president.',sysdate(),sysdate(),1)
-> ,('The Vanishing Half','NYT',2020,'27',8,'The lives of twin sisters who run away from a Southern Black community at age 16 diverge as one returns and
the other takes on a different racial identity but their fates intertwine.',sysdate(),sysdate(),2)

```

```

-> ,('Art','Spiegelman',sysdate(),sysdate())
-> ,('Margaret','Atwood',sysdate(),sysdate())
-> ,('Douglas','Adams',sysdate(),sysdate())
-> ,('David Foster','Wallace',sysdate(),sysdate())
-> ,('Philip D.','Morgan',sysdate(),sysdate())
-> ,('Hortense','Spillers',sysdate(),sysdate())
-> ,('Stephen','Best',sysdate(),sysdate())
-> ,('Anthony B.','Pinn',sysdate(),sysdate())
-> ,('Harvey','Young',sysdate(),sysdate())
-> ,('Michael A.','Gomez',sysdate(),sysdate())
-> ,('Sarah','Haley',sysdate(),sysdate())
-> ,('Daryl Michael','Scott',sysdate(),sysdate())
-> ,('Cheryl','Harris',sysdate(),sysdate())
-> ,('Frank','Herbert',sysdate(),sysdate())
-> ,('Andrzej','Sapkowski',sysdate(),sysdate())
-> ,('J.R.R.','Tolkein',sysdate(),sysdate())
-> ,('J.K.','Rowling',sysdate(),sysdate())
-> ,('Stephenie','Meyer',sysdate(),sysdate())
-> ,('George R. R.','Martin',sysdate(),sysdate())
-> ,('Diana','Gabaldon',sysdate(),sysdate())
-> ,('Robert','Jordan',sysdate(),sysdate())
-> ,('Cixin','Liu',sysdate(),sysdate())
-> ;
Query OK, 53 rows affected (0.00 sec)
Records: 53 Duplicates: 0 Warnings: 0

mysql> -- Use JAWS_DB
mysql> USE bookstore;
Database changed
mysql>
mysql> /**
/*> on your local machine, use the following:
/*> USE bookstore_db;
/*> **/
mysql>
mysql> -- Insert Books into Book Table
mysql> INSERT INTO Book(title,genre,pubYear,price,inventory,bookDescription,createdAt,updatedAt,AuthorId)
->
-> -- Values are || Title | Genre | Published Year | Price | Inventory | Description | Created DtTm | Updated DtTm | Author ID ||
-> VALUES
-> ('28 Summers','NYT',2020,'28',4,'A relationship that started in 1993 between Mallory Blessing and Jake McCloud comes to light while she is on her dea
thbed and his wife runs for president.',sysdate(),sysdate(),1)
-> ,('The Vanishing Half','NYT',2020,'27',8,'The lives of twin sisters who run away from a Southern Black community at age 16 diverge as one returns and
the other takes on a different racial identity but their fates intertwine.',sysdate(),sysdate(),2)

```

```

er. Incarcerated black women were obligated to perform both domestic labor that was gendered as female and hard physical labor that was gendered as male, but
also, as a result of this un-gendering, they were legislated out of the category of the human.',sysdate(),sysdate(),42)
-> ,('How Black Nationalism Became Sui Generis', 'Social Justice', 2012, '16.00', 2, 'Daryl Michael Scott', 's exploration of the vicissitudes of black national
ism that developed over the course of the early and mid-twentieth century negotiates a tension between black nationalism and other forms of nationalism. Whil
e nationalism on its own may be problematic, black nationalism has described everything from black separatism and sovereignty to an imperialism in imperio in wh
ich black people could self-determine without founding a new nation to a generic notion of racial solidarity among black people in a Pan-African context. The
idea continues to have critical purchase in academia.',sysdate(),sysdate(),43)
-> ,('Whiteness as Property', 'Social Justice', 1993, '20.00', 3, 'Cheryl Harris explores the extent to which whiteness became a form of property that had to
be protected by juridical and legislative means. As she argues, whiteness became valuable when white people could not be reduced to property under slavery. G
iven this, it is not only the cultural legacy of enslavement that has kept black people oppressed in the United States, but also a legal system that has cont
inued to treat whiteness as the norm. That system actively excludes people of color from the purview of equal rights and protections while also affording eco
nomic benefits exclusively or disproportionately to white Americans.',sysdate(),sysdate(),44)
-> ,('Dune', 'Fantasy', 1965, '49.95', 8, 'Set on the desert planet Arrakis, Dune is the story of the boy Paul Atreides, heir to a noble family tasked with ru
ling an inhospitable world where the only thing of value is the spice melange, a drug capable of extending life and enhancing consciousness. Coveted across t
he known universe, melange is a prize worth killing for....',sysdate(),sysdate(),45)
-> ,('The Witcher', 'Fantasy', 1994, '39.99', 8, 'The witcher is a fantasy series of novels and short stories written by Polish author Andrzej Sapkowski. The
series revolves around the titular "witcher", Geralt of Rivia. In Sapkowski's works, "witchers" are beast hunters who develop supernatural abilities at a yo
ung age to battle wild beasts and monsters.',sysdate(),sysdate(),46)
-> ,('Lord of the Rings', 'Fantasy', 1937, '35.96', 8, 'At 33, the age of adulthood among hobbits, Frodo Baggins receives a magic Ring of Invisibility from hi
s uncle Bilbo. Frodo, a Christlike figure, learns that the ring has the power to control the entire world and, he discovers, to corrupt its owner. A fellowsh
ip of hobbits, elves, dwarfs, and men is formed to destroy the ring by casting it into the volcanic fires of the Crack of Doom, where it was forged. They are
opposed on their harrowing mission by the evil Sauron and his Black Riders. The Lord of the Rings, together with The Hobbit, is considered by many to be the
start of the genre known as high fantasy, and these works have had an enormous influence on that genre as a whole.',sysdate(),sysdate(),47)
-> ,('Harry Potter', 'Fantasy', 1998, '55.99', 8, 'Harry Potter, a boy who learns on his eleventh birthday that he is the orphaned son of two powerful wizards
and possesses unique magical powers of his own. He is summoned from his life as an unwanted child to become a student at Hogwarts, an English boarding schoo
l for wizards. There, he meets several friends who become his closest allies and help him discover the truth about his parents' mysterious deaths.',sysdate(
),sysdate(),48)
-> ,('Twilight', 'Fantasy', 2005, '45.00', 8, 'The Twilight saga is a story about love. And vampires. And family. And abstinence. And racism. And the founding
of the Mormon faith. And orphans, in a really weird way.',sysdate(),sysdate(),49)
-> ,('A Song of Ice and Fire', 'Fantasy', 1991, '50.00', 8, 'A Song of Ice and Fire is a series of epic fantasy novels by the American novelist and screenwrit
er George R. R. Martin. He began the first volume of the series, A Game of Thrones, in 1991, and it was published in 1996.',sysdate(),sysdate(),50)
-> ,('Outlander', 'Fantasy', 1991, '70.00', 8, 'Outlander is a series of historical romance science fiction novels by American author Diana Gabaldon. Gabaldon
began the first volume of the series, Outlander, in the late 1980s, and it was published in 1991. She has published eight out of a planned ten volumes.',sys
date(),sysdate(),51)
-> ,('Wheel of Time', 'Fantasy', 1984, '32.00', 8, 'Written by Robert Jordan and completed posthumously by Brandon Sanderson, The Wheel of Time is one of the
greatest epics of fantasy and a #1 internationally bestselling series. Taking place both in our past and our future, the saga tells the story of a man destin
ed to face the Evil One and save the world or destroy it in the process.',sysdate(),sysdate(),52)
-> ,('Three-Body Problem Trilogy', 'Fantasy', 2008, '50.00', 8, 'From Hugo Award-winning author Cixin Lu, the most prolific and popular science fiction writer
in the People's Republic of China, the Remembrance of Earth's Past series is an action-packed, thought-provoking saga of how humanity's first contact wit
h extraterrestrial life divided Earth between those welcoming the superior beings and those fighting the alien invasion.',sysdate(),sysdate(),53);
Query OK, 54 rows affected (0.02 sec)
Records: 54 Duplicates: 0 Warnings: 0
mysql> |

```

Check if VM is created

Account ID: 3111-4154-2113 Adrina Colaco

[Alt+S]     United States (N. Virginia) 

Instances (1/1)        

Last updated less than a minute ago                  

Find Instance by attribute or tag (case-sensitive) All states 

Name = dev-epicbook-vm  Clear filters 

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☒	dev-epicbook-vm	i-0c0623d2f86d80af8	Running  	t3.micro	3/3 checks passed	View alarms	us-east-1a	-



i-0c0623d2f86d80af8 (dev-epicbook-vm)  

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Instance summary  Info

Instance ID  i-0c0623d2f86d80af8	Public IPv4 address  44.202.166.111 open address	Private IPv4 addresses  10.0.1.102
IPv6 address -	Instance state  Running	Public DNS -
Hostname type IP name: ip-10-0-1-102.ec2.internal	Private IP DNS name (IPv4 only)  ip-10-0-1-102.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding  Opt-in to AWS Compute Optimizer for recommendations.
Auto-assigned IP address  44.202.166.111 [Public IP]	VPC ID  vpc-009dc4e17d996ad23 (dev-epicbook-vpc)	

Adding port 8080 to VM sec group.

The screenshot shows the AWS Management Console interface. At the top, a green notification bar states: "Inbound security group rules successfully modified on security group [sg-0d6d4ed7a39df315c | dev-epicbook-public-sg]". Below this, the page title is "sg-0d6d4ed7a39df315c - dev-epicbook-public-sg". The "Details" section shows the security group name, ID, description, owner, and inbound/outbound rule counts. The "Inbound rules" tab is selected, displaying a table with 3 rules. A red arrow points to the rule with port 8080.

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-0d311a1f3896e44a7	IPv4	SSH	TCP	22	182.70.43.19/32
-	sgr-0261e9adfeff6eddf	IPv4	Custom TCP	TCP	8080	0.0.0.0/0
-	sgr-0218971f8b299de61	IPv4	HTTP	TCP	80	0.0.0.0/0

The screenshot shows a web browser window with the address bar displaying "44.202.166.111:8080". The website has a red header with the text "The EpicBook!". Below the header, there is a large banner with the text "Discover Your Next Great Read" and a sub-header "Whether you're seeking entertainment, pure enjoyment, or a journey that rejuvenates the mind, your next book awaits." Below the banner, there is a section titled "Browse through our collection below to find your new book!" which displays three book covers: "Behind the Corporate Curtain", "THE LITTLE ASTRONAUT", and "Journal".

terraform workspace new prod

```

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ terraform workspace new prod
Created and switched to workspace "prod"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ |

```

terraform import aws_key_pair.my_key my-keypair

```

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ terraform import aws_key_pair.my_key my-keypair
Acquiring state lock. This may take a few moments...
aws_key_pair.my_key: Importing from ID "my-keypair"...
aws_key_pair.my_key: Import prepared!
  Prepared aws_key_pair for import
aws_key_pair.my_key: Refreshing state... [id=my-keypair]
module.compute.data.aws_ami.ubuntu: Reading...
module.compute.data.aws_ami.ubuntu: Read complete after 2s [id=ami-0c398cb65a93047f2]

Import successful!

The resources that were imported are shown above. These resources are now in
your Terraform state and will henceforth be managed by Terraform.

Releasing state lock. This may take a few moments...

alanm@DESKTOP-TSHFOQK MINGW64 /c/Program files/Terraform/epicbook
$ |

```

terraform apply -auto-approve

✓ Each workspace creates separate infra (name prefix + region differ) but uses the same backend (different keys per workspace automatically).

Install project dependencies on VM

Connect the MySQL and EC2

Create the database structure

Step 10 — Locking Verification

1. Open two terminals.
2. In terminal A:

3. terraform apply

Wait at the approval prompt.

4. In terminal B:

5. terraform plan

You should see a lock message like:

"Error acquiring the state lock. LockID tfstate-locks/..."

Step 11 — Validation

- Visit `http://<vm_public_ip>` — you should see EpicBook frontend.
- API calls via `/api` should hit Node.js backend.
- MySQL only accessible within the VPC.
- Run `mysql -h <private-endpoint> -u adminuser -p` from VM to confirm connectivity.

Deliverables Checklist

Component	Deliverable
Modules	network, database, compute
Remote Backend	S3 + DynamoDB locking
Workspaces	dev, prod
App Setup	Node.js + Nginx + MySQL
NSGs	Enforced per requirements
Locking Test	Verified with dual terminal

Assignment 32 – EpicBook on Azure/AWS with Production-Grade Terraform

1. Project Structure

Create this folder layout:

```
epicbook-infra/  
├─ main.tf  
├─ variables.tf  
├─ outputs.tf  
├─ terraform.tfvars  
├─ backend.tf  
├─ modules/  
│   ├─ network/  
│   │   ├─ main.tf  
│   │   ├─ variables.tf  
│   │   └─ outputs.tf  
│   ├─ database/  
│   │   ├─ main.tf  
│   │   ├─ variables.tf  
│   │   └─ outputs.tf  
│   └─ compute/  
│       ├─ main.tf  
│       ├─ variables.tf  
│       └─ outputs.tf
```

New-Item "main.tf" -ItemType File

New-Item "variables.tf" -ItemType File

New-Item "outputs.tf" -ItemType File

New-Item "terraform.tfvars" -ItemType File

```
New-Item "backend.tf" -ItemType File
mkdir modules
cd modules
mkdir network
cd network
New-Item "main.tf" -ItemType File
New-Item "variables.tf" -ItemType File
New-Item "outputs.tf" -ItemType File
cd..
mkdir database
cd database
New-Item "main.tf" -ItemType File
New-Item "variables.tf" -ItemType File
New-Item "outputs.tf" -ItemType File
cd..
mkdir compute
cd compute
New-Item "main.tf" -ItemType File
New-Item "variables.tf" -ItemType File
New-Item "outputs.tf" -ItemType File
```

2. Backend Setup (Remote State + Locking)
Configuration in AZURE Cloudshell

Adrina Colaco

```
Microsoft Azure | Upgrade | Search resources, services, and docs (Ctrl) | Copilot | adicola13@gmail.com | RESULTS DIRECTORY ACCOUNT...

Switch to PowerShell | Restart | Manage files | New session | Editor | Web preview | Settings | Help

Requesting a Cloud Shell. Succeeded.
Connecting terminal...

Welcome to Azure Cloud Shell

Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
adriana [ ~ ]$ RESOURCE_GROUP=tfstate-rg
STORAGE_ACCOUNT=epicbooktfstate$RANDOM
CONTAINER=tfstate

az group create -n $RESOURCE_GROUP -l eastus
az storage account create -n $STORAGE_ACCOUNT -g $RESOURCE_GROUP -l eastus --sku Standard_LRS
az storage container create -n $CONTAINER --account-name $STORAGE_ACCOUNT
az storage account keys list -g $RESOURCE_GROUP -n $STORAGE_ACCOUNT

{
  "id": "/subscriptions/4aefc3f-e730-4757-944c-df7263b6fe6/resourceGroups/tfstate-rg",
  "location": "eastus",
  "managedBy": null,
  "name": "tfstate-rg",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}

{
  "accessTier": "hot",
  "accountIntegrationProgress": null,
  "allowBlobPublicAccess": false,
  "allowCrossTenantReplication": false,
  "allowSharedKeyAccess": null,
  "allowBlobKeyCaching": null,
  "azureFileIdentityBasedAuthentication": null,
  "blobStorageStatus": null,
  "creationTime": "2025-10-09T07:38:25.628839+00:00",
  "customDomain": null,
  "defaultOutboundAuthentication": null,
  "defaultEndpointType": null,
  "enableExtendedGroups": null,
  "enableHttpsTrafficOnly": true,
  "enableIMDS": null,
  "encryption": {
    "encryptionIdentity": null,
    "keySource": "Microsoft.Storage",
    "keyVaultProperties": null,
    "requireInfrastructureEncryption": null,
    "services": {
      "blob": {
        "enabled": true,
        "keyType": "Account",
        "lastEnabledTime": "2025-10-09T07:38:25.231971+00:00"
      },
      "file": {
        "enabled": true,
        "keyType": "Account",
        "lastEnabledTime": "2025-10-09T07:38:25.231971+00:00"
      }
    },
    "queue": null,
    "table": null
  },
  "extendedLocation": null,
  "failoverInProgress": null,
  "geoReplicationStatus": null,
  "id": "/subscriptions/4aefc3f-e730-4757-944c-df7263b6fe6/resourceGroups/tfstate-rg/providers/Microsoft.Storage/storageAccounts/epicbooktfstate10397",
  "identity": null,
  "immutableStorageWithVersioning": null,
  "isImmutable": null,
  "isLocalUserEnabled": null,
  "isSftpEnabled": null,
  "isShkConversionEnabled": null,
  "keyCreationTime": {
    "key1": "2025-10-09T07:38:25.216337+00:00",
    "key2": "2025-10-09T07:38:25.216337+00:00"
  },
  "keyPolicy": null,
  "kind": "StorageV2",
  "largeFileSharesState": null,
  "lastGeoFailoverTime": null,
  "location": "eastus",
  "minimumVersion": "1151.0",
  "name": "epicbooktfstate10397",
  "networkRuleSet": {
    "bypass": "AzureServices",
    "defaultAction": "Allow",
    "ipRanges": [],
    "ipRules": [],
    "resourceAccessRules": null,
    "virtualNetworkRules": []
  },
  "primaryEndpoints": {
    "blob": "https://epicbooktfstate10397.blob.core.windows.net/",
    "dfs": "https://epicbooktfstate10397.dfs.core.windows.net/",
    "file": "https://epicbooktfstate10397.file.core.windows.net/",
    "internet": null,
    "microsoft": null,
    "queue": "https://epicbooktfstate10397.queue.core.windows.net/",
    "table": "https://epicbooktfstate10397.table.core.windows.net/",
    "web": "https://epicbooktfstate10397.z13.web.core.windows.net/"
  },
  "primaryLocation": "eastus",
  "privateEndpointConnections": [],
  "provisioningState": "Succeeded",
  "publicNetworkAccess": null,
  "resourceGroup": "tfstate-rg",
  "routingReference": null,
  "sasPolicy": null,
  "secondaryEndpoints": null,
  "secondaryLocation": null,
  "sku": {
    "name": "Standard_LRS",
    "tier": "Standard"
  },
  "statusOffPrimary": "available",
  "statusOffSecondary": null,
  "storageAccountShkConversionStatus": null,
  "tags": {},
  "type": "Microsoft.Storage/storageAccounts"
}

There are no credentials provided in your command and environment, we will query for account key for your storage account.
It is recommended to provide --connection-string, --account-key or --sas-token in your command as credentials.

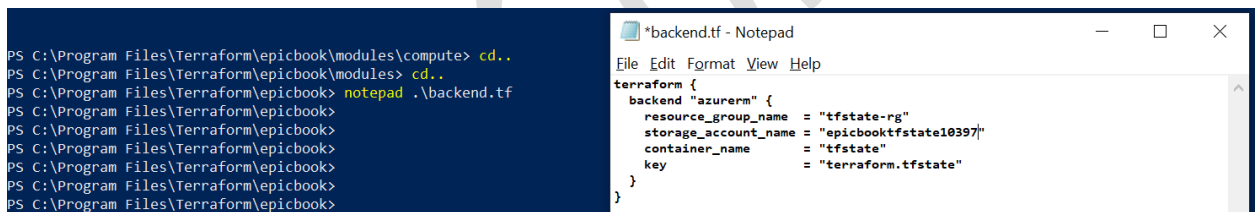
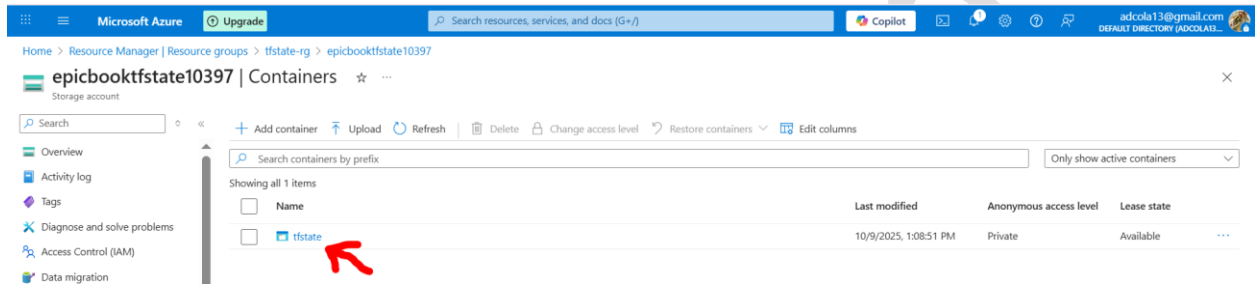
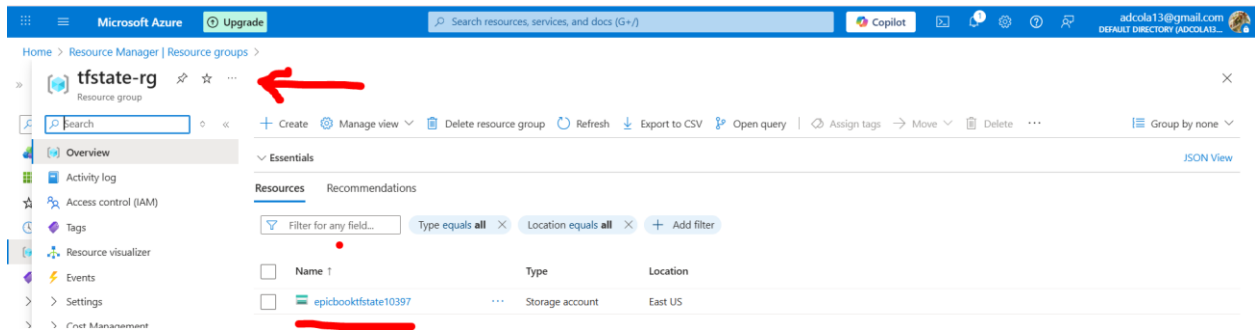
You also can add "--auth-mode login" in your command to use Azure Active Directory (Azure AD) for authorization if your login account is assigned required RBAC roles.
For more information about RBAC roles in storage, visit https://learn.microsoft.com/azure/storage/common/storage-auth-aad-rbac-cli.

In addition, setting the corresponding environment variables can avoid inputting credentials in your command. Please use --help to get more information about environment variable usage.

{
  "created": true
}

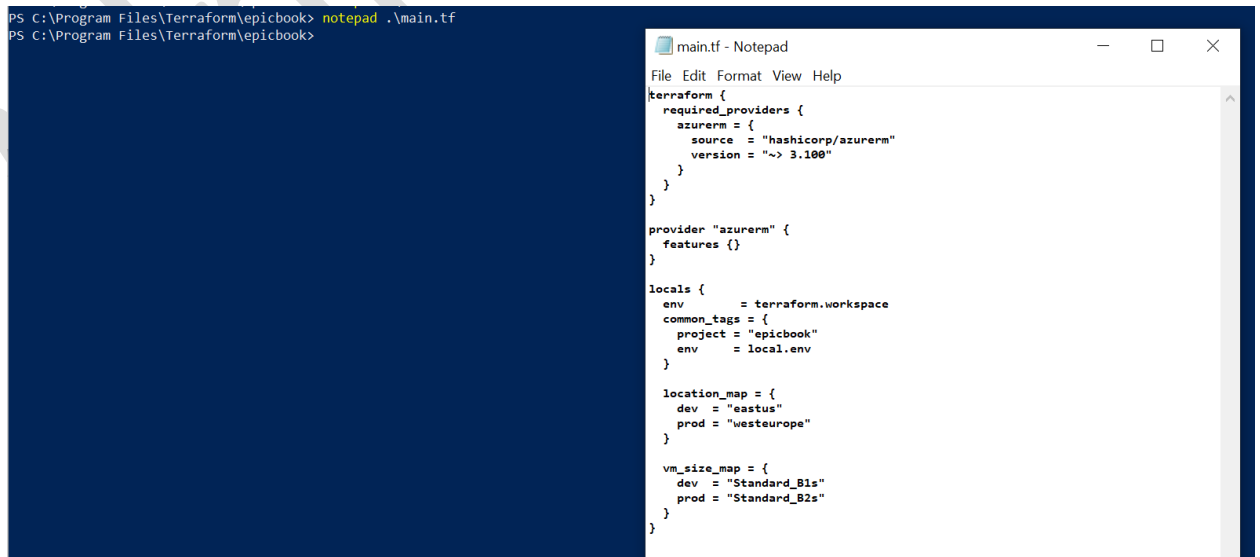
[Warning] This output may compromise security by showing the following secrets: value. Learn more at: https://go.microsoft.com/fwlink/?linkid=2258669
{
  "creationTime": "2025-10-09T07:38:25.216337+00:00",
  "keyName": "key1",
  "permissions": "FULL",
  "value": "dChk7Gz2rVqbbPMVBRV3sFz8P8Wf18G6L4BqWJFzPK+9DMJjv131PqGanZ80L3oorhuC+ASX6uMlQ=="
},
{
  "creationTime": "2025-10-09T07:38:25.216337+00:00",
  "keyName": "key2",
  "permissions": "FULL",
  "value": "6n64FPN2z6X4u6Q8B1Q21eC1Pwcf7q8C8M86u52W6VZf375ss55K4VWV7PEz5z114aewAS1NB4uA=="
}

adriana [ ~ ]$
adriana [ ~ ]$
```



3. Define Locals and Variables

Main.tf



Variables.tf

```
Administrator: Windows PowerShell (x86)
PS C:\Program Files\Terraform\epicbook> notepad .\variables.tf
PS C:\Program Files\Terraform\epicbook>

*variables.tf - Notepad
File Edit Format View Help
variable "admin_username" {
  description = "Admin username for the VM"
  type       = string
  default    = "azureuser"
}

variable "admin_password" {
  description = "Admin password for the VM"
  type       = string
  sensitive  = true
}

variable "mysql_admin_user" {
  description = "Admin username for MySQL flexible server"
  type       = string
  default    = "mysqladmin"
}

variable "mysql_admin_password" {
  description = "Admin password for MySQL flexible server"
  type       = string
  sensitive  = true
}
```

Terraform.tfvars

```
Administrator: Windows PowerShell (x86)
PS C:\Program Files\Terraform\epicbook> notepad .\variables.tf
PS C:\Program Files\Terraform\epicbook> notepad .\terraform.tfvars
PS C:\Program Files\Terraform\epicbook>

terraform.tfvars - Notepad
File Edit Format View Help
admin_username    = "azureuser"
admin_password    = "P@ssw0rd!"
mysql_admin_user  = "mysqladmin"
mysql_admin_password = "P@ssw0rd!"
```

Outputs.tf

```
Administrator: Windows PowerShell (x86)
PS C:\Program Files\Terraform\epicbook> notepad .\terraform.tfvars
PS C:\Program Files\Terraform\epicbook> notepad .\outputs.tf
PS C:\Program Files\Terraform\epicbook>

outputs.tf - Notepad
File Edit Format View Help
output "resource_group_name" {
  description = "Name of the resource group"
  value      = module.network.rg_name
}

output "vm_public_ip" {
  description = "Public IP of the EpicBook VM"
  value      = module.compute.vm_public_ip
}

output "site_url" {
  description = "URL of the EpicBook website"
  value      = "http://${module.compute.vm_public_ip}"
}

output "mysql_server_name" {
  description = "Name of the MySQL flexible server"
  value      = module.database.mysql_server_name
}

output "mysql_host_fqdn" {
  description = "FQDN of the MySQL server (for connection from the app)"
  value      = module.database.mysql_fqdn
}

output "mysql_db_name" {
  description = "EpicBook database name"
  value      = module.database.mysql_db_name
}

output "workspace" {
  description = "Terraform workspace (environment name)"
  value      = terraform.workspace
}
```

4. Networking (Module)

```
PS C:\Program Files\Terraform\epicbook> cd .\modules\network\  
PS C:\Program Files\Terraform\epicbook\modules\network> notepad .\main.tf
```

```
resource "azurerm_resource_group" "rg" {  
  name     = "${terraform.workspace}-epicbook-rg"  
  location = var.location  
  tags     = var.tags  
}  
  
resource "azurerm_virtual_network" "vnet" {  
  name            = "${terraform.workspace}-epicbook-vnet"  
  address_space   = ["10.0.0.0/16"]  
  location        = var.location  
  resource_group_name = azurerm_resource_group.rg.name  
}  
  
resource "azurerm_subnet" "public" {  
  name            = "public-subnet"  
  resource_group_name = azurerm_resource_group.rg.name  
  virtual_network_name = azurerm_virtual_network.vnet.name  
  address_prefixes   = ["10.0.1.0/24"]  
}  
  
resource "azurerm_subnet" "mysql" {  
  name            = "mysql-subnet"  
  resource_group_name = azurerm_resource_group.rg.name  
  virtual_network_name = azurerm_virtual_network.vnet.name  
  address_prefixes   = ["10.0.2.0/24"]  
}  
  
delegation {  
  name = "db-delegation"  
  service_delegation {  
    name = "Microsoft.DBforMySQL/flexibleServers"  
  }  
}  
}
```

```
# NSG for public subnet
resource "azurerm_network_security_group" "public_nsg" {
  name          = "${terraform.workspace}-public-nsg"
  location      = var.location
  resource_group_name = azurerm_resource_group.rg.name
```

```
  security_rule {
    name          = "SSH"
    priority      = 100
    direction     = "Inbound"
    access        = "Allow"
    protocol      = "Tcp"
    source_port_range = "*"
    destination_port_range = "22"
    source_address_prefix = var.allowed_ip
    destination_address_prefix = "*"
  }
}
```

```
  security_rule {
    name          = "HTTP"
    priority      = 110
    direction     = "Inbound"
    access        = "Allow"
    protocol      = "Tcp"
    destination_port_range = "80"
    source_address_prefix = "*"
    destination_address_prefix = "*"
  }
}
```

```
# NSG for MySQL subnet
resource "azurerm_network_security_group" "mysql_nsg" {
  name          = "${terraform.workspace}-mysql-nsg"
  location      = var.location
  resource_group_name = azurerm_resource_group.rg.name
```

```
  security_rule {
    name          = "MySQL"

```

Variables.tf

```
main.tf - Notepad
File Edit Format View Help

resource "azurerm_resource_group" "rg" {
  name     = "${terraform.workspace}-epicbook-rg"
  location = var.location
  tags     = var.tags
}

resource "azurerm_virtual_network" "vnet" {
  name                = "${terraform.workspace}-epicbook-vnet"
  address_space       = ["10.0.0.0/16"]
  location            = var.location
  resource_group_name = azurerm_resource_group.rg.name
}

resource "azurerm_subnet" "public" {
  name                = "public-subnet"
  resource_group_name = azurerm_resource_group.rg.name
  virtual_network_name = azurerm_virtual_network.vnet.name
  address_prefixes     = ["10.0.1.0/24"]
}

resource "azurerm_subnet" "mysql" {
  name                = "mysql-subnet"
  resource_group_name = azurerm_resource_group.rg.name
  virtual_network_name = azurerm_virtual_network.vnet.name
  address_prefixes     = ["10.0.2.0/24"]
}

delegation {
  name = "db-delegation"
  service_delegation {
    name = "Microsoft.DBforMySQL/flexibleServers"
  }
}
```

Variables.tf

```
PS C:\Program Files\Terraform\epicbook\modules> cd .\network\  
PS C:\Program Files\Terraform\epicbook\modules\network> notepad .\variables.tf  
PS C:\Program Files\Terraform\epicbook\modules\network>
```

```
*variables.tf - Notepad  
File Edit Format View Help  
#####  
# Network Module Variables #  
#####  
  
# Resource Group name  
variable "rg_name" {  
  description = "Name of the Resource Group where networking resources will be  
created."  
  type        = string  
}  
  
# Azure location / region  
variable "location" {  
  description = "Azure region where networking resources will be deployed."  
  type        = string  
}  
  
# Virtual Network CIDR range  
variable "vnet_cidr" {  
  description = "CIDR block for the Virtual Network."  
  type        = string  
  default     = "10.0.0.0/16"  
}  
  
# Public subnet CIDR (for VM, Nginx, etc.)  
variable "public_subnet_cidr" {  
  description = "CIDR block for the public subnet (VM)."  
  type        = string  
  default     = "10.0.1.0/24"  
}  
  
# Private subnet CIDR (for MySQL Flexible Server)  
variable "mysql_subnet_cidr" {  
  description = "CIDR block for the private subnet (MySQL Flexible Server)."  
  type        = string  
  default     = "10.0.2.0/24"  
}  
  
# Allowed IP for SSH (for security)  
variable "allowed_ip" {
```

Outputs.tf

```
PS C:\Program Files\Terraform\epicbook> cd .\modules\network\  
PS C:\Program Files\Terraform\epicbook\modules\network> notepad .\outputs.tf  
PS C:\Program Files\Terraform\epicbook\modules\network>
```

```
outputs.tf - Notepad  
File Edit Format View Help  
  
output "rg_name" {  
  description = "Name of the Resource Group"  
  value       = azurerm_resource_group.rg.name  
}  
  
output "location" {  
  description = "Azure location"  
  value       = azurerm_resource_group.rg.location  
}  
  
output "public_subnet_id" {  
  description = "ID of the public subnet"  
  value       = azurerm_subnet.public.id  
}  
  
output "mysql_subnet_id" {  
  description = "ID of the MySQL subnet"  
  value       = azurerm_subnet.mysql.id  
}  
  
output "dns_zone_id" {  
  description = "Private DNS Zone ID (if used)"  
  value       = try(azurerm_private_dns_zone.db_zone.id, null)  
}
```

5. Database (Module)



The screenshot shows a Notepad application window titled 'main.tf - Notepad'. The menu bar includes 'File', 'Edit', 'Format', 'View', and 'Help'. The main text area contains Terraform HCL code for two resources: 'azurerm_mysql_flexible_server' and 'azurerm_mysql_flexible_database'. The first resource is configured with various attributes including name, resource group, location, administrator login, password, SKU name, storage MB, version, delegated subnet ID, private DNS zone ID, high availability mode, backup retention days, and tags. The second resource is a database instance named 'epicdb' linked to the first server, with UTF8 charset and collation.

```

File Edit Format View Help

resource "azurerm_mysql_flexible_server" "db" {
  name                = "${terraform.workspace}-epicbook-db"
  resource_group_name = var.rg_name
  location            = var.location
  administrator_login  = var.admin_user
  administrator_password = var.admin_password
  sku_name            = "B_Standard_B1ms"
  storage_mb          = 32768
  version             = "8.0.21"

  delegated_subnet_id = var.subnet_id
  private_dns_zone_id = var.dns_zone_id
  high_availability { mode = "Disabled" }
  backup { retention_days = 7 }
  tags = var.tags
}

resource "azurerm_mysql_flexible_database" "epicdb" {
  name = "epicbook"
  server_id = azurerm_mysql_flexible_server.db.id
  charset = "utf8"
  collation = "utf8_general_ci"
}

```

Administrator: Windows PowerShell (x86)

```

File Edit Format View Help
variable "rg_name" {
  description = "Resource Group name where the DB will be created"
  type        = string
}

variable "location" {
  description = "Azure region for the DB"
  type        = string
}

variable "subnet_id" {
  description = "Subnet ID for the private MySQL Flexible Server"
  type        = string
}

variable "dns_zone_id" {
  description = "Private DNS Zone ID to link to MySQL server"
  type        = string
}

variable "admin_user" {
  description = "Admin username for MySQL"
  type        = string
}

variable "admin_password" {
  description = "Admin password for MySQL"
  type        = string
  sensitive   = true
}

variable "tags" {
  description = "Common tags to apply to resources"
  type        = map(string)
  default     = {}
}

```


The screenshot shows a Notepad window titled "outputs.tf - Notepad". The text inside is Terraform configuration code for defining outputs. It includes three blocks: one for the MySQL server name, one for the fully qualified domain name (fqdn), and one for the database name. Each block uses the 'description' and 'value' attributes to provide context and the actual resource reference.

```

output "mysql_server_name" {
  description = "Name of the MySQL flexible server"
  value      = azurerm_mysql_flexible_server.db.name
}

output "mysql_fqdn" {
  description = "FQDN of the MySQL flexible server"
  value      = azurerm_mysql_flexible_server.db.fqdn
}

output "mysql_db_name" {
  description = "Name of the database created"
  value      = azurerm_mysql_flexible_database.epicdb.name
}

```

6. Compute/App (Module)

Adriana Colaco

 main.tf - Notepad

File Edit Format View Help

```
resource "azurerm_public_ip" "vm_ip" {
  name                = "${terraform.workspace}-vm-ip"
  location            = var.location
  resource_group_name = var.rg_name
  allocation_method   = "Static"
}

resource "azurerm_network_interface" "vm_nic" {
  name                = "${terraform.workspace}-vm-nic"
  location            = var.location
  resource_group_name = var.rg_name

  ip_configuration {
    name                        = "internal"
    subnet_id                  = var.subnet_id
    private_ip_address_allocation = "Dynamic"
    public_ip_address_id        = azurerm_public_ip.vm_ip.id
  }
}

resource "azurerm_linux_virtual_machine" "vm" {
  name                = "${terraform.workspace}-epicbook-vm"
  resource_group_name = var.rg_name
  location            = var.location
  size                = var.vm_size
  admin_username      = var.admin_user
  admin_password      = var.admin_password

  network_interface_ids = [azurerm_network_interface.vm_nic.id]
  disable_password_authentication = false

  os_disk {
    caching              = "ReadWrite"
    storage_account_type = "Standard_LRS"
  }

  source_image_reference {
    publisher = "Canonical"
    offer     = "0001-com-ubuntu-server-jammy"
    sku       = "22_04-lts"
    version   = "latest"
  }

  custom_data = base64encode(file("${path.module}/cloud-init.yml"))
}
```

The screenshot shows a Windows PowerShell terminal window with the following content:

```
Directory: C:\Program Files\Terraform\epicbook\modules\compute

Mode                LastWriteTime         Length Name
----                -
-a----          10/9/2025   3:01 PM              0 cloud-init.yml

PS C:\Program Files\Terraform\epicbook\modules\compute> notepad .\cloud-init.yml
PS C:\Program Files\Terraform\epicbook\modules\compute>
```

Overlaid on the terminal is a Notepad window titled "cloud-init - Notepad" containing the following YAML configuration:

```
File Edit Format View Help
#cloud-config
package_update: true
packages:
  - nginx
  - git
  - nodejs
  - npm
  - mysql-client

runcmd:
  - cd /home/azureuser
  - git clone https://github.com/pravinmishraaws/theepicbook.git
  - cd theepicbook/frontend
  - npm install
  - npm run build
  - cp -r build /var/www/html
  - systemctl enable nginx
  - systemctl restart nginx
```

Variables.tf

The screenshot shows a Windows PowerShell terminal window with the following content:

```
PS C:\Program Files\Terraform\epicbook> cd .\modules\compute\
PS C:\Program Files\Terraform\epicbook\modules\compute> notepad .\variables.tf
PS C:\Program Files\Terraform\epicbook\modules\compute>
```

Overlaid on the terminal is a Notepad window titled "variables.tf - Notepad" containing the following Terraform variable definitions:

```
File Edit Format View Help
# modules/compute/variables.tf

variable "rg_name" {
  description = "Name of the Resource Group where the VM will be created"
  type        = string
}

variable "location" {
  description = "Azure region where the resources will be created"
  type        = string
}

variable "subnet_id" {
  description = "ID of the subnet where the VM will be placed"
  type        = string
}

variable "admin_user" {
  description = "Admin username for the VM"
  type        = string
}

variable "admin_password" {
  description = "Admin password for the VM"
  type        = string
}

variable "vm_size" {
  description = "The size of the virtual machine"
  type        = string
  default     = "Standard_B1s"
}

variable "workspace" {
  description = "Terraform workspace name (used for naming)"
  type        = string
}

variable "tags" {
  description = "Tags to apply to the VM and related resources"
  type        = map(string)
  default     = {}
}
```

Outputs.tf

```
PS C:\Program Files\Terraform\epicbook\modules> cd .\compute\
PS C:\Program Files\Terraform\epicbook\modules\compute> notepad .\outputs.tf
PS C:\Program Files\Terraform\epicbook\modules\compute>
```

```
*outputs.tf - Notepad
File Edit Format View Help
output "vm_public_ip" {
  description = "Public IP address of the EpicBook VM"
  value       = azurerm_public_ip.vm_ip.ip_address
}

output "vm_name" {
  description = "Name of the Virtual Machine"
  value       = azurerm_linux_virtual_machine.vm.name
}

output "vm_admin_user" {
  description = "Admin username for the VM"
  value       = var.admin_user
}

output "vm_fqdn" {
  description = "FQDN of the VM public IP (if DNS label is used)"
  value       = try(azurerm_public_ip.vm_ip.fqdn, null)
}
```

Add the below code to the main.tf in origin folder. It calls all other modules.

```
module "network" {
  source    = "./modules/network"
  location  = local.location_map[local.env]
  tags      = local.common_tags
  allowed_ip = "YOUR_IP_ADDRESS/32"
}

module "database" {
  source      = "./modules/database"
  rg_name     = module.network.rg_name
  location    = module.network.location
  subnet_id   = module.network.mysql_subnet_id
  dns_zone_id = module.network.dns_zone_id
  admin_user   = var.mysql_admin_user
  admin_password = var.mysql_admin_password
  tags        = local.common_tags
}

module "compute" {
  source      = "./modules/compute"
  rg_name     = module.network.rg_name
  location    = module.network.location
  subnet_id   = module.network.public_subnet_id
  vm_size     = local.vm_size_map[local.env]
  admin_user   = var.admin_username
  admin_password = var.admin_password
  tags        = local.common_tags
}
```

}

Adriana Colaco

 main.tf - Notepad

File Edit Format View Help

```
locals {
  env          = terraform.workspace
  common_tags = {
    project = "epicbook"
    env     = local.env
  }

  location_map = {
    dev = "eastus"
    prod = "westeurope"
  }

  vm_size_map = {
    dev = "Standard_B1s"
    prod = "Standard_B2s"
  }
}

module "network" {
  source      = "../modules/network"
  location    = local.location_map[local.env]
  tags       = local.common_tags
  allowed_ip = "YOUR_IP_ADDRESS/32"
}

module "database" {
  source          = "../modules/database"
  rg_name         = module.network.rg_name
  location        = module.network.location
  subnet_id      = module.network.mysql_subnet_id
  dns_zone_id    = module.network.dns_zone_id
  admin_user     = var.mysql_admin_user
  admin_password = var.mysql_admin_password
  tags           = local.common_tags
}

module "compute" {
  source          = "../modules/compute"
  rg_name         = module.network.rg_name
  location        = module.network.location
  subnet_id      = module.network.public_subnet_id
  vm_size        = local.vm_size_map[local.env]
  admin_user     = var.admin_username
  admin_password = var.admin_password
  tags           = local.common_tags
}
```

Terraform configuration

terraform init

```
PS C:\Program Files\Terraform\epicbook> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 3.100"...
- Installing hashicorp/azurerm v3.117.1...
- Installed hashicorp/azurerm v3.117.1 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Program Files\Terraform\epicbook>
```

terraform workspace new dev

```
PS C:\Program Files\Terraform\epicbook> terraform workspace new dev
Created and switched to workspace "dev"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
PS C:\Program Files\Terraform\epicbook>
```

```
Select Administrator: Windows PowerShell (x86)

PS C:\Program Files\Terraform\epicbook> terraform plan
Acquiring state lock. This may take a few moments...
module.network.azure_rm_resource_group: Refreshing state... [id=/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/dev-epicbook-rg]
module.network.azure_rm_virtual_network: Refreshing state... [id=/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/dev-epicbook-rg/providers/Microsoft.Network/virtualNetworks/dev-epicbook-vnet]

Note: Objects have changed outside of Terraform

Terraform detected the following changes made outside of Terraform since the last "terraform apply" which may have affected this plan:

# module.network.azure_rm_virtual_network.vnet has been deleted
resource "azure_rm_virtual_network" "vnet" {
  id = "/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/dev-epicbook-rg/providers/Microsoft.Network/virtualNetworks/dev-epicbook-vnet" -> null
  name = "dev-epicbook-vnet" -> null
  # (9 unchanged attributes hidden)
}

Unless you have made equivalent changes to your configuration, or ignored the relevant attributes using ignore_changes, the following plan may include actions to undo or respond to these changes.

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  create

Terraform will perform the following actions:

# module.compute.azure_rm_linux_virtual_machine.vm will be created
resource "azure_rm_linux_virtual_machine" "vm" {
  admin_password = (sensitive value)
  admin_username = "azureuser"
  allow_extension_operations = true
  bypass_platform_safety_checks_on_user_schedule_enabled = false
  computer_name = (known after apply)
  custom_data = (sensitive value)
  disable_password_authentication = false
  disk_controller_type = (known after apply)
  extensions_time_budget = "PT1H30M"
  id = (known after apply)
  location = "westeurope"
  max_bid_price = -1
}
```

terraform apply -auto-approve

```
PS C:\Program Files\Terraform\epicbook> terraform apply -auto-approve
Acquiring state lock. This may take a few moments...
module.network.azure_rm_resource_group: Refreshing state... [id=/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/dev-epicbook-rg]
module.network.azure_rm_virtual_network: Refreshing state... [id=/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/dev-epicbook-rg/providers/Microsoft.Network/virtualNetworks/dev-epicbook-vnet]

Note: Objects have changed outside of Terraform

Terraform detected the following changes made outside of Terraform since the last "terraform apply" which may have affected this plan:

# module.network.azure_rm_virtual_network.vnet has been deleted
resource "azure_rm_virtual_network" "vnet" {
  id = "/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/dev-epicbook-rg/providers/Microsoft.Network/virtualNetworks/dev-epicbook-vnet" -> null
  name = "dev-epicbook-vnet" -> null
  # (9 unchanged attributes hidden)
}

Unless you have made equivalent changes to your configuration, or ignored the relevant attributes using ignore_changes, the following plan may include actions to undo or respond to these changes.

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  create

Terraform will perform the following actions:

# module.compute.azure_rm_linux_virtual_machine.vm will be created
resource "azure_rm_linux_virtual_machine" "vm" {
  admin_password = (sensitive value)
  admin_username = "azureuser"
  allow_extension_operations = true
  bypass_platform_safety_checks_on_user_schedule_enabled = false
  computer_name = (known after apply)
  custom_data = (sensitive value)
  disable_password_authentication = false
  disk_controller_type = (known after apply)
  extensions_time_budget = "PT1H30M"
  id = (known after apply)
  location = "westeurope"
}
```

terraform workspace new prod
terraform apply -auto-approve

7. Terraform Platform Features

portal.azure.com/#@adcola13gmail.onmicrosoft.com/resource/subscriptions/4aaefc3f-e730-4757-944c-df72e636bfe6/resourceGroups/dev-epi...

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot adcola13@gmail.com DEFAULT DIRECTORY (ADCOLA13...)

Home > Resource Manager | Resource groups << >>

Resource Manager | Resource groups << >> Default Directory

Search

+ Create Manage view <> ...

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Help

You are viewing a new version of Browse experience. Click here to access the old experience.

Name 1

dev-epicbook-rg

NetworkWatcherRG

tfstate-rg

Showing 1 - 3 of 3. Display count: auto

Add or remove favorites by pressing Ctrl+Shift+F

dev-epicbook-rg Resource group

Search

+ Create Manage view <> Delete resource group Refresh ...

Group by none <>

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Deployments

Security

Deployment stacks

Policies

Properties

Locks

Cost Management

Monitoring

Automation

Help

Essentials

Resources Recommendations

Filter for any field... Type equals all Location equals all + Add filter

Name 1 Type Location

dev-epicbook-vnet Virtual network East US

dev-mysql-nsg Network security group East US

dev-public-nsg Network security group East US

mysql.database.azure.com Private DNS zone Global

Showing 1 - 4 of 4. Display count: auto

Give feedback

```
Administrator: Windows PowerShell (x86)

This is a bug in the provider, which should be reported in the provider's own issue tracker.

Error: updating Public IP Address (Subscription: "4aaefc3f-e730-4757-944c-df72e636bfe6"
Resource Group Name: "dev-epicbook-rg"
Public IP Address Name: "dev-vm-ip"): performing CreateOrUpdate: unexpected status 400 (400 Bad Request) with error: IPv4BasicSKU public IP addresses for this subscription in this region.
with module.compute.azure_rm_public_ip.vm_ip,
on modules\compute\main.tf line 1, in resource "azure_rm_public_ip" "vm_ip":
Resource Group Name: "dev-epicbook-rg"
Virtual Network Name: "dev-epicbook-vnet"): unexpected status 404 (404 Not Found) with error: ResourceNotFound: The Resource 'Microsoft.Network/virtualNetworks/dev-epicbook-vnet' under resource group 'dev-epicbook-rg' was not found. For more details please go to https://aka.ms/ARMResourceNotFoundFix
with module.network.azure_rm_subnet.public_subnet,
on modules\network\main.tf line 14, in resource "azure_rm_subnet" "public_subnet":
14: resource "azure_rm_subnet" "public_subnet" {

Error: creating Subnet (Subscription: "4aaefc3f-e730-4757-944c-df72e636bfe6"
Resource Group Name: "dev-epicbook-rg"
Virtual Network Name: "dev-epicbook-vnet"
Subnet Name: "mysql-subnet"): performing CreateOrUpdate: unexpected status 404 (404 Not Found) with error: ResourceNotFound: The Resource 'Microsoft.Network/virtualNetworks/dev-epicbook-vnet' under resource group 'dev-epicbook-rg' was not found. For more details please go to https://aka.ms/ARMResourceNotFoundFix
with module.network.azure_rm_subnet.mysql_subnet,
on modules\network\main.tf line 21, in resource "azure_rm_subnet" "mysql_subnet":
21: resource "azure_rm_subnet" "mysql_subnet" {

Error: creating/updating Private Dns Zone (Subscription: "4aaefc3f-e730-4757-944c-df72e636bfe6"
Resource Group Name: "dev-epicbook-rg"
Private Dns Zone Name: "mysql.database.azure.com"): performing CreateOrUpdate: unexpected status 409 (409 Conflict) with response: {"code":"Conflict","message":"Another operation is pending for requested object. Operation group '\operations\groups\id\subscriptions\4aaefc3f-e730-4757-944c-df72e636bfe6\resourceGroups\dev-epicbook-rg\privateDnsZones\mysql.database.azure.com' was created simultaneously by another operation."}
with module.network.azure_rm_private_dns_zone.db_zone,
on modules\network\main.tf line 89, in resource "azure_rm_private_dns_zone" "db_zone":
89: resource "azure_rm_private_dns_zone" "db_zone" {

Releasing state lock. This may take a few moments...
```

Note to Mentor: I tried my level best to bypass the Public IP issue , also faced by a lot of other team members.

Message: "Cannot create more than 0 IPv4 Basic SKU public IP addresses for this subscription in this region."

A lot of the resources were getting created.

I deleted all the older resource groups. However, nothing seemed to make it work. I will give this assignment a shot at a later time.

LinkedIn Post1:

[https://www.linkedin.com/posts/adrina-colaco_terraform-multi-cloud-multi-region-storage-activity-7381940869702086656-](https://www.linkedin.com/posts/adrina-colaco_terraform-multi-cloud-multi-region-storage-activity-7381940869702086656-Asuw?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM)

[Asuw?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM](https://www.linkedin.com/posts/adrina-colaco_terraform-multi-cloud-multi-region-storage-activity-7381940869702086656-Asuw?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM)

LinkedIn post2:

[https://www.linkedin.com/posts/adrina-colaco_terraform-devops-react-activity-7382100797960052736-Vb2-](https://www.linkedin.com/posts/adrina-colaco_terraform-devops-react-activity-7382100797960052736-Vb2-?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM)

[?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM](https://www.linkedin.com/posts/adrina-colaco_terraform-devops-react-activity-7382100797960052736-Vb2-?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM)

Blog 1:

<https://medium.com/@adcola13/terraform-multi-cloud-multi-region-storage-setup-1ce0e8f4ec80>

Blog 2:

<https://medium.com/@adcola13/deploy-a-react-app-using-terraform-workspaces-dev-prod-environments-985181f3d949>

The End

P.S. This document is part of the **FREE DevOps Micro Internship (DMI) Cohort** run by [Pravin Mishra](#). You can start your DevOps journey for free from his [YouTube Playlist](#).